

Equities

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A Guide to the World of Metals & Mining

Mining Made Simple

- **World of Metals & Mining** — The report aims to provide a simple, yet comprehensive, overview of the major components of the mining world. We explore the full spectrum of the metals & mining world, from the formation of ore bodies, through to exploration methods, extraction & processing techniques, and including the end use of metals.
- **Metal by metal guide** — Each of the key base, bulk and precious metals are covered with analysis focusing on the global resource base, extraction methods, processing techniques, key mines & producers, end consumption and supply/demand dynamics.
- **Other key issues** — Along with the overview of the major components of the mining world we also touch on some of the major issues facing the mining world as we know it.
 - **Mining a global industry:** Mining is a truly global industry and the fortunes of many countries have been developed on the back of their mining industry. Mining stocks also dominate many of the world's largest bourses and as such are an integral part of the financial system. From a natural wealth perspective (the dollar value of reserves in the ground), South Africa is the richest nation with over US\$2.5 trillion of non-energy monetary reserves. Its number 1 position is largely driven by PGM's, where South Africa accounts for nearly 90% of global reserves which are valued at cUS\$2.2 trillion.
 - **Evolving face of global mining:** Resource rich countries no longer control as much of the world's natural wealth as they used to. In 2004, the developed mining markets (Australia, UK, USA, Canada and South Africa) accounted for 84% of global mining market cap. Today that figure has fallen to 60%. A growing Chinese mining market has been the prime driver. In 2004, China had 8 listed mining companies with a total EV of US\$19bn. In 2011, it has 33 companies with over US\$320bn of EV.
 - **Rising royalties a risk:** The continued rise in commodity prices and miners profitability coupled with the struggling economies of many countries has resulted in higher taxation of miners emerging as a popular political platform. We do not see this theme disappearing and believe that the risk is firmly to the upside for royalties and taxation of resource companies. Over the past year, Australia, Brazil, China, Peru and Russia (to name a few) have all looked to increase the tax burden of miners.

■ Industry Overview

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See Appendix A-1 for Analyst Certification, Important Disclosures and non-US research analyst disclosures.

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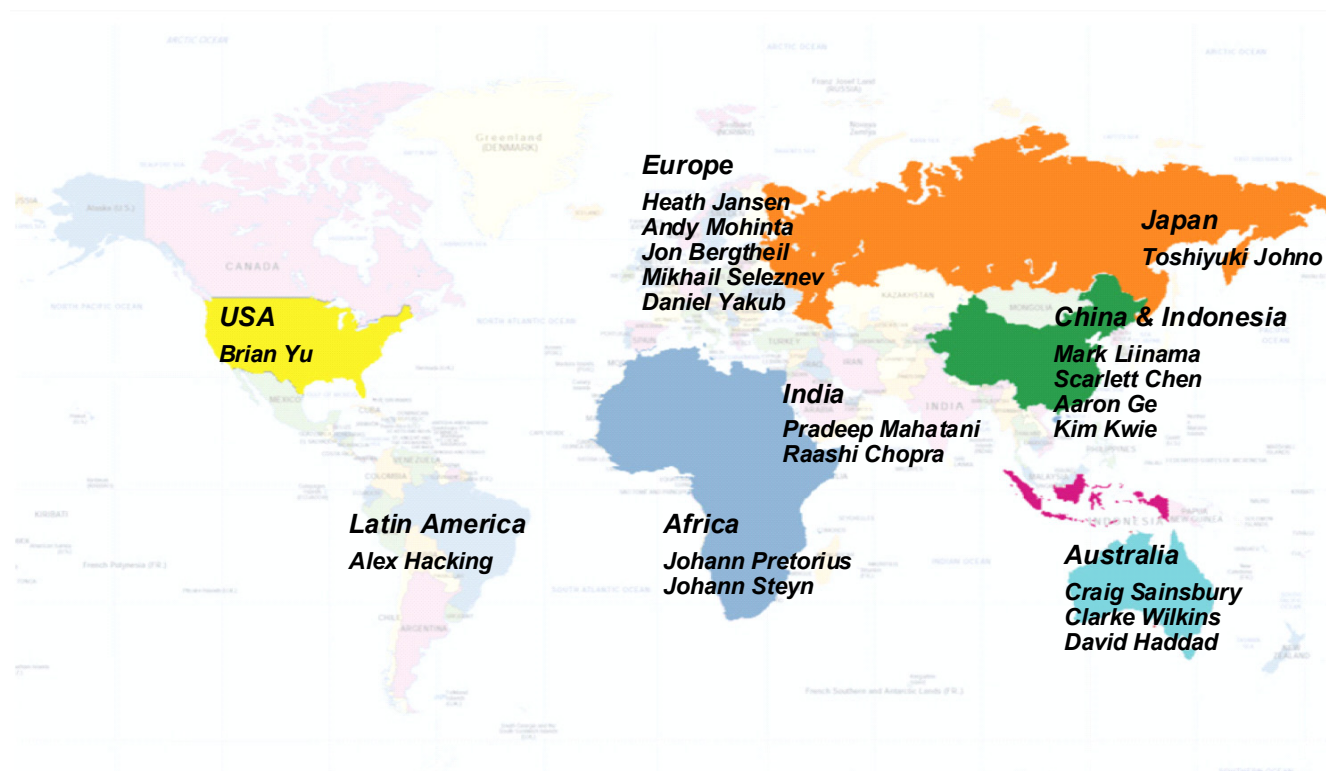
Introduction

This report aims to provide a simple, yet comprehensive, overview of the major components of the mining world. We explore the full spectrum of the mining world, from the formation of ore bodies, through to exploration methods, extraction and processing techniques, and including the end use of metals.

We also examine some of the major macro trends (royalties, tax, sovereign risk, M&A) that are impacting the stocks in the mining world.

The report is introductory in nature and represents a small part of the industry expertise that resides amongst Citi's Global Mining Research Team. Our team is prepared to assist with any further information or investment decisions

Figure 1. Citi's Global Mining Research Team



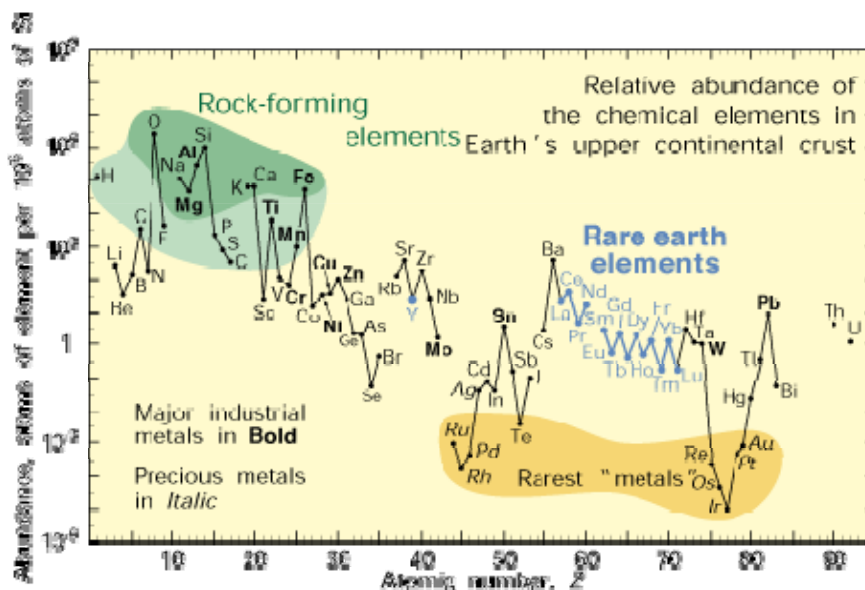
Source: Citi Investment Research and Analysis

Section 1: Geology, Reserves and Resources

Mineral Formation

The earth's crust is essentially one big mineral deposit, but economic concentration of metals are a rarity and their existence is a function of the geological activity of the earth. Essentially all minerals have been deposited in the earth's crust from the intrusion of the magma in one way or another. The formation of economic concentration of metals, and the presence of one mineral versus another is dependent predominantly on the geochemistry of the magma and circulating fluids as well as the associated temperature and pressure. The table below looks at the abundance of the various minerals in the earth's crust.

Figure 2. Crustal abundance of industrial metals



Source: USGS and Citi Investment Research and Analysis

Ore Types

Minerals and metals are deposited in to the host rock in varying ways. The way in which the metals are deposited is very significant in determining the quality of the ore body and essentially the profitability and value of the deposit. There are four main ways in which the ore can be deposited into the host rocks.

Massive: A deposit where large proportion of the host rock contains mineralization. The 'massive' essentially refers to the uniformity of the mineralisation through that host rock. Commonly associated with sulphide deposits.

Disseminated: Similar in nature to a massive deposit, but where the deposit lacks the uniformity of a massive deposit.

Veined: Mineralisation that is concentrated along fractures and faults in the rock. The fractures allowed the magma to intrude into the rock where it cooled and deposited the minerals. Vein type geology is commonly associated with gold.

Stockwork: A deposit of intense vein mineralisation over a large area. Veins are generally small and intersect each other making the removal of individual veins impossible.

Coal is an outlier as it has been formed by sedimentary drivers and not magmatic forces.

The above ore emplacement methods lead to a multitude of further 'ore deposit types' that we associate with the mines we know. The table below provides an overview of the major metals and the geological complex that they can be associated with.

Figure 3. Major Metals and their deposit style

Metal	Deposit Type	Snapshot	Mine example
Copper	IOCG	Provincial style deposit where several deposits of similar 'style' are found in same region. Hosted in igneous rock and form in pipe like manner.	Olympic Dam, Prominent Hill, Carrapeteena
	VMS	Volcanic Massive Sulphide deposits (VMS) tend to be small and relatively high grade. Deposits tend to be found in clusters within regions.	Kidd Creek, Brunswick, Sandfire
	Porphyry	Associated with tectonic action and large scale magmatic intrusion in to the earths crust.	Grasberg, Escondida, Cadia, Chuquibambilla
Nickel	Laterite	Formed in tropical climates through the weathering process.	Goro, Ravensthorpe, Murrin Murrin
	Sulphide	More common nickel deposit which is associated with the base interaction of flowing Lava.	Mt Keith, Sudbury
Iron Ore	Banded Iron Formation	Layers of sedimentary rock containing iron ore.	Mt Tom Price, Brockman
	Channel Iron Deposits	Formed through erosion of sedimentary rocks which is then deposited in river environments.	Yandi
Gold	Epithermal vein	Magmatic intrusions into the host rock.	Lihir, Co-O
	Placer deposits	Concentrated deposition of gold through weathering of rock and marine transportation of the gold.	Witswatersrand region in South Africa
	Shear Hosted	Gold concentrated in quartz veins through mechanical forces of faulting and shearing.	Higginsville

Source: Citi Investment Research and Analysis

Copper

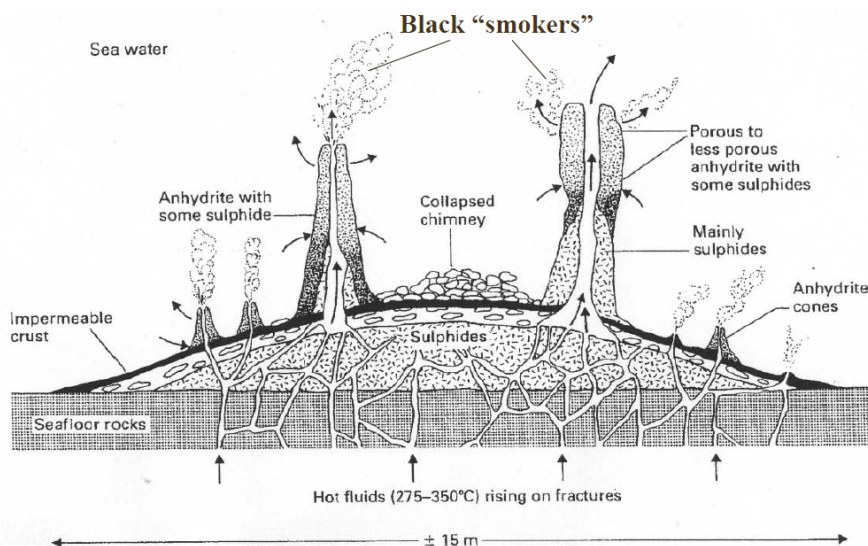
■ Iron Oxide Copper Gold (IOCG)

So named because of the association with iron oxides, rather than the rock formation process, IOCG deposits tend to be large scale ore-bodies which can occur regionally many times across similar geological formations. The iron association provides high magnetic and gravity association making geophysical exploration methods important. Formed by magmatic and hydrothermal activity.

■ Volcanogenic Massive Sulphide (VMS)

Created by metal-rich submarine hydrothermal solution discharges around seafloor volcanoes, VMS deposits commonly occur in a number of lens shaped deposits in close proximity to each other.

Figure 4. VMS Deposit Formation

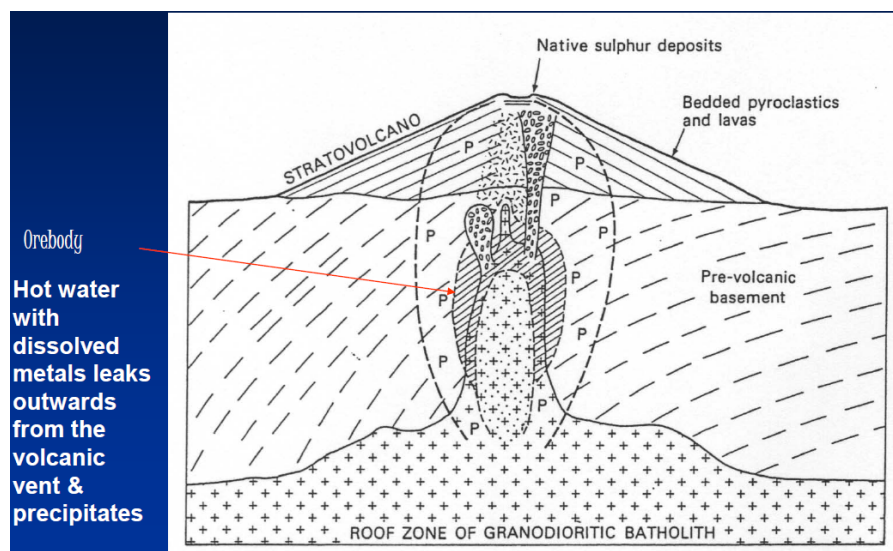


Source: Western Australian School of Mines

■ Porphyry

Formed by the cooling of magma to rock of a volcanic intrusion. Magma plumes ascend through the earth's surface typically near active tectonic plate collisions. Currently the largest source of copper ore, porphyry deposits are typically mined using an open cut mining method. Deposits tend to be large low grade deposits but with uniform (massive) mineralization.

Figure 5. Porphyry Deposit Formation



Source: Western Australian School of Mines

Nickel

■ Laterite

Accounting for ~73% of global nickel resources, laterites are formed from the weathering of nickel sulphide ores to oxide materials. The concentration takes place as host rock is eroded away. Lateritic ore deposits tend to be very large, low grade, flat in shape and close to the surface, driven by the process of weathering over millions of years. Due to their proximity to the surface these deposits tend to be mined using open-cut methods. While easy to mine, the disadvantage of laterite deposits is that chemical separation is a more difficult and costly process, necessitating large economies of scale to provide economic viability.

■ Sulphides

While accounting for a lower portion of global nickel resources (~27%) these deposit types account for the majority of current nickel production. Sulphide deposits tend to be higher grade and as such are more economically attractive deposits. These deposits are formed in lava flows and are typically found deeper underground. Chemical separation of nickel sulphides is a simple process using a physical separation process called flotation.

Iron Ore

■ Banded Iron Formations (BIF)

An iron rich deposit type consisting of alternate layers of iron minerals and a silica rich rock type called chert. The structures are formed as sedimentary layers on the seafloor with the varied layers a result of varied climatic conditions as each new layer of sediment was deposited. These formations are common to the Pilbara.

■ Channel Iron Deposits (CID)

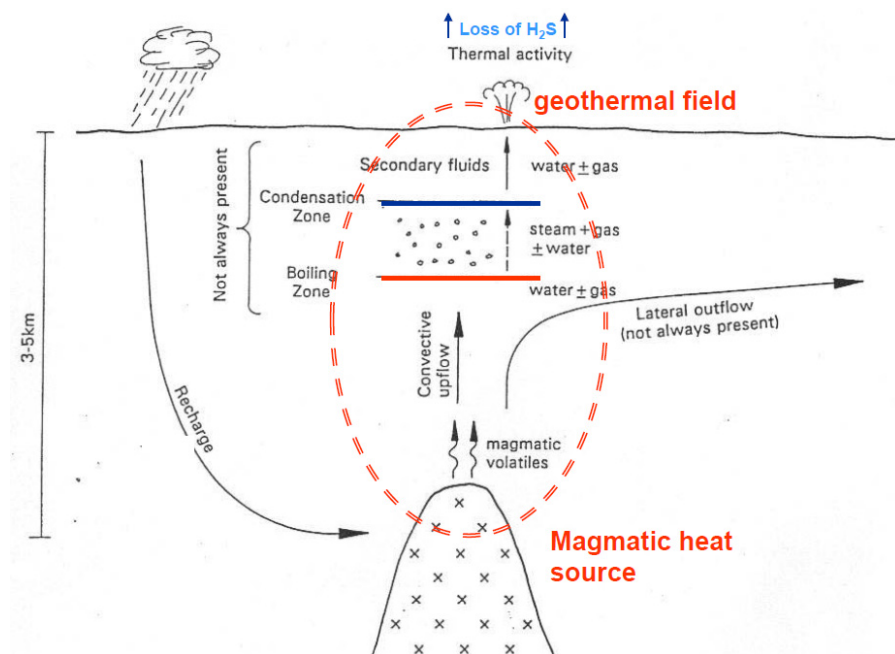
CID deposits are formed via weathering of BIF deposits leading to concentrations of iron minerals typically in river channels. Believed to be unique to Western Australia, the deposits usually consist of low flat-topped hills concealed under a cover of more ancient rocks.

Gold

■ Epithermal

Typically shallow deposits formed by geothermal systems at or near the earth's surface. Volcanic fluids flowing through fractures in host rocks and minerals precipitate as fluids cool when they approach the surface. Given the mechanics of the deposit formation these deposits occur in veins with stockworks and disseminations. These deposits are not always gold but are typically so.

Figure 6. Basic Epithermal System

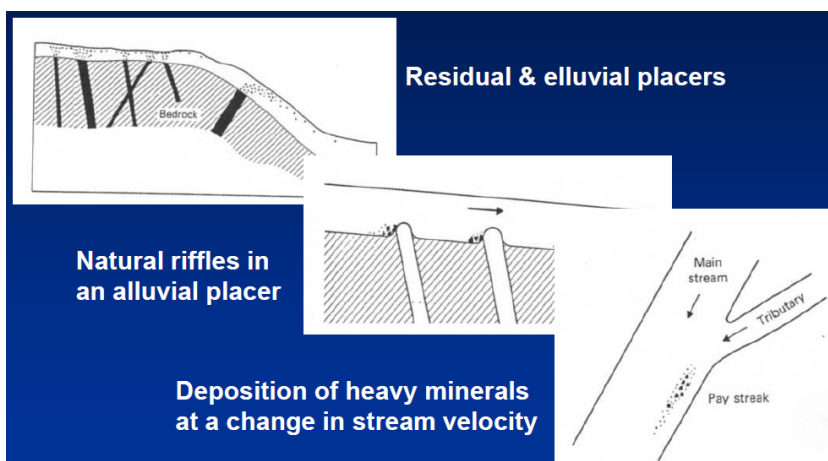


Source: Western Australian School of Mines

■ Placer Deposits

Weathering of mineralized rock over millions of years erodes surrounding material and transports mineral particles. Mineralisation then accumulates in topographical features that capture the heavier gold particles such as natural features along a watercourse.

Figure 7. Placer Deposits



Source: Western Australian School of Mines

■ Shear and Structural Hosted

These deposits are typically located in areas where the Earth's crust has undergone significant stresses and strains resulting in folding and faulting of the rocks. The process of gold mineralisation involves fluids travelling through a source rock and

dissolving a suite of metals depending on the temperature, pH and geochemistry. The metal bearing fluids travel through structural weaknesses in the rock formed during metamorphism and given the right conditions (temperature, pressure and geochemistry) the gold precipitates out leaving a concentration of metals within the rock. The mineralisation is classified as shear/structural hosted as the concentration of gold is within a structural zone and the orientation of the deposit is typically dependent on the features of the original structure (orientation and other structural features). Examples of these deposits can be found in Achaean greenstone belts such as the Yilgarn and Pilbara Cratons in Western Australia.

Associated Mineral assemblage

As explained above, the various elements are deposited into the earth's crust in a myriad of ways. As the formation process can differ, so too can the main ore minerals that host the economic elements of the underlying commodity.

Figure 8. Main ore minerals

Mineral	Main Ore Minerals
Bauxite	Gibbsite ($\text{Al}(\text{OH})_3$), Diaspore ($\text{AlO}(\text{OH})$), Boehmite ($\text{AlO}(\text{OH})$)
Copper	Chalcocite (Cu_2S), Chalcocite (Cu_2S), Bornite (Cu_5FeS_4), Malachite ($\text{Cu}_2\text{Co}(\text{OH})_2$)
Nickel	Pentlandite ($\text{Fe,Ni}_9\text{S}_8$), Pyrrhotite (FeS), Niccolite (NiAs)
Iron Ore	Magnetite (Fe_3O_4), Hematite (Fe_2O_3), Siderite (FeCO_3)
Rare Earths	Bastnasite ($\text{Ce,Lu}(\text{CO}_3)\text{F}$), Monazite ($\text{Ce,Lu,Nd,Th}(\text{PO}_4)$)
Zinc	Sphalerite (ZnS), Smithsonite (ZnCO_3)
Uranium	Uranite (UO), Coffinite ($\text{U}(\text{SiO}_4)$)
Molybdenum	Molybdenite (MoS)

Source: Citi Investment Research and Analysis

What's a good grade

The grade profile of a similar type of deposits vary massively given the different formation process, host rocks, size, etc. Thus it is very difficult to classify what is a good or bad grade for a deposit or commodity type, as even low grades can deliver economical returns in favorable geological settings. The table below attempts to loosely assign what is a good and poor grade for a deposit.

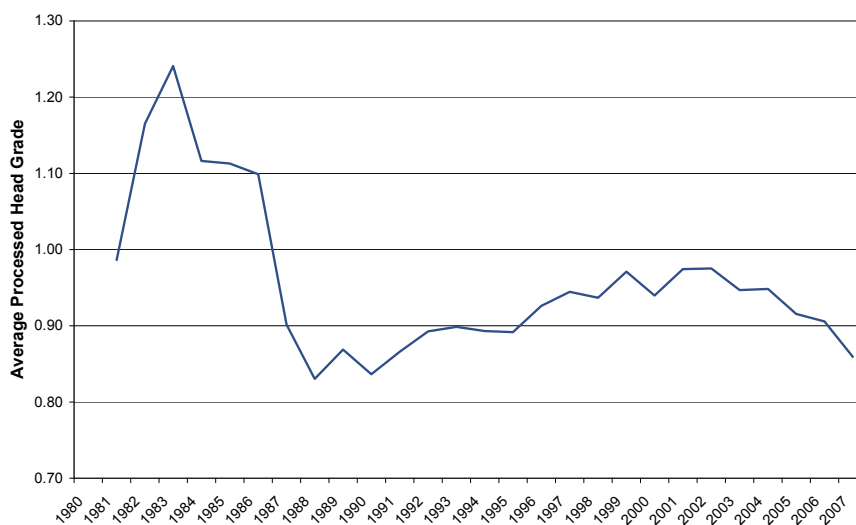
Figure 9. Grade profile of various commodity and deposit types

Deposit Type	Unit	Low	Average	Good
Iron Ore	%	<50%	50%	>60%
Copper - Porphyry	%	<0.5%	0.50%	>1%
Copper - IOCG	%			
Copper - Open Pit	%	<0.5%	0.50%	>1%
Copper - Underground	%	<1%	1%-2%	>2%
Gold - epithermal	g/t	1g/t	4g/t	>6g/t
Gold - sulphide	g/t			
Gold - oxide	g/t			
Nickel - laterite	%	<1%	1-2%	>2%
Nickel - sulphide	%	<1%	2-4%	>4%
Uranium	ppm	400		1500

Source: Citi Investment Research and Analysis

Whilst the table above gives an indicative guide to grades at various commodities and ore types, there is a clear trend towards falling grades across most operations. The grade decline has been derived through a combination of higher grade operations being mined first and the higher commodity prices making lower grade operations more economic. The chart below looks at the decline in the average copper grade.

Figure 10. Average processed head grade



Source: Brook Hunt, company reports and Citi Investment Research and Analysis

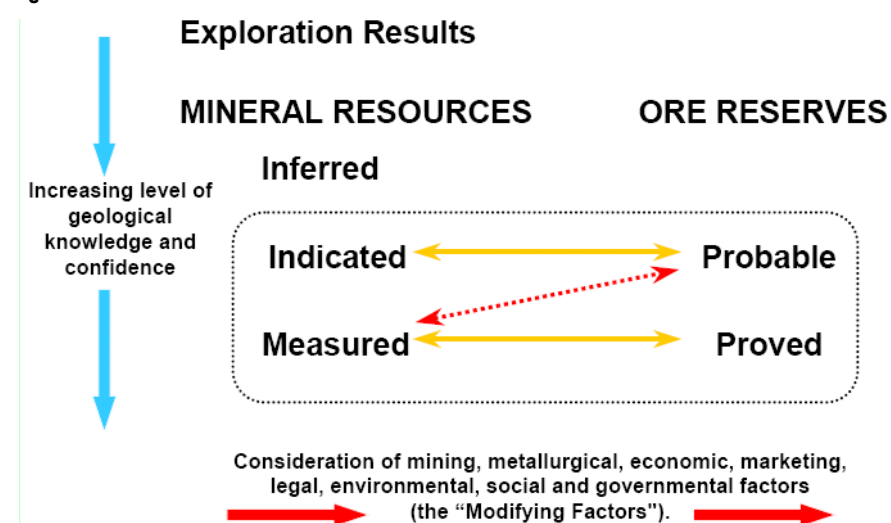
Reporting Reserves and Resources

The international mining community has been working towards a set of uniform reporting standards since 1994. Since 2002, the Combined Reserves International Reporting Standards Committee (CRIRSCO) has been coordinating the development of standards for the definition and reporting of exploration results, mineral resources and mineral reserves between member countries. The international definitions have been accepted as part of national reporting codes by Australia, New Zealand, South Africa, Canada, Peru and Chile. The national codes essentially provide a framework for the classification of tonnage and grade estimates according to geological confidence, technical factors and economic viability. One of the key requirements of the codes is that the public reports must be based on work performed by a recognised competent person. The codes provide extensive guidelines on the criteria to be considered when classifying reserves and resources. However, ultimately, the classifications are determined by the professional judgment of the competent person.

The JORC Code

The Australian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves (the JORC Code) has been widely adopted by the global mining industry and used as a model for other national codes. An overview of the different JORC classifications is provided below.

Figure 11. Classification of Resources and Reserves



Source: MEGWA, JORC and CIRA

The classification of mineral resources is based purely on the level of geological knowledge and confidence:

- Measured Resources are where the tonnage, density, shape, physical characteristics, grade and mineral content can be estimated with a high level of accuracy. Measured resources must therefore be supported by extensive drilling and sampling so that the shape and characteristics of the ore body can be firmly established.
- Indicated and Inferred resources are as above but where the estimates can only be provided with a reasonable and low level of confidence respectively.

Reserve classifications are based on a combination of the type of resource (as defined above) and a range of factors (known as 'Modifying Factors') that affect extraction including mining, metallurgical, marketing, legal, environmental, social and governmental considerations:

- Proved Reserves are the economically mineable part of a Measured Resource.
- Probable Reserves are as above but apply to Indicated Resources (and in some cases measured resources where there are uncertainties over some of the Modifying Factors).
- Inferred Resources cannot be converted to reserves.

Exploration Results

In respect of the reporting of exploration targets, the JORC Code provides the following guidance:

- The terms resources and reserves must not be used.
- Quantity and grade must be expressed as ranges rather than exact amounts.
- There must be appropriate explanatory notes and caveats with regards to the basis of the estimates and the uncertainty surrounding them.

NI 43-101

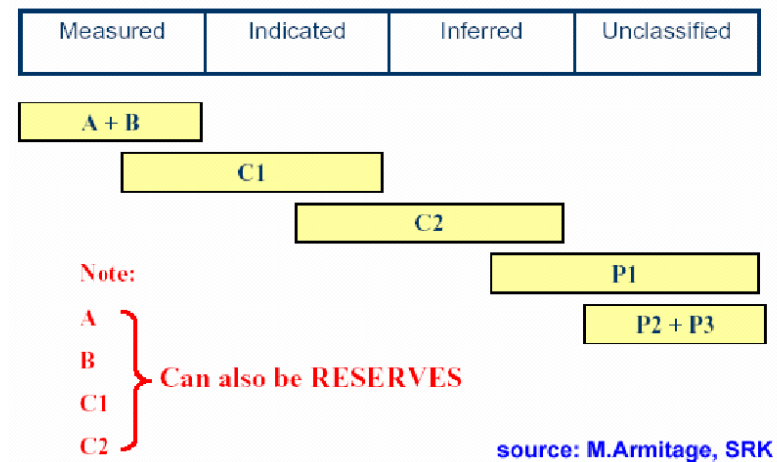
National Instrument 43-101 (NI 43-101) is effectively Canada's equivalent of the JORC code. The NI is a strict set of rules for how public companies disclose information about mineral projects that govern companies listed on the Canadian exchange. Whilst it is effectively interchangeable with the JORC system in Australia (and SAMREC in South Africa) there are few subtle differences.

- JORC is effectively a code for the reporting of mineral reserves and resources whilst the 43-101 requires substantially more technical disclosures.
- 43-101 is a requirement of the Canadian Securities Authority. JORC is separate from any financial exchange and is a system laid down by the Joint Ore Reporting Committee which is independent to the ASX.
- 43-101 information disclosure exceeds that of JORC.

Russian Resource / Reserve System

In Russia an alternative system (GKZ), which is not entirely comparable with the JORC Code, is used to classify reserves and resources. Mining companies in Russia must report reserves and resources to the state authority (GKZ) under the Russian code of reporting. Below we present a broad reconciliation between the two methods:

Figure 12. Russian vs. JORC Reporting System



Source: Peter Hambro Mining

Mineral Exploration

The process of finding new deposits involves several sequential stages, where each subsequent stage becomes more extensive and expensive than the previous stage:

Mapping

The first stage involves locating a potential new resource and creating a geological profile of the area. To do this, land areas of high mineral potential are targeted and geological 3D maps are created. An example of a 3D model map is provided below:

This mapping process can be approached from several different angles (often in combination):

- Geophysicists fly air surveys over areas to build up electro-magnetic patterns. Once complete, the area map will show anomalous patterns that can be interpreted in terms of mineral distributions and can then be targeted for further investigation.
- Geochemists attempt to identify deposits by taking samples from streams, drainages and soil. Once areas of interest have been identified more detailed sampling will follow.
- Geologists use published maps, air photographs and satellite images in order to identify potential deposits for further exploration based on their knowledge of rock formations and mineralisations.

Surveys

The early stage of an exploration program is focused on finding a target to drill. Most early stage exploration programs focus on magnetics or geophysics.

- Magnetic Survey: Measurements of the earth's magnetic field (captured via airborne or ground surveys) seeking to reveal areas where the field is not uniform in intensity. Useful where an orebody may be associated with magnetic rocks (eg. Magnetite) or is potentially under cover.
- Electrical Methods & Induced Polarisation: Measures the conductivity & resistivity of rocks via an induced electro-magnetic field or by injecting an electrical current and measuring the decay of current. Particularly useful in exploration for disseminated sulphide mineralization and porphyry's. Native metals, metal oxides and metal sulphides are strongly conductive.
- Gravity survey: Variations in the density of near-surface rocks generate variations in the gravitational field which can be measured to provide information on sub-surface geology.
- Radiometric: Often combined with aeromagnetic surveys, seeks to measure radiation levels which can be useful in determining rock types or searching for radioactive mineral resources.
- Seismic: Using acoustic waves to map the structure of rocks below surface by measuring reflection/refraction of the pulses back to surface. Boundaries between rock formations will generate these reflections, and the wave's velocity can indicate the materials density, rigidity or hardness.
- Topographic: The shape of the ground surface which provides information on the historical formation of regional structures and indicate areas where mineralization may have been more likely to accumulate.

- **Nearology:** The assumption that similar structures/rock types will exist in close proximity to each other and thus focusing exploration on areas which are closely located to existing ore bodies.

Drilling

Drilling is used to form a detailed picture of the ore body. Drilling methods fall into two broad categories:

- **Percussion drills** break up the rock as the hole is drilled producing rock chips which are then flushed to the surface by circulating fluids (wet ground) or by using compressed air (dry ground). Percussion drilling can be used to depths of up to 300m but is typically used for short holes of less than 50m. RC (Reverse Circulation drilling) is a sub set of percussion drilling.
- **Core drills** cut around a central core which is then retrieved forming a continuous rock sample. Core drilling can be used to depths of over 3,000m. Commonly referred to as diamond drilling.

The samples are collected and taken away for testing. Drilling programs will progress from widely-spaced drill holes to tightly-spaced drill holes that can more accurately determine the shape and grade of the ore body.

Figure 13. Drill rig in action



Source: Licensed from Shutterstock

To the lab

After the exploration target has been drilled the samples are sent to a laboratory where a series of physical and chemical tests determine the composition of the sample and the concentration of mineral. The process is referred to as assaying. The assaying process varies in length depending on whether the sample is from an exploration program or from on site reconciliation of an ore body. But typically it takes a period of weeks to process a batch of assays.

Whilst all assay methods are focused on determining the composition of metal in the rock, there are several different types of processes used.

- **Microscopic analysis:** Thin slices of the rock are studied under microscope to determine the type of mineralogy
- **Spectroscopy:** Looks at electromagnetic radiation wavelength to determine mineral assemblages
- **Sizing:** Uses the different metallurgical properties and weights of minerals to separate the materials
- **Fire:** Mainly used in gold where heat is used to 'melt' the mineral away from the host rock

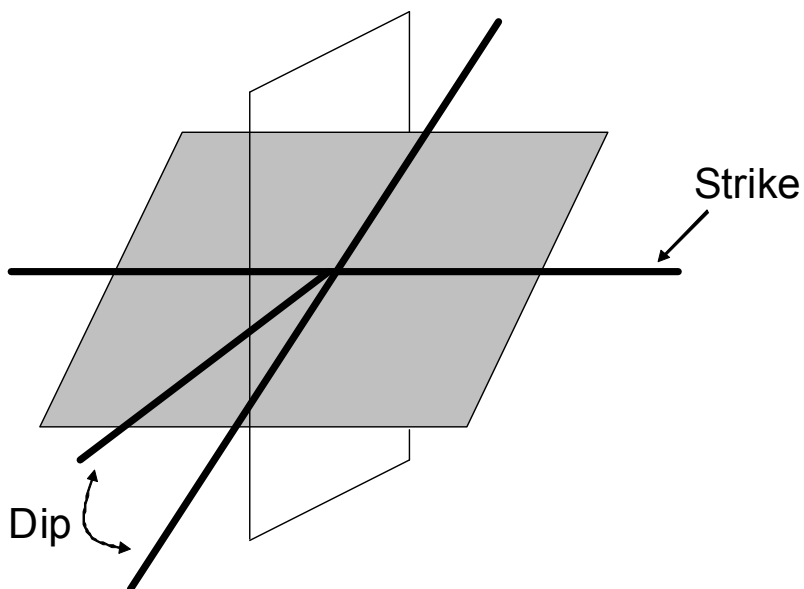
The terminology of a deposit

Most rocks are deposited horizontally, however earth movements over time tilt and fold them into an array of shapes and orientations. The following key terms are used to describe the geometry of an orebody:

Orebody Orientation

- **Strike** – The orientation of the orebody in the horizontal plane (length of an ore body)
- **Dip** – The angle of the orebody from the horizontal (perpendicular to strike)

Figure 14. Strike and Dip

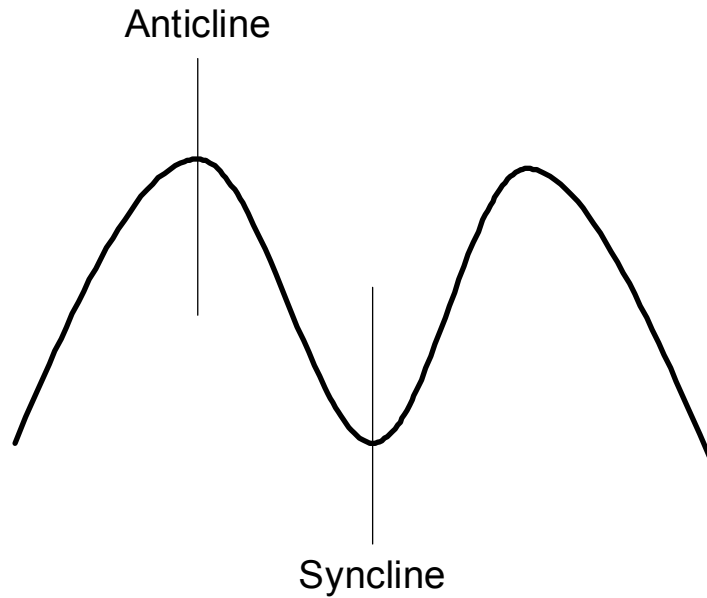


Source: Citi Investment Research and Analysis

Folds & Deformations

- Anticline – up-buckles – oldest rocks in the core
- Syncline – down-buckles – youngest rocks in the core
- Plunge – bearing and inclination of fold axes

Figure 15. Rock Folding Nomenclature



Source: Citi Investment Research and Analysis

Section 2: Mining and Metallurgy

The Mining & Metallurgy Process Flow Sheet

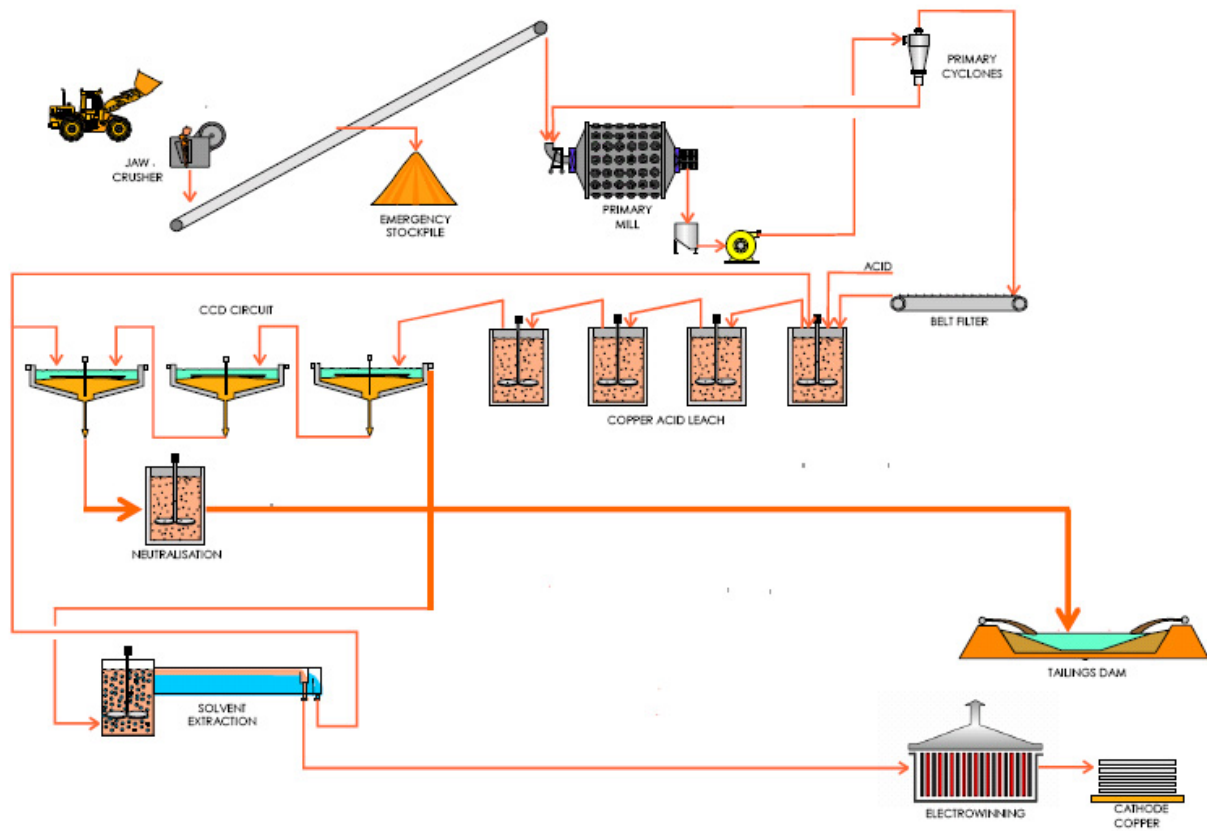
The mining and processing fundamentals for most metal deposits are very similar with only slight tweaks in process to account for the different nature of the geology or metallurgy of the material to be extracted. Iron ore and coal, due to their bulk nature have slightly different and essentially easier processing paths. The table below provides an overview of the mining & metallurgical path.

Figure 16. The Mining and Metallurgy Flowchart

Activity	Process	Definition
Mining	Grade Control	Identifies the economic concentration of the metal within the deposit and allows the miners to target the ore versus the waste. The first stage of the extraction process
	Blasting	Most deposits are mined through some method of blasting. Blasting is a precise science and allows the economic ore to be separated from the waste rock. Tight blasting parameters can reduce the amount of dilution of the ore (the amount of waste that comes along with the ore) and thus increase the economic returns of the project. The ore is transported away from the blasting area (usually through a combination of truck and shovel) and taken to the next step in the process.
Liberation	Crushing	Crushers take the recently blasted material and reduce it in size, usually by a factor of 5x-7x. Various crushing stages can occur with most mines having at least a primary and secondary crusher.
	Milling	The rock from the crusher is introduced into a mill (large steel cylinder) which rotates and crushed the ore into smaller particles. Particle sizes from the mills depend on the concentration & extraction process they will follow, but sizes typical range from 10mm to 50mm in size. The smaller size of the particle also increases surface area which provides for better processing in the concentration and extraction steps.
Concentration	Flotation	The step is focused on increasing the concentration of the mineral in the feedstock to reduce the cost of the final recovery/extraction/refining process. Most metals pass through some form of concentration phase. The most common form of concentration is the flotation process. Other forms of concentration can include gravity separation and magnetic separation.
Extraction	Pyro-Metallurgical	Involves the treatment of ores at high temperature to produce pure metals or intermediate compounds (e.g. copper concentrate). Generally sulphide ores are passed through this process.
	Hydro Metallurgical	Essentially a leaching process. Exploits aqueous chemistry for the recovery of metals from their ores. Generally oxidized ores are passed through this process.
Refining		This is the last stage in the process and takes the metal recovered from either the hydro or pyro metallurgical processes and further refines it until it's in an almost pure state (99.99% metal). The pure metal state is commonly referred to as a cathode. There are a variety of refining techniques which are dependent on the metal.

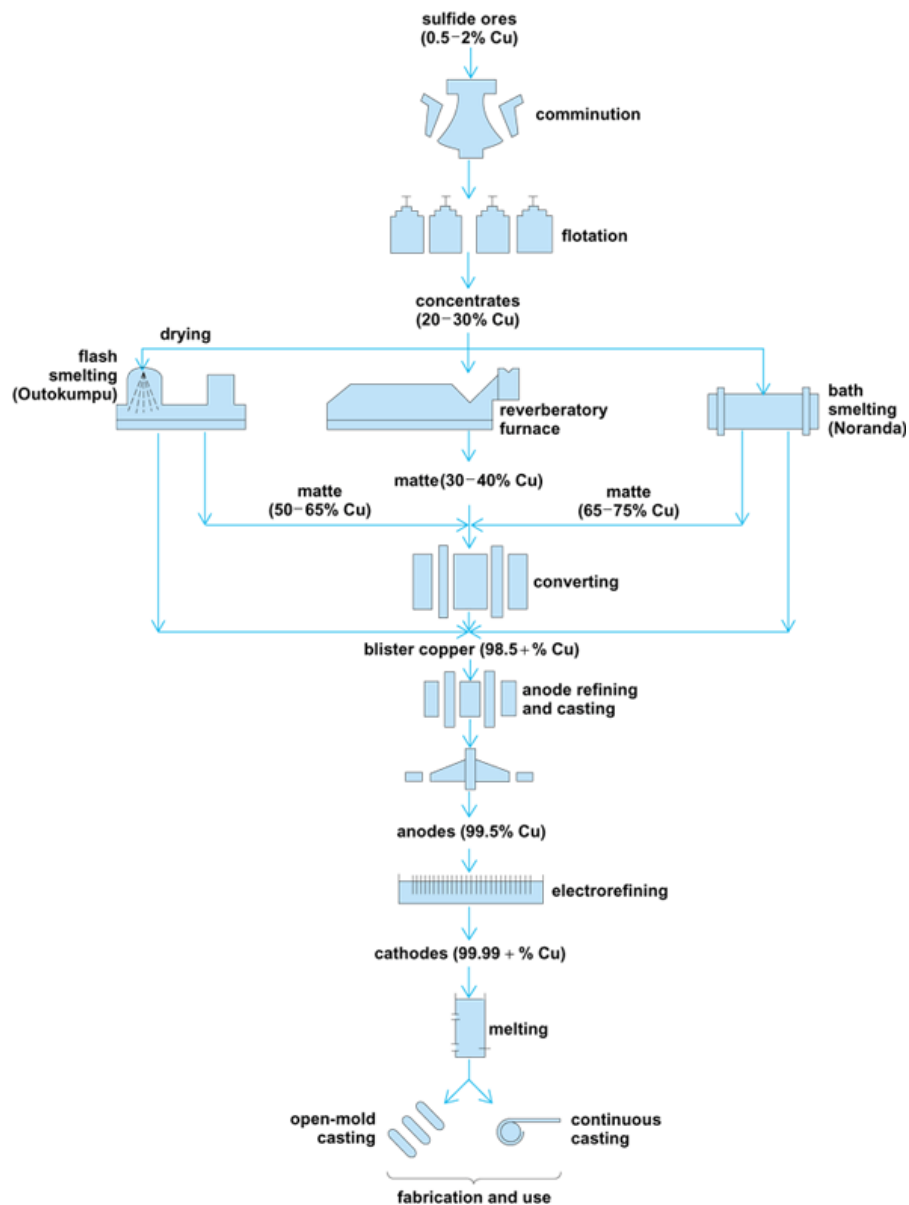
Source: Citi Investment Research and Analysis

Figure 17. Flow Chart SXEW Circuit



Source: CRU, Brook Hunt

Figure 18. Copper Sulphide refining flow sheet



Source: CRU, Brook Hunt

Mining Methods

In this section we discuss the main mining methods, both open pit and underground, as well as providing examples of each type of operation.

Open pit mining

Open pit or surface mining is only viable when the desired mineral is near the surface. Open pit mines can cover many square kilometers and use very large equipment to remove the overburden (the material that lies above the desired ore body). Compared to underground mines, open pit mines are generally simple, quick to build, have lower capex and the ore bodies are normally of a lower grade. The pit shown below is over 500m deep and 1,200m in diameter. The 220 tonne heavy dump truck shown is approximately 4m in height.

Figure 19. Heavy Dump Truck



Source: Licensed from Shutterstock

Figure 20. Open Pit Siberian



Source: Licensed from Shutterstock

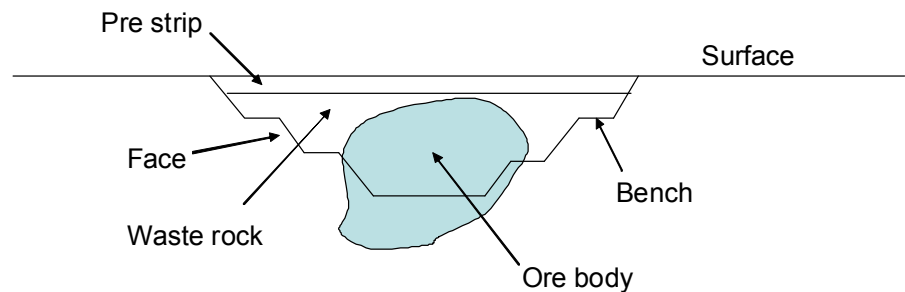
Typically, explosives are used to break up the overburden whereby a pattern of holes are drilled and filled with explosives. Trucks and shovels are then used to transport the ore to the processing facility. Below we show three of the largest open pit mines in the world for their respective commodities:

Figure 21. Examples of Open Pit Mines

Mine	Country	Ownership	Metal
Escondida	Chile	BHPB 57.5%, RIO 30%, Others 12.5%	Copper
Yanacocha	Peru	Newmont 51%, Buenaventura 5%, BRGM 44%	Gold
Trombetas	Brazil	CVRD 40%, Alcoa 16%, BHPB 15%, Alcan 12%, Others 17%	Bauxite

Source: Citi Investment Research and Analysis, Brook Hunt

Figure 22. Open pit terminology



Source: Citi Investment Research and Analysis

Drag Lines

In open pit mining, most commonly for coal mines, a large excavating machine known as a dragline can then be used to remove the overburden. The drag line has a large bucket suspended from the end of a boom which is dragged across the excavation area scooping up the overburden.

Figure 23. Drag Line



Source: Licensed from Shutterstock

Once the ore body has been reached, explosives can again be used to break up the ore into manageable pieces. The ore is then mined using front-end loaders, power shovels or hydraulic excavators and loaded onto haul trucks for processing.

Underground Mining Techniques

Underground mining is used when the desired ore is too far from the surface to be mined economically using open pit methods. Compared to open pit mines, underground mines are more technically challenging, require more capex, have longer lead times to construct and generally have higher grades to compensate.

There are five main techniques used for underground mining – hand-held mining, block caving, long-hole stoping, room and pillar and longwall. Room and pillar and longwall are used extensively in coal mining and we discuss both of these methods in the Coal section of this report.

Hand-Held Mining

Hand-held mining is still used in deep level precious metal mines, most notably in gold and platinum mines in South Africa. This is because precious metal ore bodies are much smaller (deposits can be less than 1m thick) than base or bulk metal deposits and many lie several kilometers beneath the earth's surface as the easily reached surface deposits have already been exploited. Some gold mines in South Africa are now close to 4 kilometers deep.

Working in cramped narrow conditions, miners use hand-held pneumatic drills to bore holes into the rock which are then filled with explosive. The broken ore is then hauled to the surface through a series of shafts. Mines are increasingly using mechanical and hybrid mining methods. Drilling can now be performed by specially designed, low-profile machines and transportation by trackless vehicles.

Clearly, this type of mining is hazardous due to high rock pressures that can result in ground movement, high working temperatures and cramped conditions. South African gold and platinum mines have some of the highest fatality and injury rates in the world – we discuss this further in the Sustainable Mining section later in this report.

Figure 24. Blue Steel mining



Source: Licensed from Shutterstock

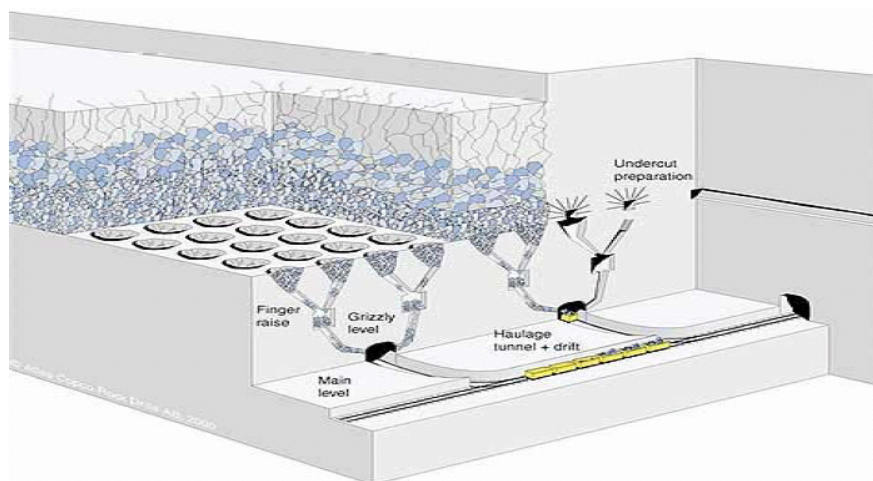
Block Caving

Block caving is commonly used to mine large volumes of ore that have a consistent grade throughout. This technique requires rock that will naturally cave when undercut and fracture into pieces small enough to be extracted (if the pieces are too large they will be difficult to extract). The process:

- Large blocks of ore, usually 60-150m in dimension, are undercut and allowed to collapse under their own weight in a controlled fashion into chutes.
- A series of haulage tunnels are constructed under the ore body so that the ore can be extracted.
- As broken ore is removed further rock collapses down. When the capping or non-mineral bearing rock begins to filter down the draw points, this indicates that all of the ore has been extracted.

Block caving can have significant environmental impact, causing large scale subsidence on the surface.

Figure 25. Block Caving



Source: Rio Colorado Reclamation Committee

Figure 26. Examples of Block Caving Mines

Mine	Country	Ownership	Metal
Northparkes	Australia	RIO 80%, Sumitomo 20%	Copper / gold
Palabora	SA	AAL 29%, Palabora 24%, RIO 47%	Copper / gold
Grasberg	Indonesia	Freeport 91%, Gov of Indonesia 9%	Copper / gold

Source: Brook Hunt, Citigroup Investment Research

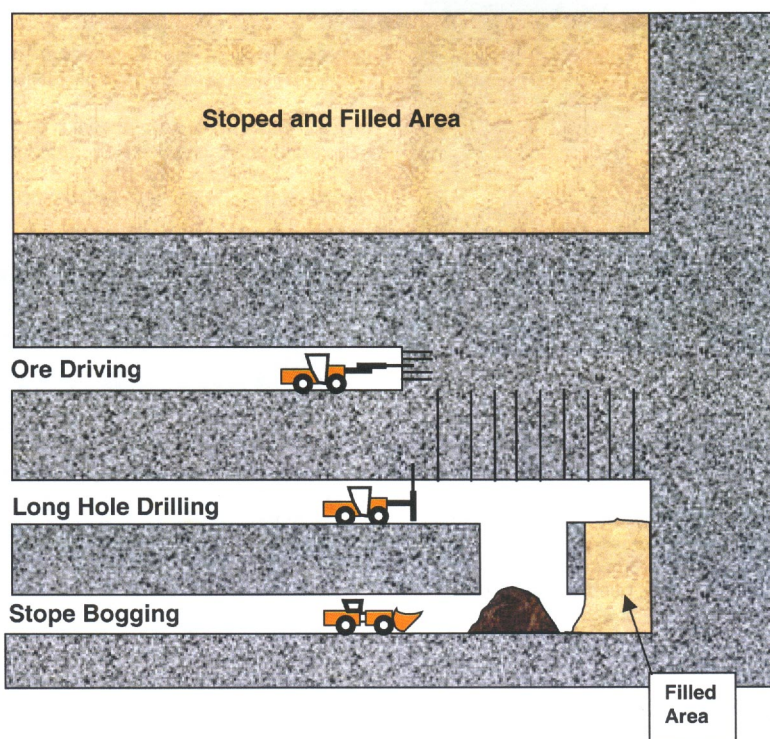
Long-hole Stopping

Long-hole stopping is a retreat mining method that involves drilling a series of long holes up into an ore body and then blasting a slice of rock that then falls into an open space below.

The broken material is then bogged and transported to the stockpile using loaders. The process is repeated by retreating back towards the crosscut and then moving to the level above.

In long-hole open stoping, small sections of the orebody are left intact to act as pillars and hold up the ground, leaving an empty void. However, this can cause ground control issues in the future. Other methods can incorporate backfill, where the void is filled using waste rock from the decline and crosscut development. This method reduces future ground control issues.

Figure 27. Long-hole Stopping



Source: Natural Resources Canada

Figure 28. Examples of Long-hole Stopping Mines

Mine	Country	Ownership	Metal
Broken Hill	Australia	Perilya Ltd 90%, Korea Zinc Group 10%	Zinc / lead
Mt Isa	Australia	Xstrata 100%	Lead / silver
Limpopo	South Africa	Lonmin 100%	PGMs

Source: Brook Hunt, Citigroup Investment Research

Board and Pillar (Room and Pillar)

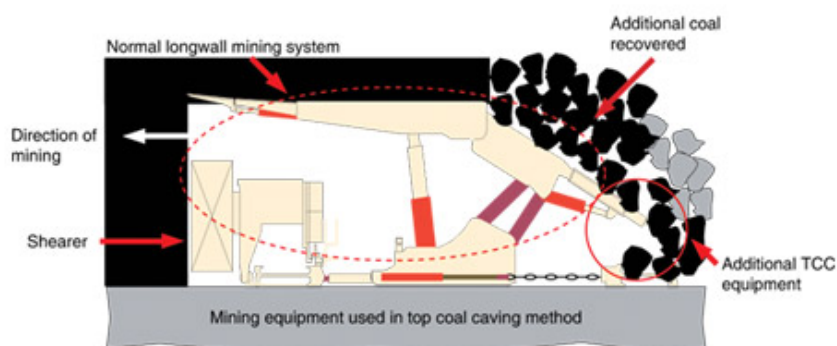
- In Room and Pillar, material is extracted across a horizontal plane leaving pillars to support the roof. Modern methods use a continuous technique where a Continuous Miner machine is used to cut the coal from the mining face (versus the conventional method of drilling and blasting).
- Used in the coal industry as well as hard rock mining where there is a vertical or shallow dipping ore body

- Room and Pillar is far less capital intensive than long wall mining and is the only viable method for mining flat narrow coal deposits, however, production rates of up to 1,000 tonnes per shift per miner unit are far lower than long wall mining rates which can be up to 30,000 tonnes per shift.

Longwall mining

Longwall mining is a type of underground coal mining that involves the use of shearers to cut away coal from the face whilst the roof is supported by self-advancing hydraulic jacks. As the coal is removed the jacks move forward in a continuous operation, leaving a cavity behind the longwall that immediately collapses once the supports are removed.

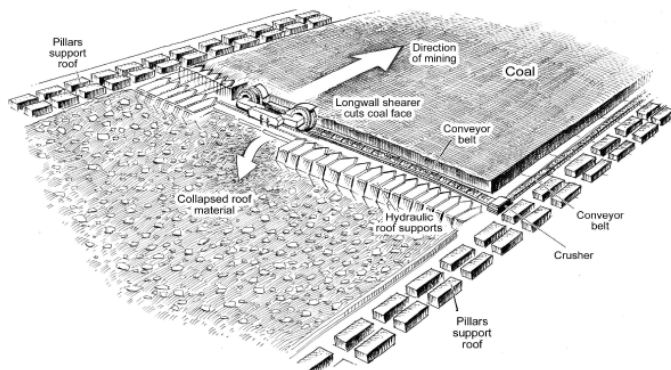
Figure 29. Cross section of a longwall



Source: Whitehaven Coal

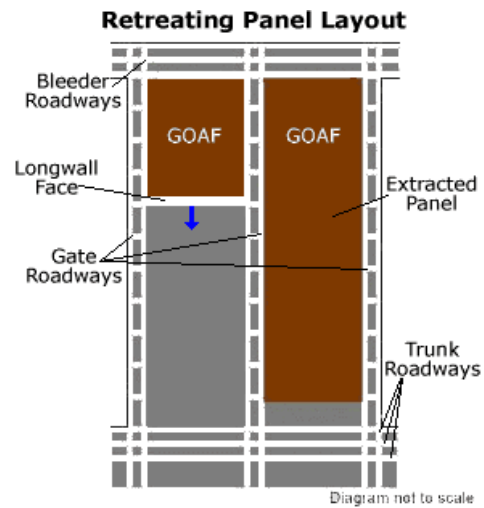
Longwall are effectively continuous miners, that mine a panel of coal. The panel's cross section is defined by the width of the longwall. In fact the longwall is defined by the mine plan and how many panels are to be mined, but effectively the width of the coal to be removed in a panel is the same as the longwall width. The widths of Australian longwalls are generally 250m wide. One of the issues with wider panels is that more power is required. The higher power requirements usually results in the need for larger drive units, and as such wider excavations in the road and gateways. This can add to the capital cost. Also with more power, more heat is usually present which can become a negative factor. The length of the panel is again dictated by the mine plan. In Australia the average panel length is 2.6km.

Figure 30. Longwall schematic



Source: International Longwall Association

Figure 31. Plan view of longwall



Source: : International Longwall Association

As the shearer passes along the length of the longwall it takes a cut out of the face of the coal. The shearer is a drum with metal 'teeth' that spins rapidly and effectively scrapes the coal off the face of the wall. A typical 'cut' of coal across the panel is around 1m, but is dependent on the longwall type, coal quality, rock conditions etc.

The height of the longwall is dependent on the height of the coal seam. The longwall heights in Australia can vary from 1.9m to 5m. In other parts of the globe, longwalls can be up to 6m high.

Mineral Extraction Methods

Mineral extraction methods are the techniques and processes used to convert mineral bearing rocks mined from the ground into saleable metals. Mineral processing, pyrometallurgy and hydrometallurgy are the three primary methods used and metal ores are often treated by a combination of these methods. We discuss the extraction methods relevant to each metal later in this report.

The metallurgical processes differ from each commodity type, but the main process that are followed to extract the mineral from the host rock are:

- **Liberation:** The ore is reduced in size through a combination of blasting, crushing, grinding and milling. Theory is that the smaller the particle size, the greater ability to disseminate the ore from the waste rock.
- **Concentration:** Separates the mineral from the waste gangue material.
- **Extraction:** The mineral is extracted from the host rock through chemical processes.
- **Refining:** Increase the purity of the mineral towards a +99% metal level.

Liberation

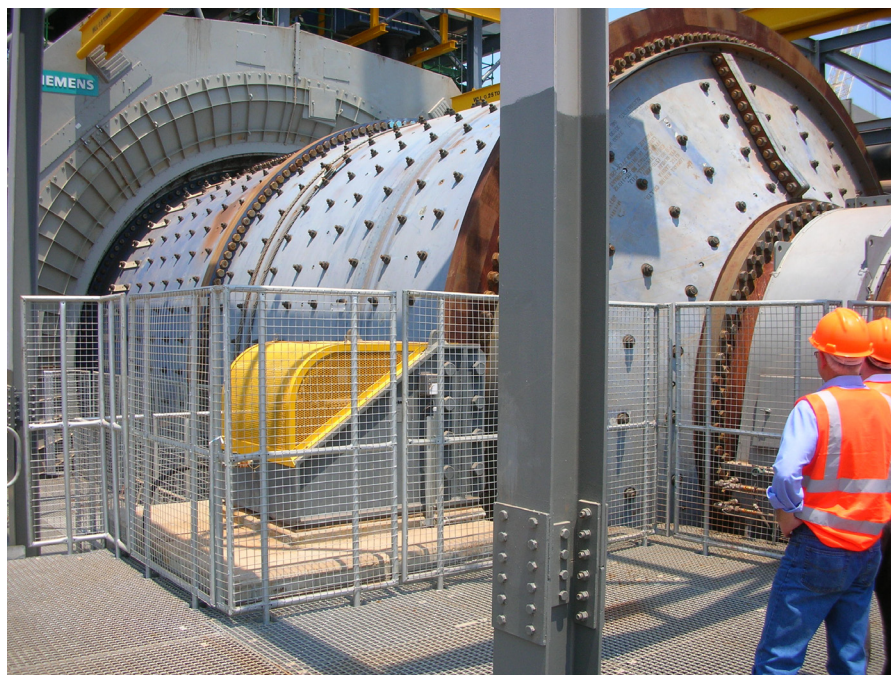
The liberation of the mineral is the first step in the metallurgical process. It is effectively a size reduction process which helps separate the mineral bearing material from its waste. The smaller size of the particle also increases surface area which provides for better processing in the concentration and extraction steps. The stages of liberation are:

Crushing: Crushers take the recently blasted material and reduce it in size, usually by a factor of 5x-7x. Various crushing stages can occur with most mines having at least a primary and secondary crusher. Some assets have a tertiary crusher. Types of crushers include jaw crusher and gyratory crusher.

Milling: The rock from the crusher is introduced in to a mill (large steel cylinder) which rotates and crushed the ore in to smaller particles. Particle sizes from the mills depend on the concentration & extraction process they will follow, but sizes typical range from 10mm to 50mm in size. There are several different types of mills which are used generally depending on the strength/hardness of the rock :

- Autogenous mill: Uses the weight of the rock in the mill to break itself. Can only be used in specific circumstances.
- Pebble mill: Uses 'pebbles' or other small stones to break up the ore.
- Ball or Rod Mill: Uses steel balls or steel rods to help crush the rock. The steel is usually about 40% of the weight of material in the mill. The steel used is called a grinding media.
- SAG mill: A semi-autogenous mill uses both steel balls and the weight of the rock to break itself up. Lower level of steel used than a ball mill.

Figure 32. Ball Mill



Source: Citi Investment Research and Analysis

Concentration

The step is focused on increasing the concentration of the mineral in the feedstock to reduce the cost of the final recovery/extraction/refining process. Most metals pass through some form of concentration phase with 'average' levels of concentration below:

- Iron ore: upgrade sub 30% ore to +60%.
- Copper: ore grading 0.5%-5% copper in to 30-40% concentrate grade.
- Nickel: ore sub 3% up to concentrate levels around 14%.

The most common form of concentration is the flotation process. A collector chemical is added to the ore to render the desired mineral hydrophobic (insoluble in water). The mixture (called pulp) is then introduced to an aerated water bath where the hydrophobic mineral bearing ore rises to the surface forming foam which is then removed for further treatment.

Other forms of concentration can include gravity separation and magnetic separation.

Extraction

The extraction stage further concentrates the mineral by removing the waste material through further chemical processes. There are two main types of extraction processes employed in the industry:

- Pyro-metallurgical processes (smelting): Involves the treatment of ores at high temperature to produce pure metals or intermediate compounds (e.g. copper concentrate). Generally **sulphide ores** are passed through this process.
- Hydrometallurgical process (leaching): Exploits aqueous chemistry for the recovery of metals from their ores. Generally **oxidized ores** are passed through this process.

Pyro-metallurgical processes

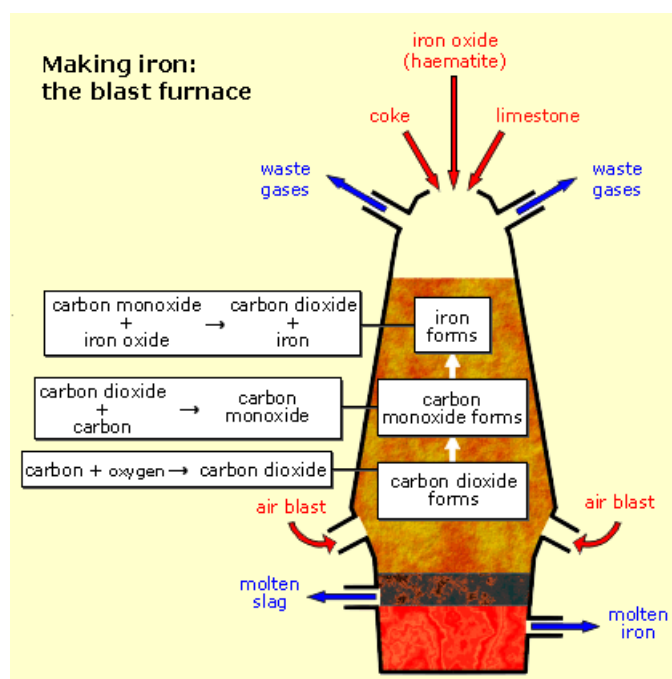
The main stages involved in this process are roasting and smelting:

- Roasting is performed prior to smelting and is often performed on sulfide ores in order to convert them into metal oxides. The metal sulfide is heated in air causing the sulphur to react with the oxygen and form sulphur dioxide and the metal to form as a solid oxide.
- In the smelting stage metal oxides are heated with coke or charcoal and a reducing agent to liberate the oxygen from the ore leaving a refined mineral. A flux is also added to remove impurities (called slag). The pure molten metal is then cast directly out of the furnace.

Smelting is used on:

- Iron
- Aluminium
- Copper (oxide ores)
- Zinc (high lead content)
- Nickel
- Molybdenum
- PGMs

Figure 33. Blast Furnace



Source: BBC

Hydrometallurgy

Hydrometallurgy exploits aqueous chemistry for the recovery of metals from their ores.

Leaching can be used on:

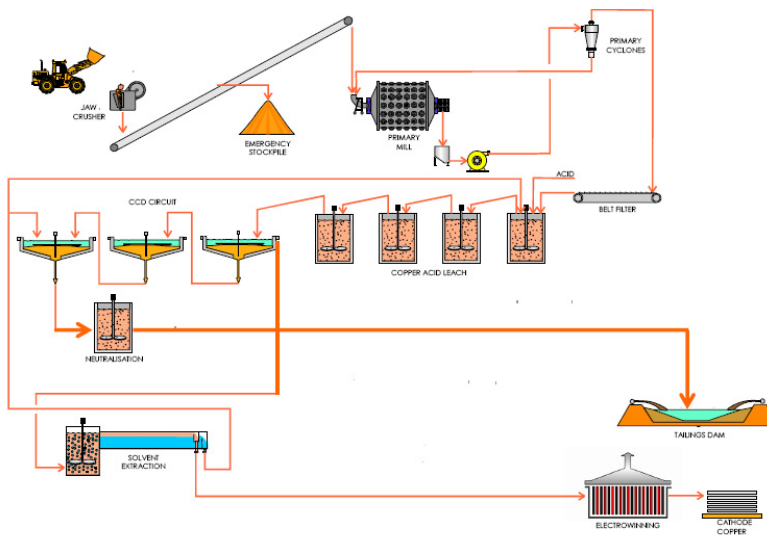
- Copper (sulphide ores)
- Gold
- Zinc
- Uranium
- Nickel

In leaching the metal ore is dissolved in an aqueous solution, allowing separation from insoluble impurities. This can be performed at the ore deposit (in-situ leaching), or the ore can be crushed and piled in a heap and the solution added (dump / heap leaching), or the leaching can take place in large tanks (vat leaching).

After leaching the pregnant solution needs to be treated in order to concentrate the solution and remove undesirable components. Common methods include precipitation; cementation, solvent extraction and ion exchange (refer to glossary for definitions).

The final hydrometallurgical step in producing saleable metals is known as electro winning. This involves passing a current through a liquid solution containing the desired metal, resulting in the metal plating at the cathode. A copper circuit is shown below:

Figure 34. Flow chart SXEW circuit



Source: CRU, Brook Hunt

Refining

This is the last stage in the process and takes the metal recovered from either the hydro or pyro metallurgical processes and further refines it until its in an almost pure state (99.99% metal). The pure metal state is commonly referred to as a cathode. There are a variety of refining techniques which are dependent on the metal.

Figure 35. Copper cathodes



Source: Citi Investment Research and Analysis

Treatment Charges and Refining Charges

The 'back end' of the metallurgical process after the concentration stages is not usually conducted by the mining companies. Independent refining companies who specialize in this area dominate the production landscape. The smelters purchase the concentrate from the mining companies, refine it for a cost, and then sell the finished product. The smelters pay the mining company the price of the metal on the day that the metal is sold. Thus in most instances the refiner doesn't take on any risk from a metal price perspective. The common terms that need to be understood from the refining and processing side of the industry are:

- Payable metal: The smelter pays a percentage of the volume of the metal that is in the concentrate. Typically this is c96% for base metals and closer to 98% for precious metals. The payable metal theoretically accounts for losses that the smelter/refiner may occur in the process. Any efficiency gains that can be made allow the refiner to 'keep' or sell back that metal.
- Treatment charge (TC): The price paid for the treatment of the concentrate. It is paid in US\$/t of concentrate processed.
- Refining charge (RC): The price paid for the refining of the metals. It is paid in US\$/lb of copper treated.

Together the TCRC's are the effective price that a smelter charges the miner for the processing of the concentrate. In periods where there are large amounts of copper concentrate hitting the market (i.e a surplus) the TCRC's are high and vice versa. Thus keeping an eye on the direction of TCRC's can provide an indication of the direction of the copper price.

New Technologies

At first glance technology and the mining industry appear to be odd bedfellows. However the sector has a long tradition of embracing new technologies, especially when faced with a need to deliver new supply, reduce costs or access stranded deposits. Major technological revolutions like the emergence of SXEW copper production in the 1970's and in-situ leach of uranium have changed the face of the sector. Other lower tech solutions like nickel in pig iron production have delivered new supply into the market and had a serious impact on commodity prices. Substitution and thrifting have also increased in times of high metal prices and low metal availability.

Whilst high metal prices are forcing some consumers of the commodities to look at ways of reducing their consumption, we believe that the looming scarcity of supply for some metals may significantly alter consumption patterns for the industry. The shortage of metal supply from traditional avenues is also seeing an increased level of investment into new technologies/methods to deliver more supply to the market. And with the high metal price environment many a technology that was historically uneconomical starts to look increasingly attractive.

Environmental legislation change, particularly carbon pricing, will also have an impact on the development of new technology in the mining space. Miners will be forced to try to cap emissions, but consumers of energy and carbon will also look to new ways of meeting their energy requirements from cleaner sources.

Rising energy demand as well as the potential for briquetted coal and coal to liquid processes are likely to see some previously un-economical coal deposits come into production potentially altering the supply demand balance in the sector.

Technological evolution is far from a certainty and many risks present themselves. High Pressure Acid Leaching (HPAL) and Hot Briquette Iron (HBI) have both been high profile cases of technology failure. Both cases saw significant overspend on capex and neither delivered economic production on a commercial scale.

Despite the failures technology will continue to change the face of the metals markets. And the major areas of technological advancement that may have near term implications on metals supply and demand balance are:

- **Upgrading low rank coal:** Low cost source of coal from previously untapped sub-economic resources. Both coal to liquids and briquetting coal could potentially add meaningful new supply into the industry and alter the cost curve.
- **New steel making technology:** Development of processes such as Corex and Finex that avoid the use of coking coal in the steel making process could be the answer for a steel world that is facing a significant shortage of premium hard coking coal.
- **Nickel laterites:** Will the HPAL technology work economically? If it is not commercially viable nearly 75% of all nickel growth projects could be mothballed and the market could face a nickel deficit mid decade.
- **Hydrometallurgical processing:** A bunch of small scale technologies like Activox and bioleaching that open up the processing of new ore bodies or the re-processing of waste dumps.

- **Undersea mining:** The next frontier which is aimed at accessing the high grade sub marine deposits. Targets have copper grade of +10%. A long way from commercialization but could be a new source of supply.

Figure 36. New technologies overview

Technology	Metal impacted	Overview	Cost	Current Supply	Impact potential on underlying commodity	Chance of success
Briquette coal	Thermal coal	Upgrades low grade coal by removing the water and increasing the energy units	Limited capital cost and a cash cost mid cost curve	Sub 2mt globally which is less than 1% of supply. Potential for 100mt by 2020	High	High
Coal to liquids	Thermal coal	Liquefies coal underground to produce liquid fuels	Most process economic at a \$50/bbl oil price	Sasol is the world largest producer.	High	Medium
Smelting Reduced Iron (Corex/Finex)	Coking coal	Reduces the need to use coking coal in the steel making process by using other sources of carbon such as thermal coal or gas	Lower capital cost of construction than a normal blast furnace given no need to build coke ovens. Operational costs on par with a blast furnace. Reliability is the issue	Limited commercialization of the process but it is rising. India the main region of growth and estimates are for sub 10mt of production currently operating	High	Medium
Direct Reduced Iron	Coking coal	Reduces the iron out of the ore in a solid state using thermal coal or gas as a reductant. Removes the need for coking coal	On par with a standard blast furnace if cheap gas available.	Global capacity is around 70mt in 2010	High	High
High Pressure Acid Leach	Nickel	High pressure and temperature to dissolve the nickel out of the ore	Significant capital costs as well as higher than anticipated operating costs. Both capex and operational costs are higher than nickel sulphides	Around 170ktpa but HPAL accounts for 75% of all nickel growth projects	High	High
Direct Nickel	Nickel	A tank leach process that operates at atmospheric pressure (with the option of mild pressure) and relatively low temperatures	Still in trial phase. But as it leaches at atmosphere should be lower cost than HPAL	Nil	Low	Low
Bio-leaching	Copper, Nickel	Bacteria are used to leach out metal from primary sulphide ores	Still in trial phase. Will be upper end of cost curve as its benefit is mainly focused on low grade ores	Limited as process is too slow. Work being done on how to accelerate the process.	Low	High
Marine technology	Copper, Gold	Sub sea mining targeting high grade gold and copper deposits. Technology derived from deep sea oil and gas industry	When out of the R&D stage each unit should have limited potential capex costs but high operational costs	0mt at present. But each 'unit' could mine c100kt of copper and 200kozpa of gold	Medium	Low
Titanium Metal Production	Titanium	Chemical process for continuous titanium powder production to produce grade 1 pure titanium	~50% reduction in cost of titanium sheet production	Nil at present however first commercial size facility currently in construction and due for completion at end 2010	High	Medium
Shale Gas	Oil, Gas	Extraction of gas from shale deposits	Cheaper than many traditional sources of gas supply	20% of all USA gas supply and may double in the next 5-10 years	High	High

Source: Citi Investment Research and Analysis

For a full overview on technology in mining, please refer to our report [Mining's Technological Revolution](#), 8 December 2010.

Section 3: Understanding Costs

The Cost of Mining

After commodity prices, the cost of mining is the single biggest factor that impacts the profitability of a mining operation. The breakdown of costs varies wildly depending on the type of mine, location, grade, processing type etc. To provide a rough overview of the cost allocation on the mining-metallurgical supply chain, we have grouped costs under four main areas: labour, maintenance (machinery/mechanical), energy (petrol, electricity), raw material (explosives, water, etc). The table below looks at a representative cost split across different types of mining style, commodities and geographical locations.

Figure 37. Cash cost breakdown

	PGM	Coal	Open pit copper	U/G Copper	Nickel	Gold - US	Iron Ore - Aust	Aluminium	Alumina
Labour	50%	39%	21%	51%	42%	40%	32%	6%	15%
Maintenance	15%	21%	12%				18%	7%	15%
Energy	10%	7%	26%	14%	18%	27%	12%	30%	22%
Raw Material	20%	23%	25%	33%	23%	23%	27%	55%	42%
Other/mining	5%	10%	16%	0%	17%	10%	11%	2%	6%

Source: Citi Investment Research and Analysis

Cost progression in the mining sector is a major issue. Whilst the commodity prices are up over 200% since the super cycle began in 2004, costs have also risen sharply. In many cases costs have risen by +100% over the same period of time. Thus whilst margins for most commodities have expanded, costs have been a continual dampener on those margins.

Figure 38. Average cost and price increases

Average		Average costs 2004	Average costs 2010	Cost Increase (‘04-‘10)	Price Increase (‘04-‘10)
Aluminium	US\$/lb	56	81	46%	28%
Alumina	US\$/t	155	240	54%	38%
Coking Coal	US\$/t	42	87	104%	240%
Thermal Coal	US\$/t	23	51	122%	140%
Copper	US\$/lb	44	103	135%	163%
Nickel	US\$/lb	201	502	150%	57%
Zinc	US\$/lb	32	47	47%	105%
Iron Ore	US\$/t	12	32	176%	337%
Gold	US\$/oz	238	513	116%	201%

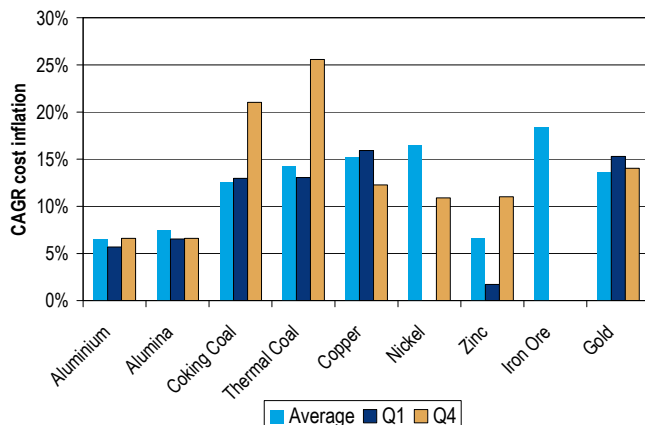
Source: Citi Investment Research and Analysis

The rate of cost pressure has been running well ahead of inflation with most commodities experiencing cost increases of +10% pa over the past 6 years. The chart below looks at the CAGR rate of cost inflation. The escalation in cash costs has been driven by a combination of factors, but the major drivers have been:

- Falling grades;
- Increasing labour prices;
- Fuel and consumable costs;
- Deeper, older mines with higher strip ratios; and
- Currency appreciation.

Whilst high prices have meant that higher cost assets have started and higher cost ore has been mined, the upwards pressure in cash costs has been uniform across all quarters of the cash cost curve. A clear indication that the cost pressures are structural issues rather than just a cyclical rising in costs due to high commodity prices.

Figure 39. Unit cost CAGR (2004-2010)



Source: Citi Investment Research and Analysis

Figure 40. Unit cost CAGR (2004-2010)

	Average	Q1	Q4
Aluminium	7%	6%	7%
Alumina	8%	7%	7%
Coking Coal	13%	13%	21%
Thermal Coal	14%	13%	26%
Copper	15%	16%	12%
Nickel	17%	-30%	11%
Zinc	7%	2%	11%
Iron Ore	18%	n/a	n/a
Gold	14%	15%	14%

Source: Citi Investment Research and Analysis

Understanding the Cash Cost Jargon

When looking at costs, there is a myriad of terms and nomenclature used to describe costs. The major terms associated with costs are:

- **Unit costs:** How much is costs to produce a tonne or ounce of material. Usually expressed in US\$ terms (US\$/oz of gold).
- **By-products:** A credit back to the company from a cost reporting perspective from the production of their secondary metals from their ore (i.e. gold by-products in a copper mine).
- **Cost curve:** The aggregation of all the producers' costs in an industry is the cost curve. The cost curve shows where an asset is placed versus its peers.

Unit Costs

There are 3 primary methods of unit cost measurement in the mining industry, referred to as C1, C2 and C3. The most commonly used measure of unit cost to compare the competitiveness of operations globally is C1 costs in US\$. Wood Mackenzie defines the various costs as follows:

$$C1 = \text{Direct Costs} = \text{Cost to concentrate} + \text{Freight} + \text{TC/RC} - \text{By-Product Credits}$$

$$C2 = C1 + \text{Depreciation}$$

$$C3 = C2 + \text{all other indirect costs (eg. royalties)} + \text{interest on financing (initial and expansion)}$$

Total Cash Cost = C1 + Royalty + Other Indirect Costs (most common for gold)

By-product Credits

Often metals are not found alone but in combination with other valuable minerals. Some common combinations are:

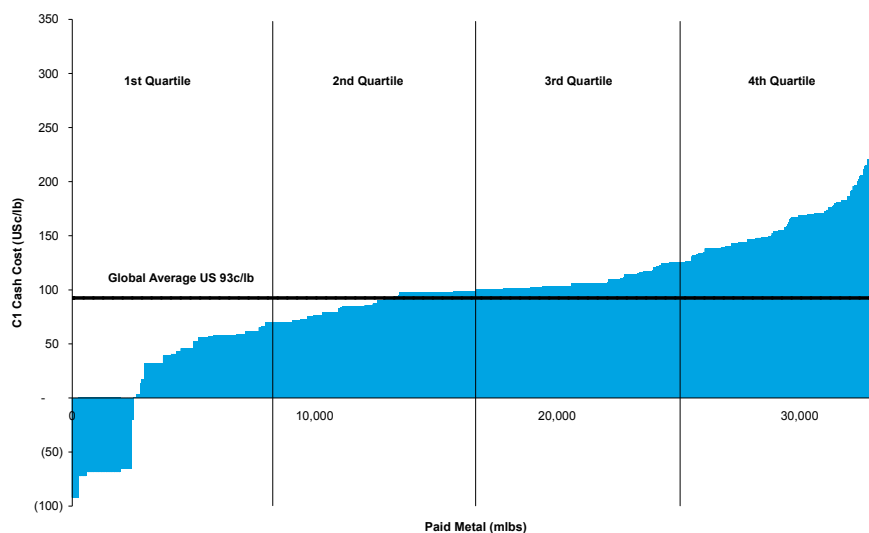
- Copper, gold, molybdenum, zinc and silver.
- The six platinum group metals (platinum, palladium, rhodium being the main three), gold and nickel.
- Zinc and lead.

From a unit cost perspective, by-product revenue is deducted from the unit cost (a by-product credit). Therefore, in some mines where there are significant by-products, C1 cash costs can be negative. Unit costs can also be reported on a co-product basis. In this method, cash costs are defined as the total cash costs multiplied by the percentage of total revenue from the respective metal, divided by the production of the metal.

Cost Curves

Cost curves are used to compare the mining costs per unit of metal mined between mining companies / countries / different mines. The curves also indicate how margins and supply will be impacted by changes in metal prices. The global 2011 copper cost curve is shown below as an example. Typically producers refer to their position on the cost curve from a quartile perspective.

Figure 41. 2011 Global Copper C1 Cost Curve



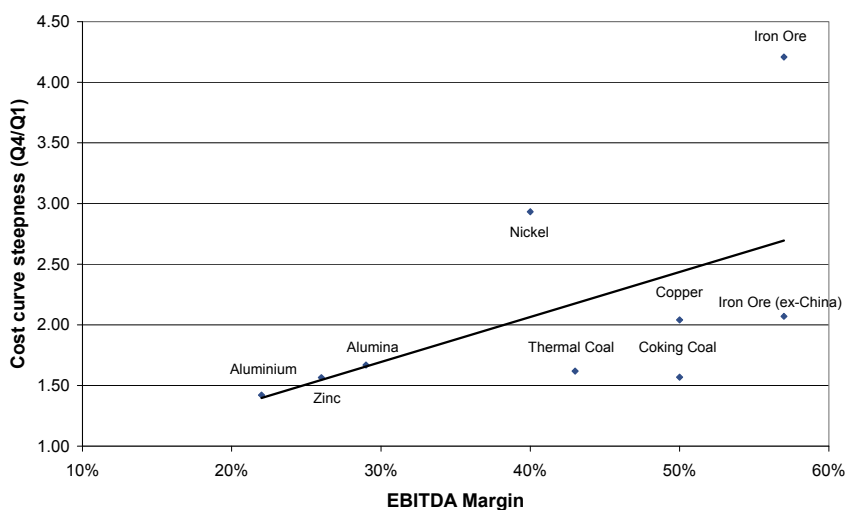
Source: Brook Hunt, Citi Investment Research and Analysis

A steep cost curve is preferable in an industry as the higher cost production should insulate the rest of the curve from serious price weakness. To assess the steepness of the cost curve we look at the 4th quartiles cash costs divided by the 1st quartiles cash cost (Q4/Q1). The top cost producers should theoretically shut production when in a loss making position, resulting in lower supply overhangs and stabilizing

price. Conversely when there is a flat cost curve, the margin compression is greater as there is no buffer of high cost supply.

Therefore industries with steeper curves should deliver better margins. And that is exactly what we see across industries as the chart below indicates. The steeper the cost curve, the better the long-run industry margins. The c2x steeper cost curve of iron ore and copper has resulted in c2-3x higher average industry margins than aluminium, zinc or alumina.

Figure 42. Cost Curve Steepness versus EBITDA Margin



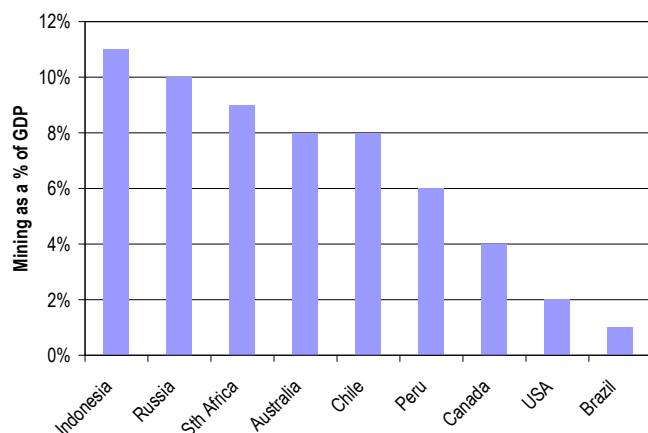
Source: Citi Investment Research and Analysis

Section 4: Mining and Its Global Reach

Mining – A Truly Global Industry

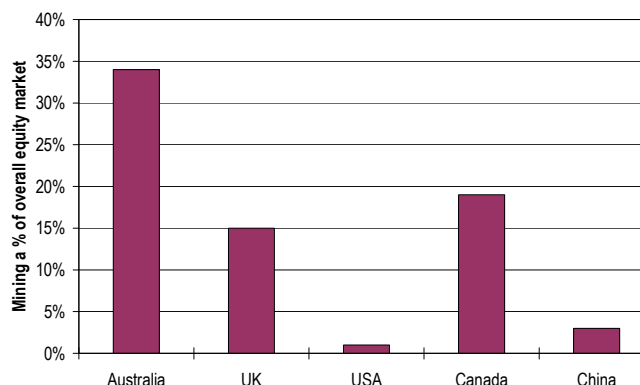
Mining is a truly global industry and the fortunes of many countries have been developed on the back of their mining industry. Mining stocks also dominate many of the globe's largest bourses and as such are an integral part of the financial system.

Figure 43. Mining as percentage of GDP



Source: Citi Investment Research and Analysis

Figure 44. Mining as percentage of overall market value



Source: Citi Investment Research and Analysis

From a natural wealth perspective (the dollar value of reserves in the ground) South Africa is the richest nation with over US\$2.5 trillion of non-energy monetary reserves. Its number 1 position is largely driven by PGM's, where South Africa accounts for nearly 90% of global reserves which are valued at cUS\$2.2 trillion.

Figure 45. Top 20 'richest' nations

Country	Value of resources - ex energy (US\$bn)	Country	Value of resources - ex energy (US\$bn)
Sth Africa	2494	India	296
Russia	1636	Kazakhstan	292
Australia	1588	Mexico	240
Canada	1000	Indonesia	227
Brazil	726	Guinea	222
China	717	Germany	128
Chile	661	Poland	127
USA	613	Sweden	125
Ukraine	516	New Caledonia	99
Peru	328	DRC	75

Source: USGS and Citi Investment Research and Analysis

To assess the value of a country's non-energy commodity reserves we determined the volume reserve position in key commodities. All data was sourced from the USGS and as such has the same methodology behind the assumptions.

Whilst the analysis is crude as it does not take into account the economic viability of the reserves, it provides a solid data benchmark to allow comparisons across countries. The table below looks at where the values of the reserves are for the top 15 countries. Australia, Russia, Canada and surprisingly China have the widest spread of value across different commodities. The high levels of reserves for China are interesting and opposite to what you would commonly think given the vast level of Chinese imports.

Figure 46. Top 15 Countries reserve position (US\$bn)

	Total	Bauxite	Copper	Gold	Iron Ore	Nickel	Zinc	PGM	Potash	Other
Sth Africa	2494	0	0	151	37	49	0	2258	0	-1
Russia	1636	6	71	125	794	87		222	324	7
Australia	1588	186	85	146	737	344	20			70
Canada	1000	0	28	25	62	54	14	11	792	14
Brazil	726	57		50	505	60			54	0
China	717	23	106	48	408	15	58	0	36	23
Chile	661		564	50					2	45
USA	613	1	123	75	119		25	32	16	222
Ukraine	516				510				5	1
Peru	328		222	35			34			37
India	296	23			255		18			0
Kazakhstan	292	11	63		187		30			1
Mexico	240		134	35	23		25			23
Indonesia	227		109	75		42				1
Guinea	222	222								0
Total		529	1505	815	3637	651	224	2523	1229	443

Source: Citi Investment Research and Analysis

On a commodity by commodity basis, iron ore has the highest dollar value of reserves. The high value of iron ore reserves is a reflection of the abundance of the material (160bt of reserves) and our high long term iron ore price.

Figure 47. Dollar value of commodities (all regions)

Commodity	\$ value of reserve (US\$bn)
Bauxite	813
Copper	1911
Gold	1179
Iron Ore	4352
Zinc	243
Silver	252
PGM	2552
Nickel	934
Potash	1552

Source: Citi Investment Research and Analysis

The influence that price assumption has on the dollar value of reserves is meaningful. For instance assuming a long term copper price of US\$1.60/lb delivers a reserve value of US\$1,911bn. If we lift that to US\$2.50/lb, then the value increases to US\$2,986bn and move Chile into a top 5 nation.

Figure 48. Countries reserve position (%)

	Bauxite	Copper	Gold	Iron Ore	Nickel	Zinc	PGM	Potash	Other
Sth Africa			6%	1%	2%		91%		
Australia	12%	5%	9%	46%	22%	1%			4%
Russia		4%	8%	49%	5%		14%	20%	
Canada		3%	3%	6%	5%	1%	1%	79%	1%
Brazil	8%	0%	7%	70%	8%			7%	
China	3%	15%	7%	57%	2%	8%		5%	3%
Chile		85%	8%						7%
USA		20%	12%	19%		4%	5%	3%	36%
Ukraine		0%	0%	99%				1%	
Peru		68%	11%			10%			11%
India	8%	0%		86%		6%			0%
Kazakhstan	4%	22%		64%		10%			0%
Mexico		56%	15%	10%		10%			10%
Indonesia		48%	33%		19%				
Guinea	100%								
Total	5%	13%	7%	31%	6%	2%	22%	11%	4%

Source: Citi Investment Research and Analysis

Royalties and Tax

The continued rise in commodity prices and miners profitability coupled with the struggling economies of many countries has resulted in higher taxation of miners emerging as a popular political platform. We do not see this theme disappearing and believe that the risk is firmly to the upside for royalties and taxation of resource companies. Over the past year, Australia, Brazil, China, Peru and Russia (to name a few) have all looked to increase the tax burden of miners.

Gathering global royalty data is a difficult exercise given how much royalties can vary in a country on a state by state level as well as whether a royalty is on revenue or profit based tax. Our analysis has attempted to gather royalties on a revenue basis and when the royalty varies on a state basis (or indeed a sliding scale on price) we have tried to determine an average. The table below looks at the key countries royalty levels for the major commodities. The data below highlights the 'headline' royalties in the countries and not the concessional royalties that may have been granted historically to induce investment in developing regions.

Figure 49. Royalty snapshot by country and commodity

	Bauxite	Copper	Gold	Iron Ore	Nickel	Zinc	PGM	Potash	Coal
Sth Africa			2%	6%			3%		3%
Russia	6%	8%	6%	5%	8%			4%	4%
Australia – pre MRRT	2%	3%		5%	8%	3%			7%
Canada		2%	2%		2%	2%		2%	
Brazil		5%	2%	3%	3%	3%		3%	3%
China	7%	5%	0%	3%					
Chile			3%					7%	
USA	5%	3%	1%		2%	1%		2%	13%
Peru		3%	3%			3%			
Kazakhstan	4%								
Indonesia		1%							14%

Source: Ernst & Young and Citi Investment Research and Analysis

We have also attempted to determine an 'average' country royalty to analyse how the various regions compare from a total royalty perspective. To do this we have taken a weighted average of the royalties by commodity and the dollar value of export of that commodity on a country by country basis. We have also weighted the royalties by the dollar value of their reserve value by country.

Figure 50. 'Average' country royalty

Country	Average Royalty (weighted by resource base)	Average Royalty (weighted by 2010 production levels)
Russia	5.50%	6.40%
Australia – pre MRRT	5.20%	4.90%
Chile	5.00%	5.00%
USA	3.20%	2.40%
Brazil	3.10%	3.10%
China	3.10%	3.00%
Peru	3.00%	3.00%
South Africa	2.50%	2.70%
Canada	2.00%	2.00%

Source: USGS, Ernst & Young and Citi Investment Research and Analysis

Whilst there is limitation with this analysis, it does provide a high level overview of the relative royalty level on a country by country basis. On this analysis Russia has the highest level of royalty rate which is a surprise to us. Australia's royalty rate is also quite high on a global basis dragged up by the high level of iron ore and coal royalties in the country.

Royalties only paint part of the picture of the total tax payable in a region. Corporate taxes along with VAT/GST also increase the tax burden of miners in various regions. The corporate tax payable by region obviously varies massively on a company by company basis given the amount of tax losses etc. The table below only looks at the 'average' royalty and headline corporate tax rate for these regions to give a clearer perspective of the total tax burden of the producers.

Figure 51. Total tax burden

Country	Average Royalty (weighted by 2009 production levels)	Headline corporate tax	Total tax burden
USA	2.40%	35%	37.4%
Brazil	3.10%	34%	37.1%
Australia – pre MRRT	4.90%	30%	34.9%
South Africa	2.70%	29%	31.7%
Peru	3.00%	27%	30.0%
China	3.00%	25%	28.0%
Russia	6.40%	20%	26.4%
Canada	2.00%	20%	22.0%
Chile	5.00%	17%	22.0%

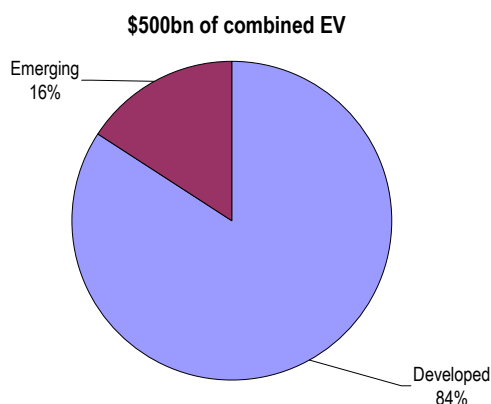
Source: USGS, Ernst & Young and Citi Investment Research and Analysis

The Changing Shape of Global Mining

Mining companies used to be the domain of 'legacy' mining regions such as Australia, South Africa, USA and Canada. Indeed at the start of the super cycle, the developed mining markets accounted for c84% of listed global mining EV.

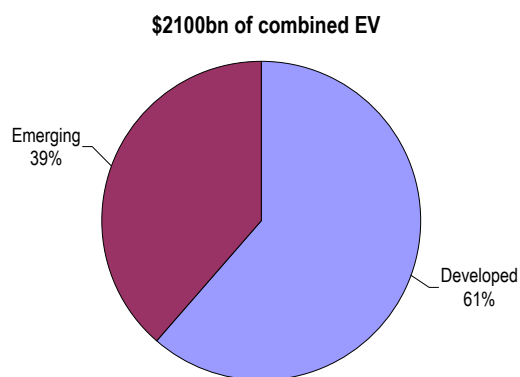
However over the course of the super cycle as high prices and strong demand have facilitated supply from emerging geographies, the domination of the developed world's ownership of mining assets has dwindled. The fall back in their position has also been fostered by the scramble for supply we have seen from regions such as China, Korea and India. Consequently as of today, the legacy mining regions' share of global mining EV has reduced to 61%.

Figure 52. Global mining EV 2004



Source: Bloomberg, Citi Investment Research and Analysis

Figure 53. Global mining EV 2010



Source: Bloomberg, Citi Investment Research and Analysis

We believe drivers in the evolution of emerging markets have been:

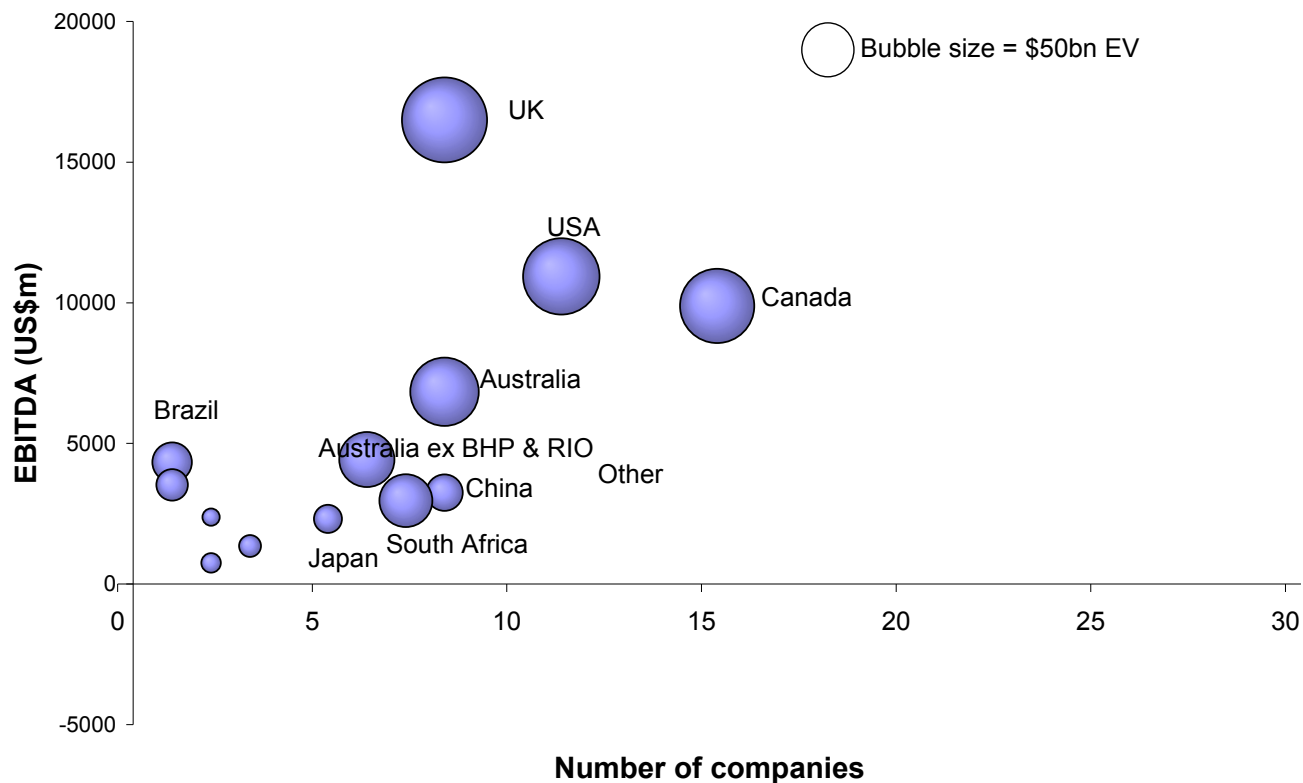
- the development of once benign resource producing regions such as China.
- the listing of equities in these regions.
- privatisation of once public assets from emerging regions such as India, Russia, Kazakhstan and China.
- the scramble for supply from large consumption regions such as China, India and Korea which has lead them to acquire resources from the developed markets.
- China's low cost of funding.

The offshore investment from regions such as China has been pronounced. Since 2004 we estimate that China has invested \$50bn in acquisition of foreign mining assets.

That investment (as well as privatisation and development of the local industry) has seen China evolve into the world's largest mining market in EV terms. China has more listed mining companies than Canada (>\$2bn EV cutoff) and has a mining EV of c\$320bn, which is second only to the UK with \$427bn of EV.

The bubble chart below look at how the landscape of the mining assets on the world's bourses has changed over the past 7 years. The x-axis looks at the number of listed companies (market cap cutoff >\$2bn), the y-axis is the EBITDA generated from the region and the bubble size represents the EV of that region.

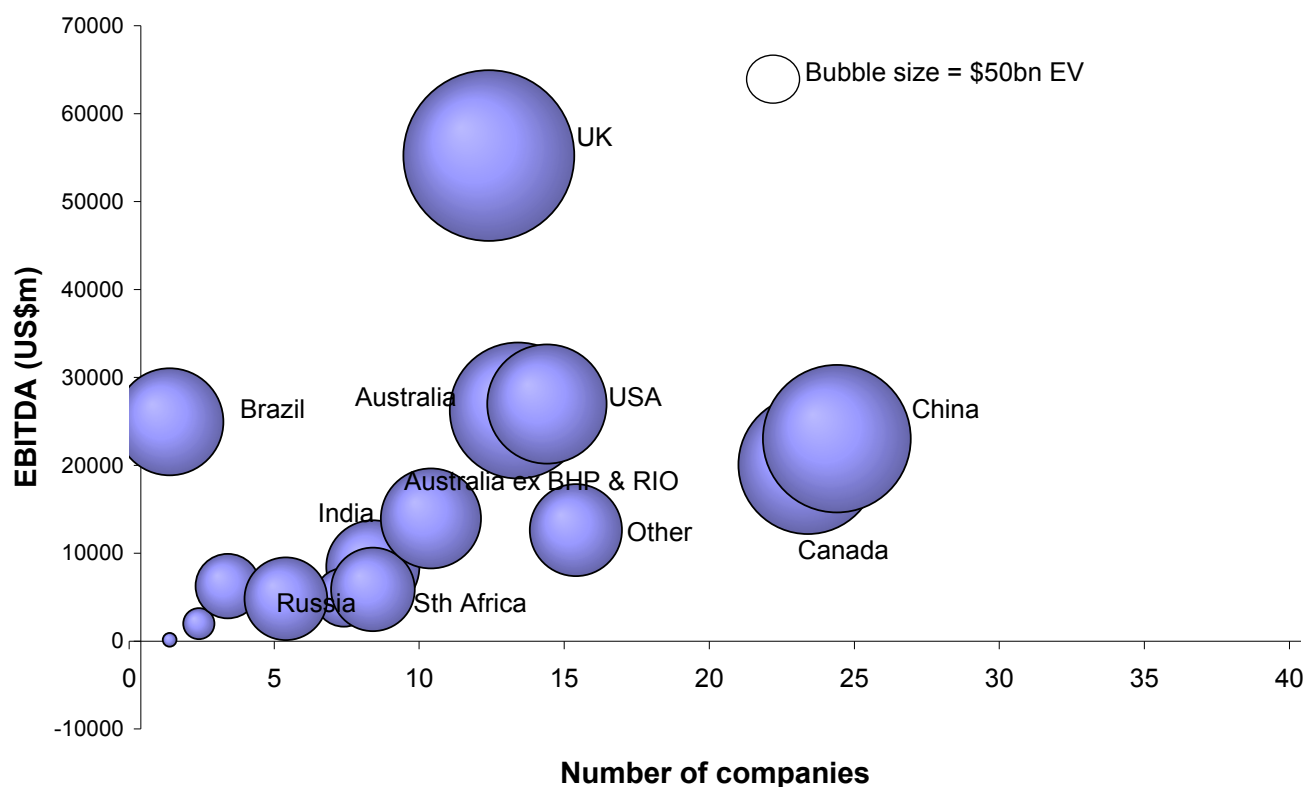
Figure 54. Global mining EV by region – 2004



Source: Bloomberg and Citi Investment Research and Analysis

*Note: BHPB's and RIO's EV are weighted according to market cap listing in UK and Australia

Figure 55. Global mining EV by region – 2010



Source: Bloomberg and Citi Investment Research and Analysis

*Note: BHPB and RIO's EV is weighted according to market cap listing in UK and Australia

As the two bubble charts indicate, the shift in mining landscape has been pronounced.

- Canada remains important despite the acquisition of large champions such as Alcan. The regeneration of its gold sector has been the major reason for the remaining presence of Canada as a major mining force.
- The USA surprisingly still remains relevant. The consolidation in its coal sector along with the development of a mid-tier mining sector with companies like Cliffs Natural Resources have kept the USA relevant.
- The UK has continued to grow due to it being the IPO region of choice for mining companies. The UK has attracted IPO's from regions such as Kazakhstan, Ukraine, Peru, India and Mexico. Add that to the natural weighting from its diversified mining companies and the UK has become the 'home' of the global mining company.
- Australia's reduction in global importance has resulted due to the decimation of the mid-cap mining space. The region has several large miners, but none in the \$10bn-\$20bn range. As a consequence the rump of the sector tends to be small exploration and development companies.

Section 5: The LME & Commodity Markets

How The Metals Are Sold

Metals are essentially sold through two methods:

- Spot traded terminal markets such as the London Metals Exchange; and
- Contracts where the volume and price is set for a specific period of time.

Historically the base and precious metals were sold on the spot market and the bulks (iron ore, coal, bauxite) were sold under long term contracts which were typically annual in nature. However with the rise in prices and the shift in demand away from Japan (who prefer contracts) to China we have seen a shift in how bulk commodities are priced with iron ore and coking coal moving to quarterly contracts and the development of a thermal coal spot market. The table below looks at how the major commodities are traded.

Figure 56. How major metals are sold

PRECIOUS METALS			Mineral Sands			Sale Process	
Gold	US\$/oz	Spot	Rutile	US\$/t			
Silver	US\$/oz	Spot	Zircon	US\$/t			
Platinum	US\$/oz	Spot	Ilmenite	US\$/t			
Palladium	US\$/oz	Spot					
BASE METALS			COAL				
Aluminium	US\$/lb	Spot	Hard coking	US\$/t		1/4ly contract	
Copper	US\$/lb	Spot	Semi soft	US\$/t		1/4ly contract	
Molybdenum	US\$/lb	Spot	Thermal benchmark	US\$/t		Spot and annual contract	
Nickel	US\$/lb	Spot	LV-PCI	US\$/t		1/4ly contract	
Cobalt	US\$/lb	Spot	IRON ORE				
Zinc	US\$/lb	Spot	Ex-Australia	US\$/t		1/4ly average price	
Lead	US\$/lb	Spot	Ex-India	US\$/t		Spot	
Tin	US\$/lb	Spot	Ex-Brazil	US\$/t		1/4ly average price	

Source: Citi Investment Research and Analysis

How the LME works

The LME provides a transparent forum for the trading of futures contracts for non-ferrous metals, steel and plastics. As a result of this trading, daily prices are 'discovered' and published by the Exchange which the physical industry around the world use as the basis of price negotiations for the physical sale or purchase of metals, steel or plastics.

Trading

As a principal-to-principal market, the only organisations able to trade contracts on the LME are its member firms, of which there are various categories. LME members provide industry with access to the market and to the delivery mechanism.

Trading of metals on the LME is conducted via three main contracts; futures, options and TAPOs.

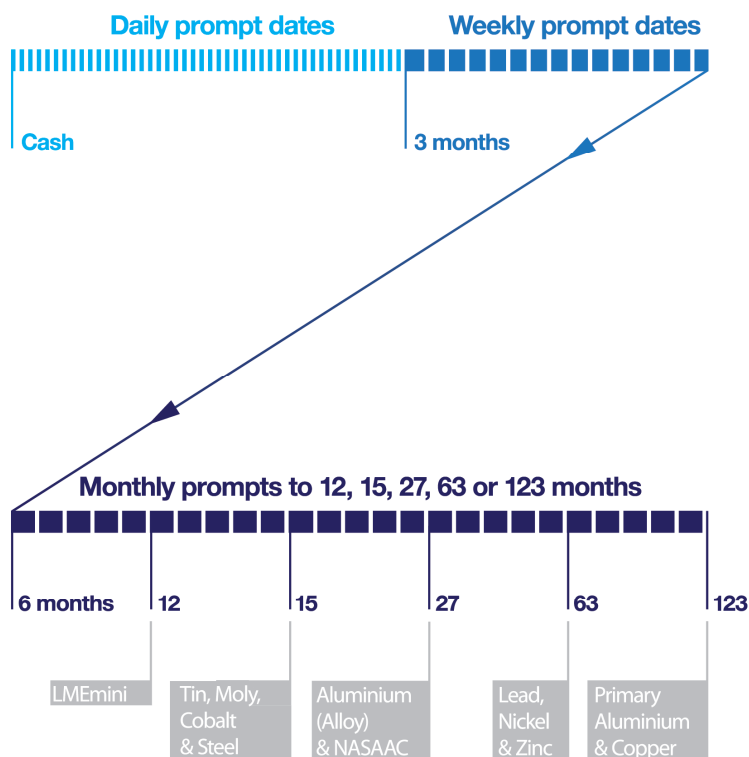
The LME's forward contracts allow producers, fabricators, merchants and consumers to insure against price risk. The LME also offers traded options contracts based on each of these futures contracts, together with traded average price options contracts (TAPOs) based on the monthly average settlement price (MASP) for all metals futures contracts.

Futures contracts are legal agreements to buy or sell goods for a specified delivery future date at a price agreed today. Futures contracts are usually closed out before maturity by an equal and opposite contract: buy/sell back. This is because futures trading usually focuses on price risk management and the offsetting of risk by hedging.

The specified settlement or delivery date of a futures contract is referred to as the prompt date, by which time either the position must be closed or a delivery will take place.

Unlike other commodity markets, which are usually based on monthly prompt dates, most LME futures contracts offer daily and weekly prompt dates (Figure 57 below). The use of this unique prompt date structure is an important difference between the LME and other futures exchanges.

Figure 57. LME Trading and Value Dates



Source: LME, Citi Investment Research and Analysis

An important aspect of LME futures contracts is that, with the exception of LMEmini and LMEX contracts, they are not settled until the prompt date. Initial margins and variation margins will be called during the term of a contract and settlement of net positions will result in either receipt or delivery of warrants on the prompt date.

Trade is conducted in lots rather than tonnes, with each lot of aluminium, copper, lead and zinc amounting to 25 tonnes. Nickel is traded in 6 tonne lots, tin in 5 tonnes and aluminium alloy and NASAAC in 20 tonne lots. Cobalt is traded in 1 tonne lots while molybdenum is traded in 6 tonne lots of Roasted Molybdenum Concentrate (RMC) and steel billet is traded in 65 tonne lots.

The contract for each metal sets out the shapes, weights and methods of strapping (metals). The contract specifications are for the quality and shape which are most widely traded and demanded by industry.

Market of last resort

As a 'market of last resort', the physical non-ferrous metals and plastics industries can use the Exchange's delivery option to sell excess stock in times of over supply and as a source of material in times of extreme shortage. The market does not replace the normal channels for the buying and selling of material and only a very small proportion of contracts actually result in delivery. The presence, or 'threat', of physical delivery plays a vital role in creating price convergence between the futures and the physical market.

Iron ore pricing

Since the 1960s representatives of the world's largest mining companies sat down with the big steel producers (predominately the Japanese) to set a benchmark price. This was typically a contract with volume and prices set for the next year.

This pricing system was challenged after China, the world's largest importer of iron ore by now, failed to accept a benchmark deal between the miners and Japan in 2009. They instead opted to buy in the spot market, a nascent market dominated between trade between India and China.

The formal collapse negotiations between iron ore producers and Chinese steel industry in 2010 resulted in quarterly contracts being developed. Initially these contracts were still negotiated between the two parties, but as the spot market grew, several spot market price indices were developed.

Now quarterly contracts are settled based on these indices. In general, quarterly contract prices are based on the previous 3 month average (for example, a 2Q contract would be priced on the average of the spot price between January and March).

With the development of a liquid spot market also came an OTC market.

Coal pricing

The majority of coal sold for electric power generation is through long-term contracts, in conjunction with spot purchases to supplement the demand.

The coal markets are distinguished between domestic markets and international seaborne markets.

For domestic markets (such as the US or South Africa), there are two main price structures:

1. Cost-plus contracts where a utility has contributed to the cost of building a mine or agrees to pay a set return to a miner that invests in a project on its own, and also pays a management fee based on a percentage of the original cost.

2. Inflation linked/commercial mines are where the miner agrees a base price with the utility that is escalated by an inflation rate over time. These contracts seem to produce a return of 15-18% over the life of the mine.

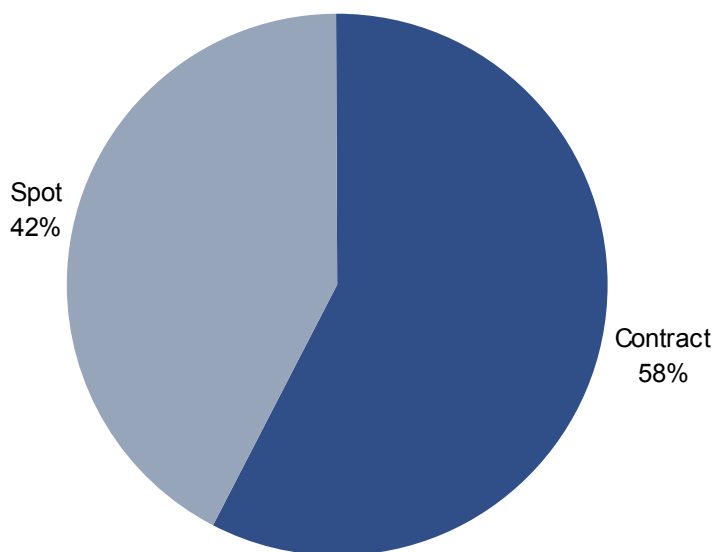
In the international seaborne market, trade is still largely conducted via long term supply contracts and negotiated prices between miner and utility. This is particularly the case in the Asia Pacific market, which is dominated by two major coal suppliers (being Australia and Indonesia) and one consumer (Japan).

Long term supply contracts are also favoured because of the variability of the commodity. The quality of coal can vary dramatically between producers, with factors such as moisture, calorific value, ash and sulphur affecting the efficiency of power utilities. Therefore utilities that are built to consume only low ash high calorific coal are have tended to negotiate long term supply contracts with trusted supply sources rather than rely on the spot market where quality may not be guaranteed at all times.

In trade between Australia and Japan, annual contracts based on the Japanese Financial Year are still the favoured contract. But other imports like South Korea and Taiwan are also keen participants in annual price contracts as well.

Elsewhere, trade between Indonesia, India and China is dominated by spot markets.

Figure 58. Coal in Asia-Pacific is still dominated by long term supply contracts with negotiated prices



Source: Citi Investment Research and Analysis

Futures

Europe, a major trading hub for thermal coal with a firmly established over-the-counter swaps market, has been relatively quick to adopt coal futures trading. Futures have been introduced to the Atlantic coal market in a drip-feed fashion, with varying degrees of success. The European Energy Exchange (EEX), based in Germany, first listed financially settled European ARA and South African Richards Bay coal futures and offered clearing services for contracts traded in the over-the-counter coal market back in May 2006 but has had very limited take-up since then.

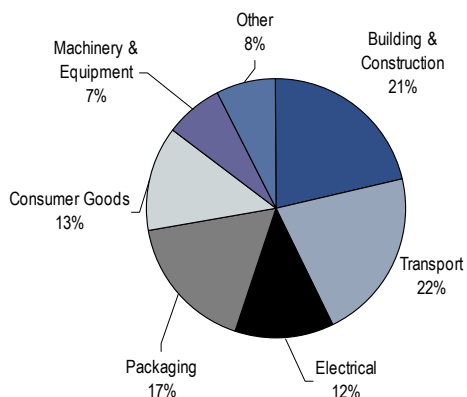
Market participants generally agree that the Asia-Pacific region already has huge potential to develop and drive a thriving coal derivatives market, whether in futures or OTC contracts. It has the largest physical coal market in the world. Market growth projections for the Asia-Pacific region are impressive, with the developing economies of China and India both rapidly increasing their coal consumption in recent years. Despite these positive ingredients, the Asia-Pacific market has been slow to embrace derivatives trading, with the market still heavily influenced by long-term supply contracts negotiated by producers and end-users each year.

Section 6: Metals and Their Uses

Metal Applications

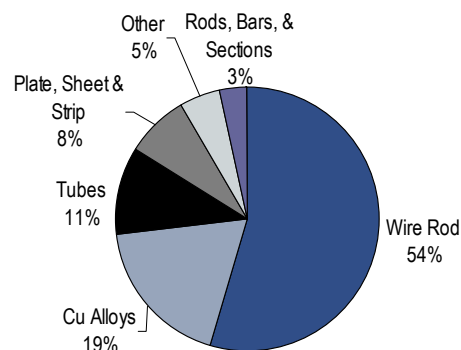
The charts below summarise the major uses of each of the main metals. We provide further details on applications within the second part of this report.

Figure 59. Aluminium



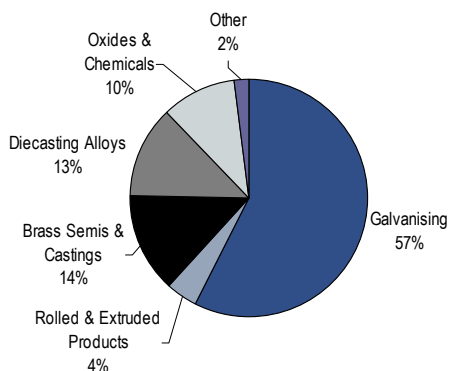
Source: Brook Hunt

Figure 60. Copper



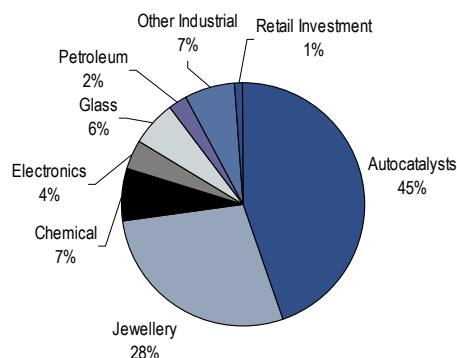
Source: Brook Hunt

Figure 61. Zinc



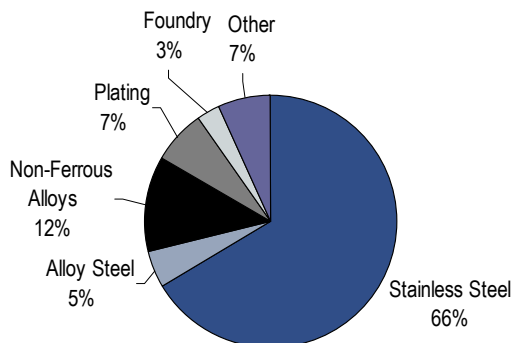
Source: Brook Hunt

Figure 62. Platinum



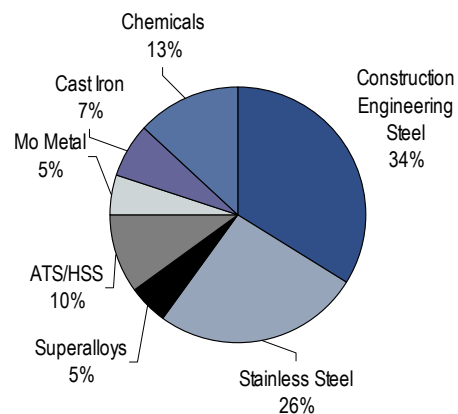
Source: GFMS

Figure 63. Nickel



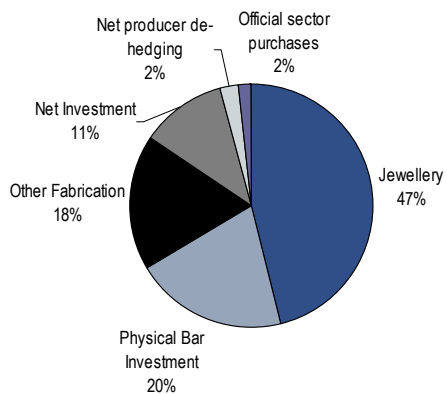
Source: Brook Hunt

Figure 64. Molybdenum



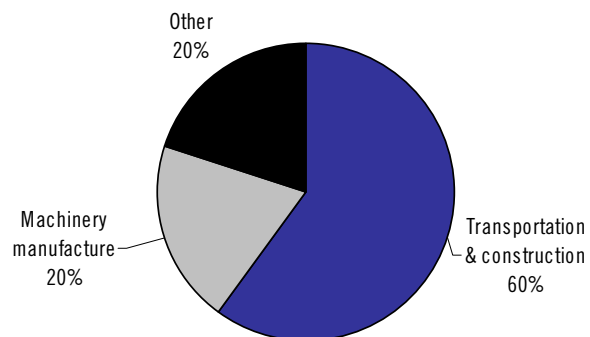
Source: Brook Hunt

Figure 65. Gold



Source: Brook Hunt

Figure 66. Iron & Steel



Source: Citi Investment Research and Analysis

Aluminium

The Aluminium Process

Aluminium is the most abundant metallic element in the Earth's crust, constituting 7.3% by mass, and is the third most plentiful element. In its natural form, aluminium only exists in stable combinations with other materials due to its high chemical reactivity, most commonly as bauxite (essentially an impure aluminium oxide). Bauxite is formed by the weathering of sedimentary rocks and is normally found in strata 3-6m thick located about a meter below the topsoil.

Over 80% of global bauxite production is from open cast mines. The production of aluminium involves the refining of bauxite into alumina (known as the Bayer Process) and then the smelting of alumina into aluminium metal. The process is very energy intensive, with around a third of the cost coming from power. A critical issue for aluminium smelters is therefore having access to cheap and secure energy sources such as hydro power or long-term power contracts.

Aluminum Supply and Demand

Production and Reserves

Aluminium ores occur mainly in tropical and sub-tropical regions, such as Australia, China, Brazil and Africa. Aluminium is a relative newcomer to the base metals family and has only been produced in commercial quantities for just over 100 years. At the current rate of production there will be enough bauxite for over 130 years of production based on current known reserves and 2010 production. Australia is the dominant supplier of bauxite, accounting for 33% of 2010 production. Other significant producers include Guinea, China, India and Brazil.

Figure 67. Global Bauxite Reserves and 2010 Production

	Bauxite Reserves (mt)	2010 Mine Production (kt)	Mine Life (yrs)
Guinea	7,400	17,400	425
Australia	5,400	70,000	77
Brazil	3,400	32,100	106
Other countries	3,300	4,440	743
Vietnam	2,100	30	n/m
Jamaica	2,000	9,200	217
India	900	18,000	50
Guyana	850	1,800	472
China	750	40,000	19
Greece	600	2,000	300
Suriname	580	3,100	187
Kazakhstan	360	5,300	68
Venezuela	320	2,500	128
Russia	200	4,700	43
	28,160	210,570	134

Source: USGS, Citi Investment Research and Analysis

Bauxite Mines

Per Brook Hunt information, there were only 38 operational bauxite mines in 2011, the largest five of which accounted for 55% of total production.

Figure 68. Top 10 Bauxite Mines 2011

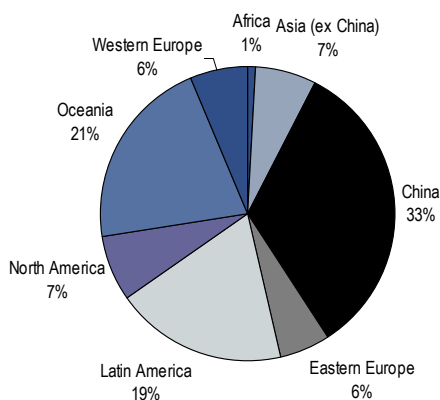
Rank	Mine	Country	2011 Production (kt)
1	Huntly	Australia	22,200
2	Weipa	Australia	19,000
3	Trombetas	Brazil	15,500
4	Boddington	Australia	12,800
5	Boké	Guinea	12,600
6	Willowdale	Australia	8,900
7	Paragominas	Brazil	8,800
8	Gove	Australia	7,100
9	Panchpatmali	India	5,300
10	Discovery Bay	Jamaica	4,800

Source: Brook Hunt, Citi Investment Research and Analysis

Alumina and Aluminium Production

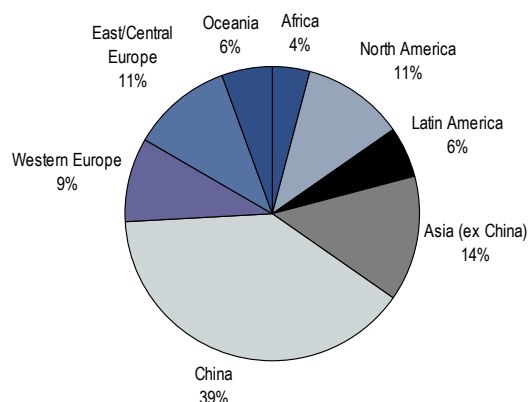
Further downstream, China, Russia and North America become the dominant producers of aluminium.

Figure 69. 2010 Global Alumina Refinery Capacity (98mt)



Source: Brook Hunt, Citi Investment Research and Analysis

Figure 70. 2010 Global Aluminium Production (42mt)



Source: Brook Hunt, Citi Investment Research and Analysis

Mining Companies

Rusal, Alcan and Alcoa are the three dominant suppliers of aluminium. The acquisition of Alcan by Rio Tinto will form the largest aluminium producer in the world.

Figure 71. Top 10 Aluminium Producers 2010

	Attrib. Production (mt)
UC Rusal	4.1
Rio Tinto	3.8
Alcoa	3.4
Chalco	3.2
Hydro	1.4
BHP	1.2
Dubai	1.1
Alba	0.9
East Hope	0.7
Henan Zhongfu	0.7

Source: Citi Investment Research and Analysis

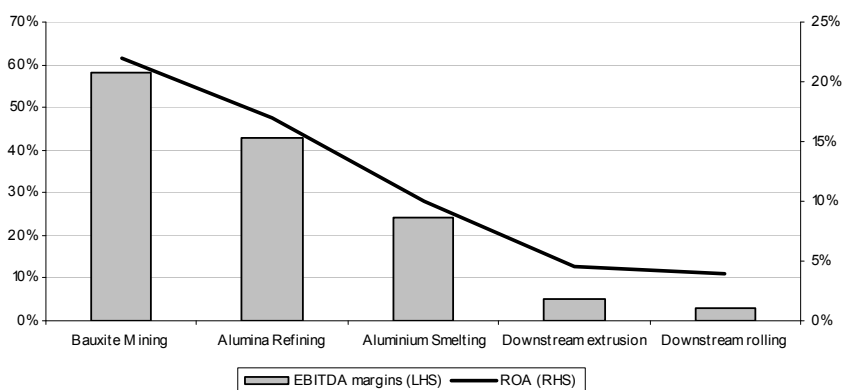
Figure 72. Top 10 Alumina Producers 2010

	Production (kt)
Chalco	12,151
Alcoa	9,850
Rio Tinto	8,856
UC Rusal	7,857
Alumina Ltd	6,069
Xinfa	5,165
BHP	4,036
Vale	3,359
Weiqiao	3,125
Hydro	2,004

Source: Citi Investment Research and Analysis

The production of aluminium involves five broad stages. Returns in the aluminium industry generally favor the upstream operations of bauxite mining and alumina refining:

Figure 73. EBITDA Profitability (%) and Return on Assets (%) for the aluminium industry

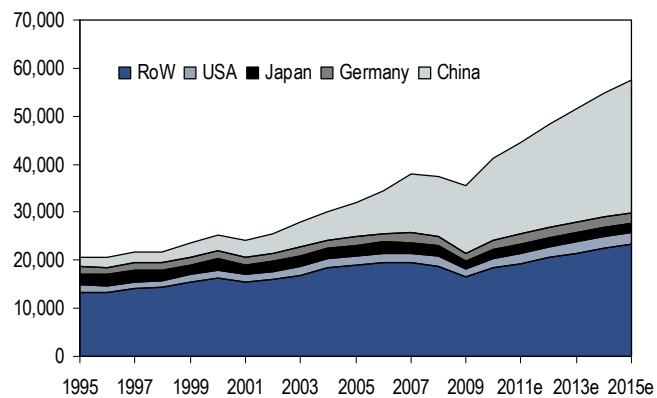


Source: Citi Investment Research and Analysis

Applications and Consumption

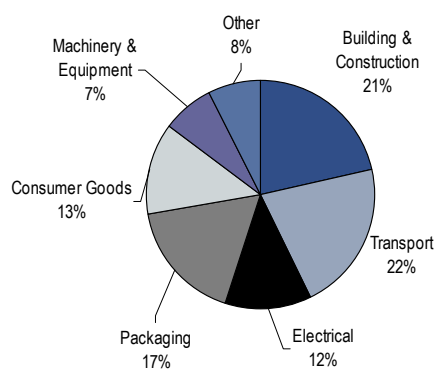
Despite only being commercially produced for approximately 150 years, more aluminium is produced than all other non-ferrous metals combined. Aluminum's high resistance to corrosion, light weight, strength and non-toxic properties lends itself to a wide range of industries, in particular transportation, packaging and construction.

Figure 74. Aluminium Consumption by Geography – 1995-2015e



Source: Brook Hunt, Citi Investment Research and Analysis

Figure 75. Western World End Use 2011



Source: Brook Hunt, Citi Investment Research and Analysis

Copper

The Copper Process

Copper deposits are often found with other metals, commonly gold, lead, zinc and nickel, and are mostly extracted from large open pit mines. Underground mining of copper usually involves block caving. Copper can be found in many forms, however, commercially viable copper is found either as sulfide ores (primary ores), or as oxide ores (secondary ores). Sulfide ores formed when molten solution flowed into the earth's crust as a result of volcanic activity. Oxide ores are found near the surface and are formed from the oxidation of sulfide ores.

The earth's crust contains an average of 60 ppm of copper, although estimates vary. For commercial exploitation a grade in excess of 0.5% is usually required and a typical grade is 2% for underground and 0.5-1.0% for open pit. The production method is dependent on the ore type; sulfide ores are beneficiated in floatation cells whereas oxide ores are generally leached.

Copper Sulfide Ores (Concentrates)

The copper ore is crushed and ground into a fine powder and sent to a floatation machine for concentration. At the concentration stage the ore is mixed with a collector chemical and introduced to a water bath. Copper sulfides form as a froth at the surface and are skimmed off. The dried froth contains between 20-40% copper and can be traded as an intermediate product in its own right, known as copper concentrate.

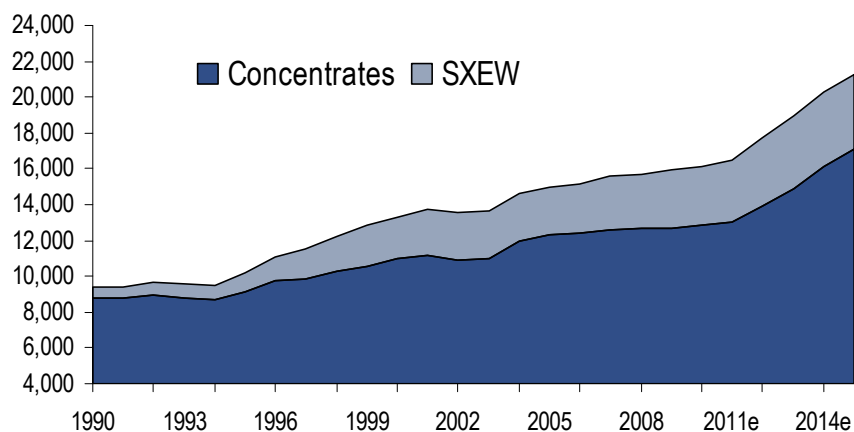
Copper Oxide Ores (SXEW)

Sulphuric acid is used to dissolve the copper in either a heap leach or dump leach process (where the crushed ore is piled in a heap and irrigated with the leaching solution). The dissolved copper solution is collected and transferred to a solvent extraction plant where an organic extractant removes the copper from the solution. The resulting electrolyte solution is transferred to the electrolytic cells for recovery as copper plates (cathodes). This process is known as Solvent Extraction Electrowinning (SXEW)

Concentrate vs. SXEW

Concentrate is the dominant source of copper and expected growth in copper production:

Figure 76. Concentrate and SXEW Supply



Source: Citi Investment Research and Analysis, Brook Hunt

Copper Supply

Production and Reserves

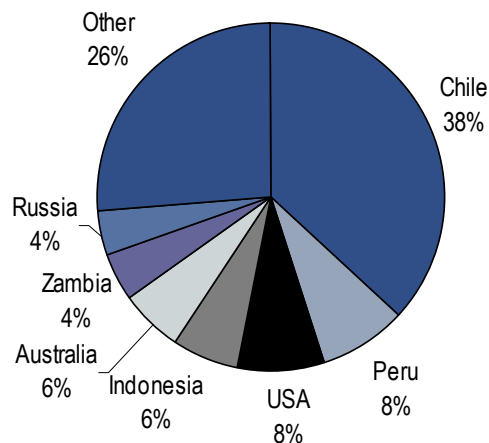
Copper is found throughout the world; most significantly in South America where close to half of the world's copper is currently produced. At the present rate of production, current global reserves will be depleted in approximately 40 years (ignoring potential new deposits and recycling of scrap). Chile is by far the largest copper producer in the world, accounting for approximately 38% of global production in 2010, followed by Peru & the US at 8%.

Figure 77. Global Copper Production and Reserves

	Reserves (kt)	2010 Copper Production (kt)	Mine Life
Chile	150,000	5,520	27
Peru	90,000	1,285	70
Australia	80,000	900	89
Other	80,000	2,300	35
Mexico	38,000	230	165
USA	35,000	1,120	31
China	30,000	1,150	26
Indonesia	30,000	840	36
Russia	30,000	750	40
Poland	26,000	430	60
Zambia	20,000	770	26
Kazakhstan	18,000	400	45
Canada	8,000	480	17
Total	635,000	16,175	39

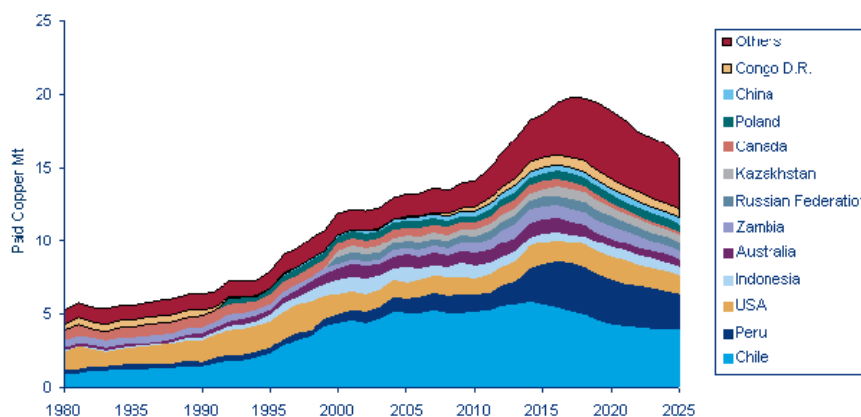
Source: Citi Investment Research and Analysis, USGS

Figure 78. Global Copper Production by Country 2010



Source: Citi Investment Research and Analysis, Brook Hunt

Figure 79. Mine Supply Growth



Sources: Brook Hunt, a Wood Mackenzie Company

Source: Brook Hunt

Copper Mines

Five of the ten largest copper mines by production in 2011 are in Chile:

Figure 80. Estimated Top 10 Copper Producing Mines 2011

Rank	Mine	Production (kt)	Country
1	Escondida	905	Chile
2	Chuquibambilla Mill	880	Chile
3	PT Freeport Indonesia	510	Indonesia
4	Collahuasi	491	Chile
5	Los Pelambres	423	Chile
6	El Teniente	413	Chile
7	Antamina	355	Peru
8	Norilsk	352	Russia
9	Cerro Verde	300	Peru
10	Morenci	273	USA

Source: Brook Hunt, Citi Investment Research and Analysis

Copper Mining Companies

Chilean state copper miner Codelco is the world's largest copper producer, producing over 10% of global production in 2011:

Figure 81. Estimated Top 10 Copper Producers 2011

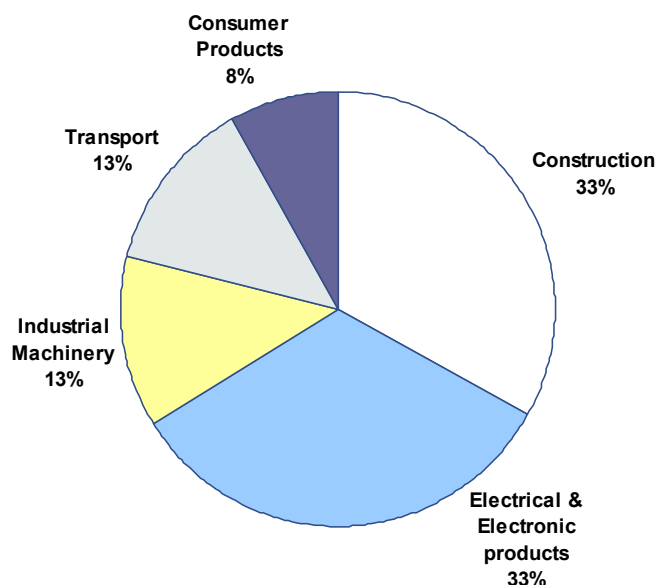
Company	Production (kt)
Codelco	1826
F-McM Copper & Gold	1500
BHP Billiton	1094
Xstrata AG	877
Anglo American plc	694
Southern Copper (ex SPCC)	607
Rio Tinto	560
KGHM Polska Miedz	430
Antofagasta plc	414
RAO Norilsk	395

Source: Brook Hunt, Citi Investment Research and Analysis

Applications

Copper use dates back to ancient civilisations over 5,000 years ago and was widely used by the Greeks and Romans. Copper is now one of the major industrial metals, ranking only behind iron and aluminium in terms of consumption by mass.

Figure 82. 2009 Consumption by End Use



Source: Brook Hunt and CIRA

Copper's malleability, thermal and electrical conductivity (copper has the second highest electrical conductivity of all the metals behind silver), and its resistance to corrosion, give it wide-ranging uses. Consumption can be divided into three main product groups:

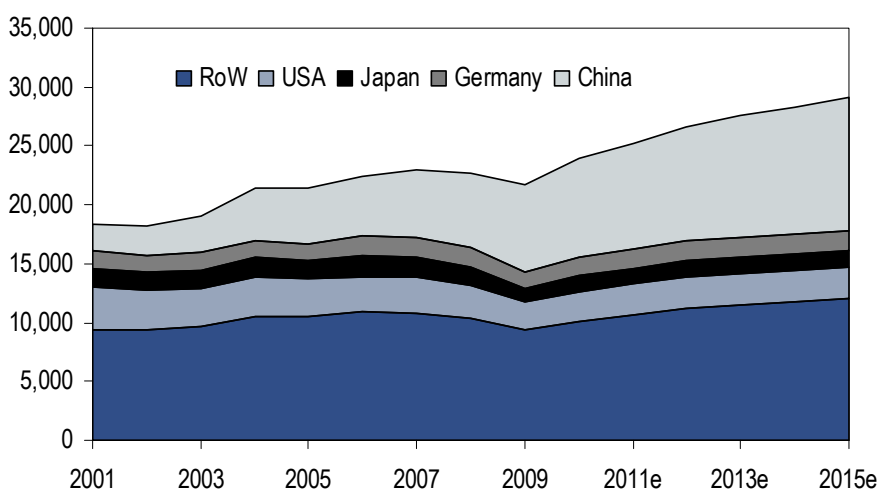
- **Copper wire rod:** This is the largest group, accounting for c.54% of consumption in 2010, encompassing building wire, power transmission, telecommunications and winding wire.
- **Copper products:** Copper tube is the largest of the copper product markets. It is used in plumbing, heating and air conditioning systems.
- **Copper alloy products:** Bronze and brass are two of the most well-known copper alloys, both used for building fixtures and fittings. Copper sheet and strip is used for roofing, gutters and drain pipes. Copper alloys are also used in coinage.

Copper by-products from manufacturing and obsolete copper products are readily recycled.

Consumption

Chinese demand has grown rapidly since the mid-1990s, more than trebling between 2001 and 2010. China is now the world's largest consumer, having overtaken the US, accounting for ~35% of world consumption in 2010.

Figure 83. Global Copper Consumption – 2001 to 2015e



Source: Brook Hunt, Citi Investment Research and Analysis

In contrast, US demand has declined from 3.0mt in 2000 to c. 2.5mt in 2010. US copper demand is closely linked to the housing cycle and demand remains lackluster from both construction and autos sectors.

Zinc

The Zinc Process

Zinc is the 23rd most abundant element in the Earth's crust (average level of 70 ppm) and is mined in over 50 countries. It can be found in several different forms and is often associated with other metals such as copper, gold and lead. Approximately 64% of zinc is mined from underground mines, 21% from combined underground and open pit mines, and the remaining 15% from open pits.

Typical zinc ores contain 5-15% zinc. The most commonly mined ore is Zinc Blende (a zinc sulfide), accounting for over 95% of global production. Once the ore has been extracted, it is crushed and ground in order to separate it from other minerals and then concentrated using the froth floatation method. This is often done at the mine site to reduce transportation costs.

Zinc can be recovered using either pyrometallurgical or hydrometallurgical techniques. However, before this can be done the sulphur must be removed by roasting the zinc concentrate at more than 900°C, converting it to zinc oxide.

Over 90% of zinc is produced using hydrometallurgical processes:

- Firstly, the zinc is leached using sulphuric acid. Zinc dust is then added to remove impurities.
- The purified solution then undergoes an electrolytic process with pure zinc forming at the cathode.
- The zinc is then stripped off and cast into various shapes, grades and alloys depending on customer requirements. High Grade zinc is 99.95% pure and Special High Grade zinc is 99.99% pure.

Pyrometallurgical techniques are used when the ore has a high lead content:

- The zinc and lead oxides are smelted in an Imperial Smelting furnace along with coke at temperatures of up to 1,500°C. The coke converts to carbon monoxide which then allows the zinc and lead oxides to reduce to metallic zinc and lead.
- Zinc evaporates out of the top of the furnace where it is quenched with a molten lead condenser. The metals are then cooled and the zinc separated.
- The zinc produced is of a lower grade than that produced from electrolytic refining and impurities are removed via thermal refining.
- This method of zinc processing is used less and less due to rising energy costs and fewer concentrates with large amounts of lead.

Zinc Supply and Demand

Production and Reserves by Geography

China is the world's largest zinc producer, accounting for ~30% of production in 2010, followed by Peru (12%) and Australia (12%). These same three countries account for ~47% of the global reserve base.

Figure 84. Global Reserve Base and 2010 Production

	Zinc Reserves (kt)	2010 Mine Production (kt)	Mine Life (yrs)
RoW	62,000	1,580	39
Australia	53,000	1,450	37
China	42,000	3,500	12
Peru	23,000	1,520	15
Kazakhstan	16,000	480	33
Mexico	15,000	550	27
US	12,000	720	17
India	11,000	750	15
Bolivia	6,000	430	14
Canada	6,000	670	9
Ireland	2,000	350	6
Total	248,000	12,000	21

Source: USGS, Citi Investment Research and Analysis

Figure 85. Top 10 Mines by Production 2010

Rank	Mine	kt Zn	Country
1	Rampura-Agucha	730	India
2	Red Dog	559	US
3	Century	505	Australia
4	Mount Isa Pb/Zn	345	Australia
5	San Cristobal	265	Bolivia
6	Antamina	215	Peru
7	Brunswick	214	Canada
8	Lanping	200	China
9	Tara	179	Ireland
10	Vazante	172	Brazil

Source: Brook Hunt, Citi Investment Research and Analysis

Major Producers

London-listed Xstrata is the world's largest zinc producer followed by Canadian listed Teck Cominco:

Figure 86. Estimated Top 10 Zinc Producers 2010

Company	kt Zn
Xstrata AG	1068
Hindustan Zinc	853
China Minmetals Corp	643
Teck	628
Glencore	430
Votorantim	367
Vedanta Resources	361
Boliden	310
Minera Volcan	279
Sumitomo Corp	268

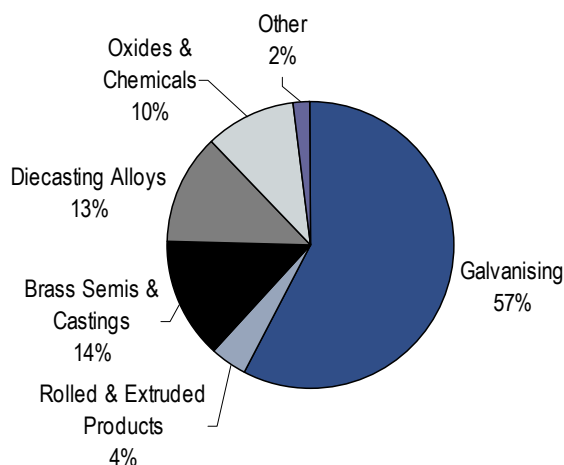
Source: Brook Hunt, Citi Investment Research and Analysis

Applications and Consumption

Zinc use dates back to the Roman Empire where it was used to make brass and today it is the fourth most common metal in use behind iron, aluminium and copper. Zinc's most significant application is in the manufacture of galvanised steel. A zinc coating is added to steel to protect against corrosion. Other uses include the production of brass in combination with copper, die casting and the manufacture of batteries.

Galvanised steel has a zinc coating that protects it against corrosion, increasing the performance and life span of steel products.

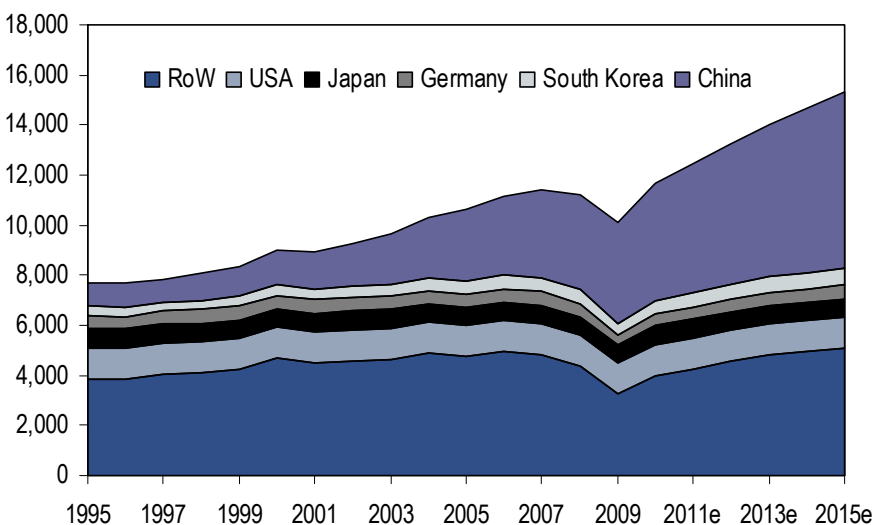
Figure 87. Consumption by End Use (USA)



Source: Brook Hunt & CIRA

China is the dominant consumer of zinc accounting for 40% of global consumption in 2010 followed by the US at 11%.

Figure 88. Global Consumption (kt) – 1995 to 2015e



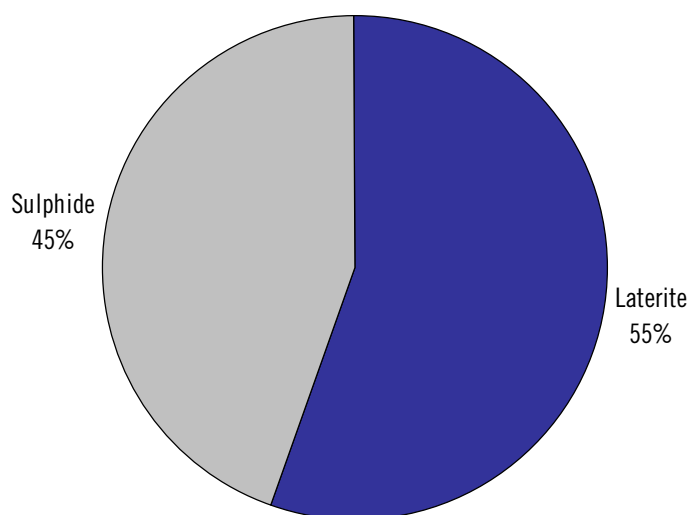
Source: Brook Hunt, Citi Investment Research and Analysis

Nickel

The Nickel Process

Nickel has an abundance of c.75 ppm in the earth's crust and is often found with sulphur, iron, copper and smaller amounts of gold, silver and PGMs. The most commonly mined ores are laterites (oxide ores) and pentlandites (sulfide ores).

Figure 89. Nickel Reserves – Sulphides and Laterites



Source: USGS, Citi Investment Research and Analysis

Nickel ores are usually concentrated using the froth floatation method. The concentrate is then roasted and reduced to produce 75%+ pure nickel. A purification technique known as the Mond process, created by Ludwig Mond in 1899, is then used to produce nickel that is 99.99% pure, known as carbonyl nickel. In this process the nickel is reacted with carbon monoxide to form nickel carbonyl. The gas is then passed at high temperature into a chamber, causing the carbonyl nickel to decompose and form pure nickel in pellet or powder form.

Nickel can also be produced using hydrometallurgical techniques (leaching and electrowinning). Oxide ores can be leached directly; however, sulfide ores must first be treated with bacteria to break down the sulfide minerals to make them amenable to leaching. Leaching techniques are being increasingly used for nickel recovery, but the use of hydrometallurgical processing is yet to be widely, commercially proven over a sustained period and a full commodity cycle. A number of major new hydrometallurgical plants will be commissioned over the next few years. Time will tell how successful they are, but it is clear they will not have the industry transforming characteristics of low capital and operating costs and huge by-product revenues which the technology promised ten years ago.

Nickel in pig iron

Nickel in pig is a process that took off in 2007 with high nickel prices. China sourced abundant low grade lateritic nickel supplies from Indonesia and the Philippines for feed in unused blast furnaces. The concentrate produced is low in nickel (1-2%). Technology has evolved and improved to reduce the costs associated with this style of nickel processing. Blast furnaces are gradually being replaced with EAF.

The process

Nickel laterite ore is finely crushed then mixed with coke, limestone and other material in mixing machines and transported to the sintering plant. The raw material mix is then placed on a steel conveyor belt in the sintering plant where it is heated by gas fired furnace to form sinter.

The sinter is mixed with coke and limestone and feed into the furnace where hot gas heats the mixture reducing it from metallic oxides to liquid metal. The resultant molten nickel pig iron flowing out of the furnace is cast into moulds and ready for delivery.

The resulting nickel pig iron is then mixed with chromium and other materials to produce 200 and 300 series stainless steel.

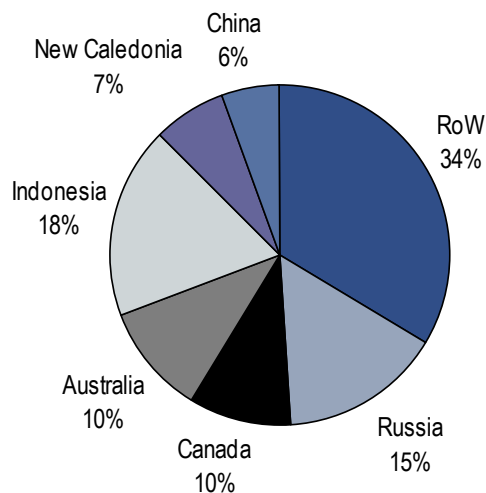
An alternative to blast furnaces are electric arc furnaces (EAF). In China, the output from EAF is increasing due to the lower associated operating costs.

Nickel Supply

Production and Reserves

Indonesia is the largest producer in the world, generating 18% of global output in 2011, followed by Russia (15%). However, Indonesia is only 7th in terms of reserve base and at the current rate of production, its reserve base will be fully depleted within 17 years, the fastest rate of the top ten producing countries.

Figure 90. 2011 Nickel Production



Source: Brook Hunt, Citi Investment Research and Analysis

Figure 91. Nickel Reserves

	Nickel Reserves (kt)	2010 Mine Production (kt)	Mine Life (yrs)
Australia	24,000	139	173
Brazil	8,700	66	131
New Caledonia	7,100	138	51
Russia	6,000	265	23
Cuba	5,500	74	74
Other	4,500	78	58
Indonesia	3,900	232	17
Canada	3,800	155	25
South Africa	3,700	42	89
China	3,000	77	39
Colombia	1,600	70	23
Madagascar	1,300	8	173
Philippines	1,100	156	7
Dominican Republic	960	3	310
Botswana	490	23	21
Venezuela	490	14	34
Total	76,140	1,540	49

Source: Citi Investment Research and Analysis

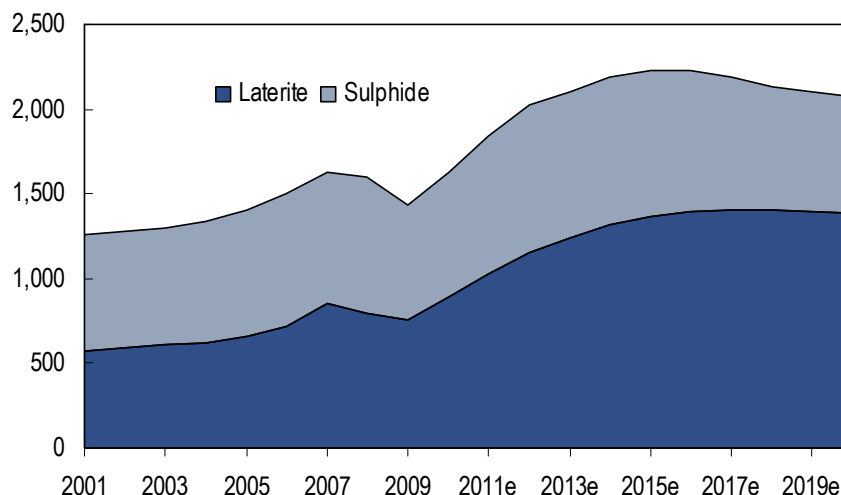
Figure 92. Top 10 Nickel Operations by 2011 Production

Rank	Operation	kt Ni	Country
1	Norilsk	187	Russia
2	Soroaka	80	Indonesia
3	Jinchuan	71	China
4	Vale - Sudbury	65	Canada
5	Antam High Grade Ore	61	Peru
6	SLN in house mines	60	New Caledonia
7	Indept. & SLN Ore	56	New Caledonia
8	Murrin Murrin	43	Australia
9	Pechenga	41	Russia
10	Punta Gorda	40	Cuba

Source: Brook Hunt, Citi Investment Research and Analysis

Most new supply is expected to come from laterites. Between 2010 and 2015 production from laterites will account for ~80% of new supply, and beyond that sulphide production is scheduled to decline.

Figure 93. Nickel Mine Production from Sulphides and Laterites (kt)



Source: Brook Hunt, Citi Investment Research and Analysis

Mining Companies

Russia's Norilsk is the world's largest nickel producer, followed by two global players, Vale and BHP.

Figure 94. Estimated Top 10 Nickel Producers 2011

Rank	Company	kt Ni	% Global
1	RAO Norilsk	262	14.2
2	Vale	196	10.6
3	BHP Billiton	101	5.5
4	PT Aneka Tambang	92	5.0
5	Xstrata AG	77	4.2
6	Jinchuan	71	3.9
7	Anglo American plc	56	3.0
8	Union del Niquel	54	2.9
9	Eramet	36	2.0
10	Talvivaara Mining	34	1.9

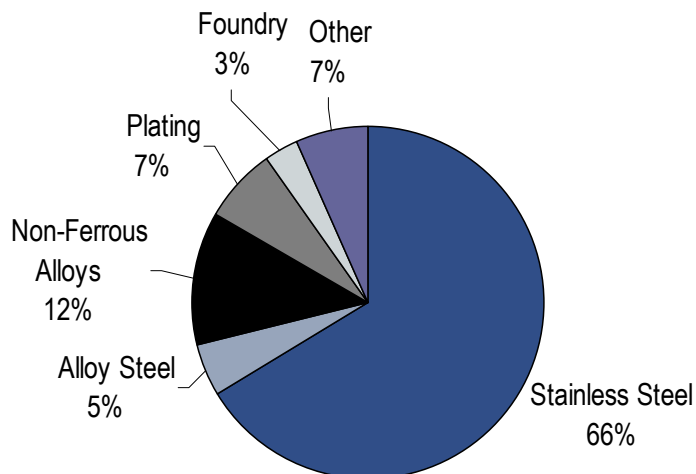
Source: Brook Hunt, Citi Investment Research and Analysis

Nickel is a highly consolidated industry compared to most other base metal commodity markets – the five largest firms accounted for ~40% of total production in 2011. The consolidation increased following several large mergers / acquisitions: BHPB's acquisition of WMC; CVRD's acquisition of INCO; and Xstrata's takeover of Falconbridge.

Applications

Nickel's primary use is in the manufacture of stainless steel and other alloys. Stainless steel is used in a variety of industries, including chemical and food processing equipment, transportation and construction. Nickel use in alloys stems from its resistance to corrosion, strength and heat resistant properties. Nickel is also used in coinage, batteries and fuel cells.

Figure 95. Nickel Consumption by End Use 2011

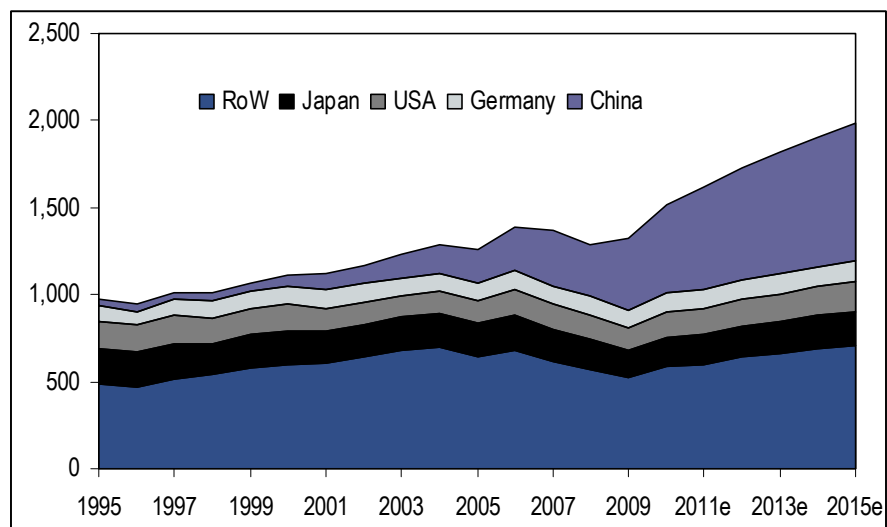


Source: Citi Investment Research and Analysis

Consumption

Over recent years demand for stainless steel (and thus nickel) has been fuelled by China. Nickel demand grew by 14% in 2010, driven primarily by stainless production growth.

Figure 96. Global Consumption – 1995 to 2015e



Source: Brook Hunt, Citigroup Investment Research

Uranium

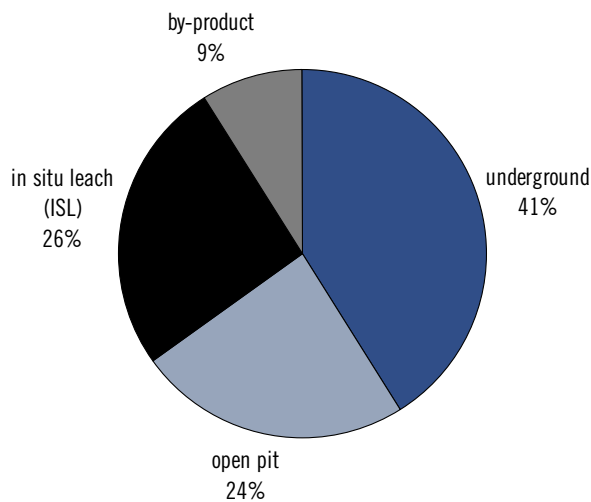
The Uranium Process

Uranium is one of the heaviest of all the naturally-occurring elements (c.70% denser than lead), occurring naturally in low concentrations in soil, rock and water (average of c.1-2 ppm). Uranium is weakly radioactive and occurs most commonly as uranium-238 (99.3% of all uranium), but also as uranium-235 (0.7%) and in very small amounts as uranium-234. The important isotope is uranium-235 (U-235) as it is this isotope that is the most fissile and can be used in nuclear reactions.

Uranium is only found combined with other elements, most commonly as uraninite (a uranium oxide). Commercial ores typically contain 0.1-0.25% of uranium oxides. There are three main methods used for mining uranium – block caving, open pit and in situ leaching. In situ leaching is an increasing source of uranium, particularly when the ore grade is too low to be economically processed via other methods. Details on ISL mining are provided in the Mining Extraction Methods section of this report.

ISL has lower capex and lower operating costs compared to conventional mining. As an example, in the US, uranium ISL cash costs are roughly 50% that of conventional mining (c.US\$20/lb vs. US\$40/lb).

Figure 97. 2006 Uranium Production methods



Source: WNA

Once the ore has been extracted, it is crushed and ground to a fine slurry and then leached using sulphuric acid. The pregnant solution then goes through several more stages of processing, including wash tanks, solvent extraction, precipitation and thickening. The solution is finally placed in a centrifuge to separate the uranium from the solution and the uranium is then dried to produce uranium concentrate known as Yellow Cake that contains c.90% uranium oxide. The Yellow Cake is packaged in steel drums and shipped to a refinery for further processing.

Uranium Enrichment

Enrichment involves boosting the U-235 content. Prior to enrichment, Yellow Cake must be converted to a fluoride; firstly it is converted to uranium dioxide and then to gaseous uranium hexafluoride. U-235 has a slightly lower mass than U-238 and it is this physical difference that is exploited to produce enriched uranium. There are currently two commercial methods for enriching uranium, gaseous diffusion and centrifuge enrichment. Both use gaseous UF_6 as feed, exploiting the mass differential between U-235 and U-238, and both are highly energy intensive:

- Gaseous diffusion, generating c.40% of global enriched uranium, involves forcing UF_6 through a series of porous membranes. The lighter U-235 molecules move faster through the membranes and the UF_6 that diffuses through is slightly enriched. This process must be repeated up to 1,400 times to produce 3-4% U-235 (reactor grade).
- In centrifuge enrichment, UF_6 is fed into a vacuum tube that spins at up to 500 m/s, causing the heavier U-238 to increase in concentration towards the cylinders outer edge and U-235 concentration to increase near the centre. The enriched gas is separated and the process repeated 10-20 times.

Gaseous diffusion enrichment is being phased out and replaced by centrifuge enrichment. Centrifuge processes are lower cost than diffusion, less capital and energy intensive and can be built in a modular form.

After enrichment UF_6 is converted to UO_2 and made into fuel pellets that are encased in metal tubes to form fuel rods up to 4 m long. These rods are then ready to be fed into the nuclear reactor.

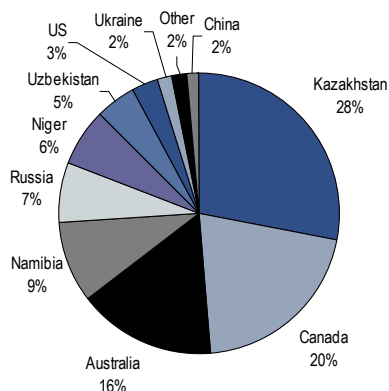
Uranium-238 can be used to produce plutonium-239. It is this element that is used in most modern nuclear weapons and to power space probes. Plutonium is virtually non-existent in nature and is formed by bombarding uranium-238 with neutrons in a nuclear reactor.

Uranium Supply and Demand

Production and Reserves

Canada and Australia are the two dominant uranium producers, accounting for almost half of global production in 2009. These two nations along with Kazakhstan also account for over 50% of global reserves.

Figure 98. 2009 Uranium Production



Source: Citi Investment Research and Analysis

Figure 99. Global Uranium Reserves and Estimated Mine Life

	Uranium Reserves (kt)	2009 Mine Production (t)	Mine Life (yrs)
Australia	714	7,982	89
Kazakhstan	344	14,020	25
Canada	329	10,173	32
South Africa	206	563	366
Russia	172	3,564	48
Brazil	157	345	456
Namibia	145	4,626	31
Ukraine	127	840	151
US	99	1,453	68
Uzbekistan	55	2,429	23
China	44	750	59
Niger	44	3,234	14
Other	0	793	1
	2,438	50,772	48

Source: Citi Investment Research and Analysis

Uranium Mines

The McArthur River Mine, located in Canada, is the world's largest high-grade uranium mine, producing 7.3 Kt of uranium in 2009; however, the largest single uranium resource is Olympic Dam (owned by BHPB), located in Australia.

Figure 100. Top 5 Uranium Mines 2009

Rank	Operation	Country	Majority Owner	2009 Production (t)
1	McArthur River	Canada	Cameco	7,339
2	Ranger	Australia	Energy Resources of Australia	4,444
3	Rossing	Namibia	Rio Tinto	3,520
4	Priargunsky	Russia	ARMZ	3,004
5	Olympic Dam	Australia	BHPB	2,955

Source: Citi Investment Research and Analysis

Secondary Uranium Supplies

Approximately 40% of global uranium supply comes from secondary sources. There are three main sources:

- Inventories account for the bulk (c.75%) of the secondary market and were built up during the 1950-70s when there was excess supply of uranium. Inventories have been drawn down and supply from this source is diminishing. Little inventory data is publicly available, and none is current. Estimates of Western inventories by the World Nuclear Association were 168kt at the end of 1997, and 112 kt at the end of 2002. Inventories have been drawn down at about 5-8ktpy since the mid-1990s. Minimum working inventory is commonly estimated at two-thirds of annual consumption, although policies at individual utilities vary with

their appetite for risk. There are indications that some utilities may have already drawn inventories down to their minimum respective working levels. Others are believed to be building inventories in response to rising prices.

- Highly enriched uranium contains at least 20% U-235 and can be down-blended into reactor-grade fuel. The majority of this source is from Russian weapons grade uranium (over 90% U-235). Russia currently delivers the equivalent of 9,300t of uranium per annum to the US annually under an agreement ("Megatons to Megawatts") struck in 1997 that expires in 2013. President Putin recently reaffirmed that the programme will not be renewed or extended. What Russia intends to do with its huge remaining stockpiles is unclear. HEU from the US nuclear weapons programme is also being declared as surplus and is in a marketable form. Although there are large stocks of HEU remaining in Russia and the US, it seems unlikely that supply will be increased to meet a shortfall. Both Russian and US authorities have recently reaffirmed this.
- The enrichment process creates 5-10kg of depleted uranium for every 1kg of enriched uranium produced. These depleted uranium stockpiles along with depleted fuel can go through re-enrichment programs. Supply from these sources could increase in response to higher prices (similar to other metals such as gold and copper where scrap supply increases as prices rise).

Mining Companies

Canada-based Cameco is the world's largest uranium producer accounting for c.20% of global production in 2006. Industry ownership is relatively consolidated with the top 7 companies accounting for c.78% of global production.

Figure 101. Top 10 Uranium Producers

	2009 Production (tU)	World Share (%)
Areva	8623	17%
Cameco	8000	16%
Rio Tinto	7963	16%
KazAtomProm	7467	15%
ARMZ	4624	9%
BHP	2955	6%
Navoi	2429	5%
Uranium One	1368	3%
Paladin	1210	2%
General Atomics	583	1%

Source: World Nuclear Association

Applications

Uranium is a fissile element, meaning that an atom can easily be broken apart to form lighter elements (fission fragments) when bombarded with neutrons (a nuclear reaction). When bombarded with neutrons U-235 becomes U-236 which immediately divides into smaller nuclei releasing binding energy. The fission fragments have a combined lower mass due to the ejection of neutrons and the release of energy. The ejected neutrons go on to collide with other heavy atoms causing a chain reaction. These properties enable uranium to be used in power generation and nuclear weapons.

Power Generation

In power generation, the chain reaction is controlled to produce usable energy. The energy released is used to convert water into steam, which is used to turn a turbine and generate electricity. There are over 400 nuclear power reactors in the world; the

majority located in North America and Europe. Nuclear power provides c.16% of global electricity generation.

One kilogram of enriched uranium can theoretically produce as much energy as 1,500 tonnes of coal. Most commercial power plants use fuel that is enriched to c.3% uranium-235.

Nuclear Weapons

During the Cold War thousands of nuclear weapons that used enriched uranium and plutonium were produced. At the break-up of the Soviet Union, it is estimated that there was enough weapons grade uranium to make 40,000 nuclear warheads. Weapons grade uranium contains c.90% U-235.

Nuclear weapons have been used only twice in warfare, both in August 1945 at the end of World War II. Estimates of the immediate combined death toll from Hiroshima and Nagasaki vary between 100,000 and 200,000.

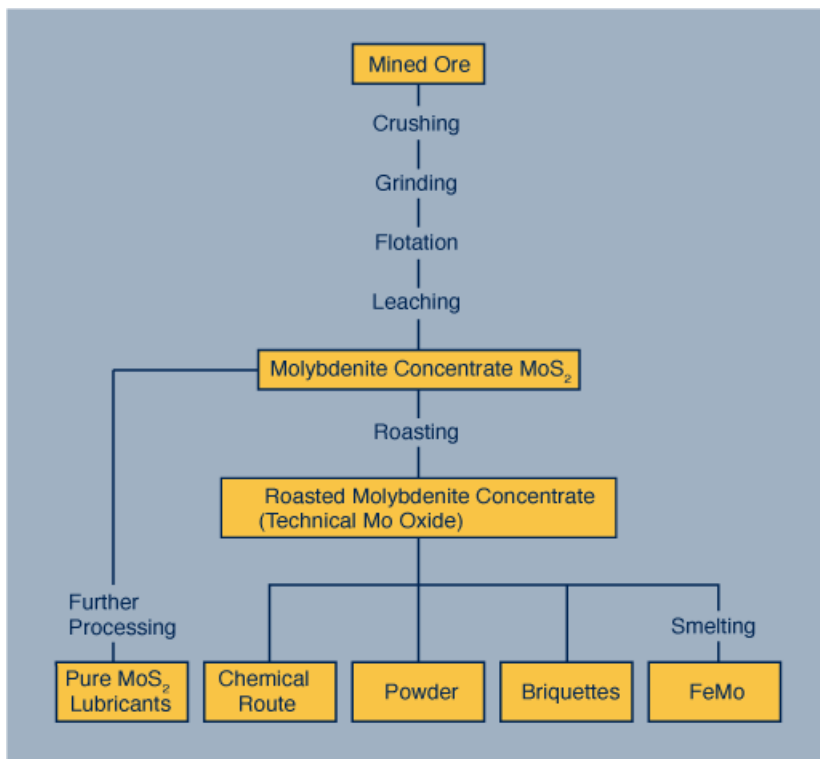
Depleted uranium when alloyed with other elements is also used by the military as high-density penetrators and as armour on combat vehicles.

Molybdenum

The Molybdenum Process

Molybdenum has a concentration of 1-2 ppm in the earth's crust. The only commercial source of molybdenum is molybdenite (a metal sulfide). Ore grades range from 0.01-0.25%. Molybdenum is mostly produced as a by-product of copper but is also produced in large-scale (due to the relatively low grade of most Molybdenum ores) primary projects.

Figure 102. The Molybdenum Process



Source: International Molybdenum Association

- The molybdenum ore is passed through a series of crushers and rotating balls or rod mills to break it into tiny particles less than a millimeter in diameter. The ore is then concentrated in aerated flotation tanks.
- Acid is used in the leaching process to remove impurities such as copper and lead.
- Roasting converts the sulfide concentrate to molybdic oxide by forcing air through the system at temperatures of approximately 650°C. This 10-hour process produces technical grade molybdic oxide which contains a minimum of 57% molybdenum.
- Further processing to increase the Molybdenum content can take place, depending on the intended end-use.

Molybdenum Supply and Demand

Production and Reserves by Geography

The three largest molybdenum producers, US, China and Chile, together account for close to 80% of global production and over 80% of global reserves. As the Molybdenum market is tiny compared to other base metals and because it is largely produced as a by-product, prices are volatile and tend to be highly sensitive to supply-side changes.

Figure 103. Molybdenum Reserves, 2010 Production and Mine Life

	Molybdenum Reserves (kt)	2010 Mine Production (t)	Mine Life (yrs)
China	4,300	94,000	46
US	2,700	56,000	48
Chile	1,100	39,000	28
Peru	450	12,000	38
Russia	250	3,800	66
Armenia	200	4,200	48
Canada	200	9,100	22
Mongolia	160	3,000	53
Kazakhstan	130	400	325
Mexico	130	8,000	16
Kyrgyzstan	100	250	400
Uzbekistan	60	550	109
Iran	50	3,700	14
Total	9,830	234,000	42

Source: USGS, Citi Investment Research and Analysis

Mining Companies

Phelps Dodge of the US and Codelco of Chile, both two of the world's largest copper producers, are the two largest molybdenum producers in the world:

Figure 104. Estimated Top 10 Molybdenum Producers 2007

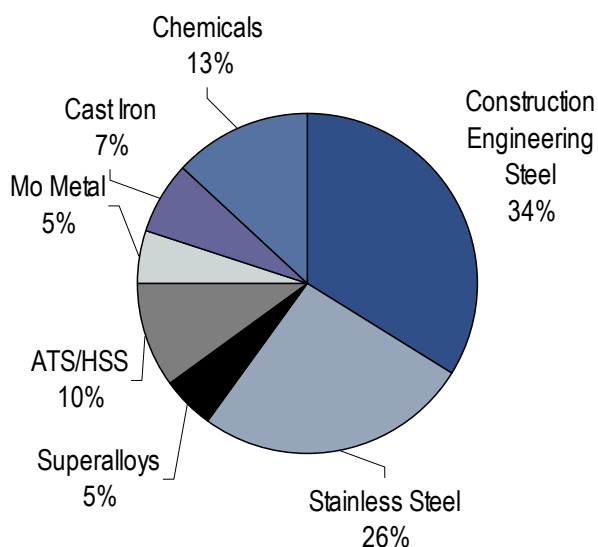
Company	Production (kt)
Freeport	32.9
Codelco	29.5
Kennecott	15.9
SPCC	11.8
Thompson Creek	9.5
Antamina	7.9
Antofagasta	6.8
Endako	5
Collahuasi	4.5
Montana Resources	3.6

Source: Tex, company reports, Citigroup Investment Research

Applications and Consumption

Molybdenum's primary use is in the production of stainless steel and alloy steels, providing strength and corrosion resistant properties. Different types of stainless steel contain different quantities of molybdenum up to a maximum of 7%. It is also used as a catalyst in the petroleum industry, in paint pigments, flame retardants and light bulb filaments.

Figure 105. Consumption by End-use 2009



Source: IMOA

Whilst there are potential substitutes for Molybdenum, for example vanadium, chromium, columbium and boron in alloy steels, tungsten in tool steels, graphite, tungsten and tantalum for refractory materials in high-temperature furnaces, and chrome-orange, cadmium-red, and organic-orange pigments for dies, they are rarely used mainly due to the availability of molybdenum.

Iron Ore and Steel

Iron ore is the largest single contributor to global mining EBITDA. It is a commodity that has delivered decades of strong EBITDA margins and has been an area of significant volume growth and returns for miners over the past decade.

Historically Japan was the largest importer of iron ore to feed their steel requirements. However in the past years China has overtaken Japan as the largest consumer and importer of iron ore. The emergence of China has effectively seen the iron ore market be segregated in to two areas: domestic Chinese production of iron ore and the seaborne traded market.

Production and Reserves by Geography

China, Brazil and Australia are the dominant suppliers of iron ore, accounting for almost 70% of total global production in 2010. Russia and the Ukraine also stand out with significant reserves and exploitation rates far lower than the three aforementioned countries. The production figures provided below are usable ore figures. The average grade of iron ore sold is c.63% therefore the iron content of 2010 iron ore production of 2.4 bn tonnes is ~1.5 bn tonnes.

Figure 106. Iron Ore Reserves and 2010 Production

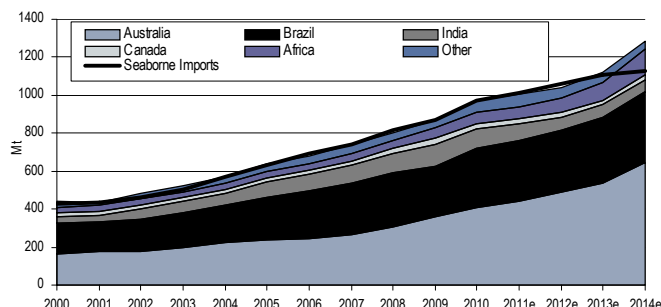
	Iron Ore Reserves (mt)	2010 Mine Production (mt)	Mine Life (yrs)
Ukraine	30,000	72	417
Brazil	29,000	370	78
Russia	25,000	100	250
Australia	24,000	420	57
China	23,000	900	26
Other	11,000	50	220
Kazakhstan	8,300	22	377
India	7,000	260	27
US	6,900	49	141
Canada	6,300	35	180
Venezuela	4,000	16	250
Sweden	3,500	25	140
Iran	2,500	33	76
Mauritania	1,100	11	100
South Africa	1,000	55	18
Mexico	700	12	58
	183,300	2,430	75

Source: Citi Investment Research and Analysis

The Seaborne Iron Ore Trade

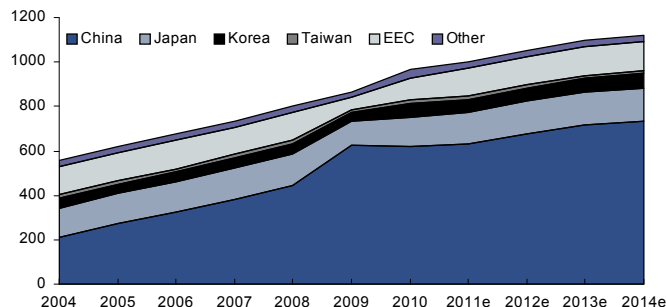
Australia and Brazil are the dominant exporters of iron ore, accounting for over 74% of total exports in 2010. India is also a significant supplier at c.10% in 2010, but Government intervention is likely to see a reduction in future export levels. The major iron ore producers remain Vale, Rio Tinto, BHP Billiton, Fortescue and Kumba.

Figure 107. Seaborne Iron Ore Exporters



Source: Citi Investment Research and Analysis, ICR, Tex Report

Figure 108. Seaborne Iron Ore Importers



Source: Citi Investment Research and Analysis, ICR, Tex Report

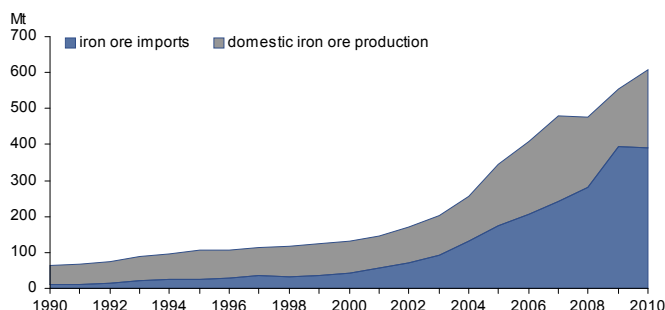
Chinese Domestic Production

Chinese domestic share of production is on declining trend, driven by strong demand and lower cost of imported ore versus domestic cost of production. However we have seen a recent rise in the domestic/import split driven by robust demand and a lack of supply response from the large export markets. As such, domestic producers have been forced to increase production to meet demand from Chinese steel mills. These producers have merely been acting as the industry's swing producer.

Estimating the grade of ROM ore is fraught with difficulties but on the ground estimates and statistics both point to a current average grade of around 20%. Increasing production is likely to result in a further decline in average grade. Notably, increasing production and declining grade have roughly balanced to keep production of iron units stable at around 200Mt.

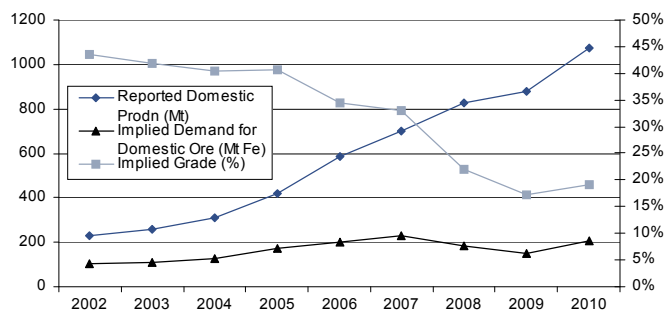
In our forecasts we assume that increasing ROM production is balanced by declining grade, maintaining output of iron units at around 200Mt. Increased domestic production, supported by high prices, is one of the most likely sources of additional supply which will close the projected market deficit in 2011 and 2012.

Figure 109. China's Iron Ore Consumption (Contained iron)



Source: China Customs, Citi Investment Research and Analysis

Figure 110. Domestic Production

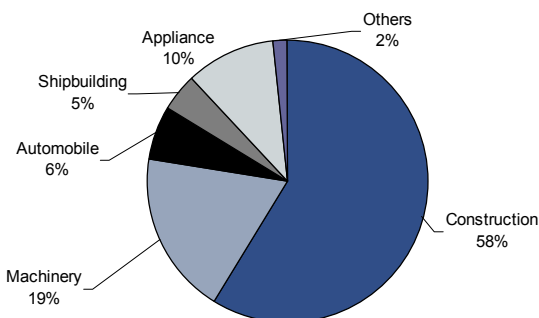


Source: CISA, Citi Investment Research and Analysis

Applications

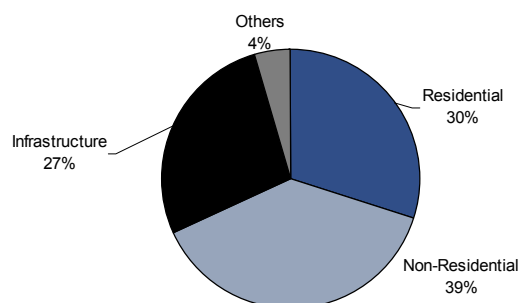
Iron ore is exclusively used in the steel making process and it is by far the most used of all the metals. Iron's low cost and strength make it an essential part in structural components and transportation. Approximately 60% of iron and steel products are used in transportation and construction, 20% in machinery manufacture and the remainder in containers, appliances and equipment.

Figure 111. Steel Consumption – 58% in Construction



Source: Citi Investment Research and Analysis

Figure 112. Steel Consumption in Construction – Residential, Non-Residential, Infrastructure



Source: Citi Investment Research and Analysis

Iron ore types and iron ore mining

The concentration of iron varies between different layers of the earth, from over 80% at the inner core to 5% in the outer crust. Iron is the second most abundant metal in the earth's crust behind aluminium. Common iron ores are hematite and magnetite, both iron oxides.

Hematite (also called Direct Ship Ore): Accounts for the bulk of global production but is actually less common than magnetite ore bodies. Generally high grade with iron content of over 60%. The benefit of hematite ore is that its high grade and mineralogical composition mean that it does not need beneficiation and can be past directly in to the blast furnace process. Thus its cost of production is cheaper.

Magnetite: More common than hematite but lower grade (often sub 50% iron) and thus needs to go through a beneficiation process. The beneficiation process involves crushing, grinding, screening and magnetic separation which significantly increases the cost of production. A growing source of new supply especially from Western Australia.

The steel making process

Iron oxide is reduced to iron in a blast furnace; however, before the ore can be fed into the furnace it must undergo treatment processes – two common methods are pelletising and sintering:

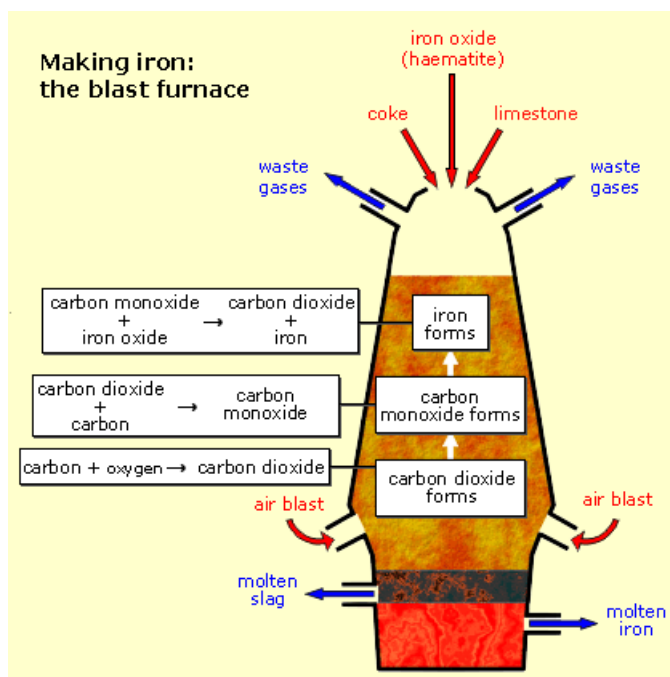
- Pelletising is a treatment process used for very fine or powdery ores. Pellets are an ideal blast furnace feed because they are hard and of regular size and shape.
- Sintering is a process used to agglomerate iron ore fines in preparation for blast-furnace smelting, and is usually carried out at iron and steelmaking centers. It involves the incorporation of crushed limestone, coke and other additives available from iron and steelmaking operations. These additives include wastes extracted from furnace exhaust gases, scale produced during rolling mill operations, and coke fines produced during coke screening.

The diagram below summarises the process flow of the iron making stage.

Blast Furnace

The most common blast furnace feed is lumps or pellets. The iron ore, coke and limestone are added at the top of the furnace whilst heated air is forced through the bottom. Temperatures in the furnace are around 2000°C. The coke reacts with the oxygen forming carbon monoxide which then reacts with the iron ore forming iron and carbon dioxide. The limestone is added to melt impurities present in the iron ore such as sand and sulphur forming silicon dioxide (known as slag). The denser iron sinks to the bottom of the furnace where it is run off and cooled (known as pig iron). The slag is run off separately and can be used in road construction or agriculture.

Figure 113. Blast Furnace



Source: BBC

Pig iron produced from the blast furnace is an intermediate product containing c.5% carbon and impurities such as silicon. It is subsequently converted to steel or a purer form of iron such as cast or wrought iron.

The Steel Process

Iron's main use is to make steel, a strong and versatile metal. Iron making is therefore only the first stage in the production process of an integrated steel mill. The iron making stage is where integrated mills incur up to 70% of their total cash costs. The cost of iron ore is the biggest contributor, particularly for companies that are exposed to market prices (without captive supply of iron ore).

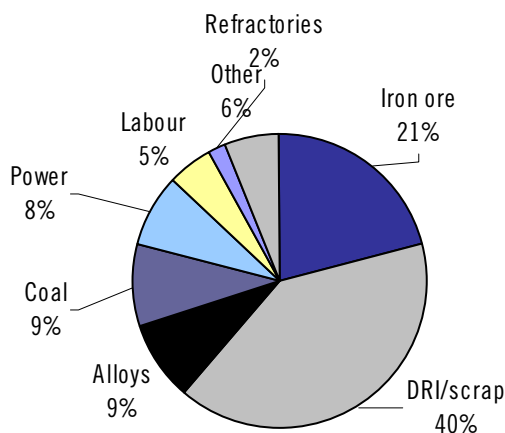
There are many different forms of steel depending on the carbon content (usually under 1%) and the addition of other metals to form steel alloys. Pure iron is relatively soft, however, the addition of a small amount of carbon makes it significantly stronger and harder. Other metals used in steel alloys include chromium, manganese, nickel and molybdenum.

Costs at this stage of the process are primarily determined by the input cost of iron, Direct Reduced Iron (DRI) and scrap. This is the stage where scrap and/or DRI are introduced and depending on availability, costs and production technology/method the consumption level differs from plant to plant. For integrated mills hot metal from the blast furnace is normally the biggest input and is therefore the key determinant of competitiveness. Mini mills could either be scrap-based or DRI-based. The figure below shows the steelmaking process.

Basic Oxygen Furnace (BOF) is normally charged with significantly higher volume of hot metal whereas the Electric Arc Furnaces (EAF) are charged with higher volumes of scrap and/or DRI. Generally the cost of scrap is higher than that of hot metal and DRI. It is normally more expensive to convert scrap to liquid steel via EAF than it is to convert hot metal to liquid steel using oxygen converter.

The chart below shows the cash cost per tonne of liquid steel in the global steel industry as per Metal Bulletin cost data. The cost of iron units (i.e. iron ore / DRI / scrap) make up 61% of total cash costs per tonne of liquid steel produced. It is clear therefore that access to cheap iron units or internal raw material supply is a key differentiator between steel making companies.

Figure 114. Cash cost per tonne of liquid steel



Source: James F. King as published in Metal Bulletin Research: Global Steel Cost Service, Citi Investment Research and Analysis

Casting Stage

Once the liquid steel has been produced, it must now be cast into shapes so that it can be rolled. This is done by continuous casting machines that mould the liquid steel into different shapes called slabs, blooms, and/or billets. These are the first products in the steelmaking process and are classified as semi-finished products. The process is continuous because liquid steel is continuously poured into a 'bottomless' mould at the same rate a continuous steel casting is extracted.

Around 85% of operating costs at this stage are accounted for by the input cost of liquid steel. Other cost components are alloys (3-5%), treatment (3-5%), yield loss and other (5-9%).

Rolling stage

This is the main method used to shape steel into different products after it has been cast. The rolling process (both hot and cold) consists of passing the steel between two rollers revolving at the same speed but in opposite directions. The gap between the rollers is smaller than the steel being rolled, so that the steel is reduced in thickness and at the same time lengthened. For Hot Rolling, slabs, blooms and billets are heated in a furnace before they go through the rolls.

Certain types of steel are cold rolled after hot rolling. Before cold rolling the steel is cleaned with acid (pickled) through a continuous pickling line to remove scale. Cold rolling is carried out at room temperature and is rolled at very fast speeds. Cold rolling increases strength, makes steel thinner and produces a bright smooth surface.

Coating and Painting Stage

Both iron and steel corrode (rust) quickly in water and air. However, there are several methods that can be used to protect against this:

- **Coating:** Some cold rolled products, usually flat steel, can be coated with other metals or paint to protect the steel surface or give it special characteristics. The strip enters a coating bath that contains either a zinc or aluminium mixture, tin or even plastic coating, which is heated. Some coatings are done through an electrolytic process like the electro-galvanised sheets. Hot-dipped galvanised, electro-galvanised and aluminised products are types of coated sheet. These are also the highest value-added sheet products because they require the greatest degree of processing and tend to have the strictest quality requirements.
- **Stainless steel:** Steel can be alloyed with other metals such as chromium and nickel that protect the steel from corrosion. Alloy steels contain varying amounts of carbon as well as other metals, such as chromium, vanadium, molybdenum, nickel and tungsten. They are often used for structural purposes.

Coal

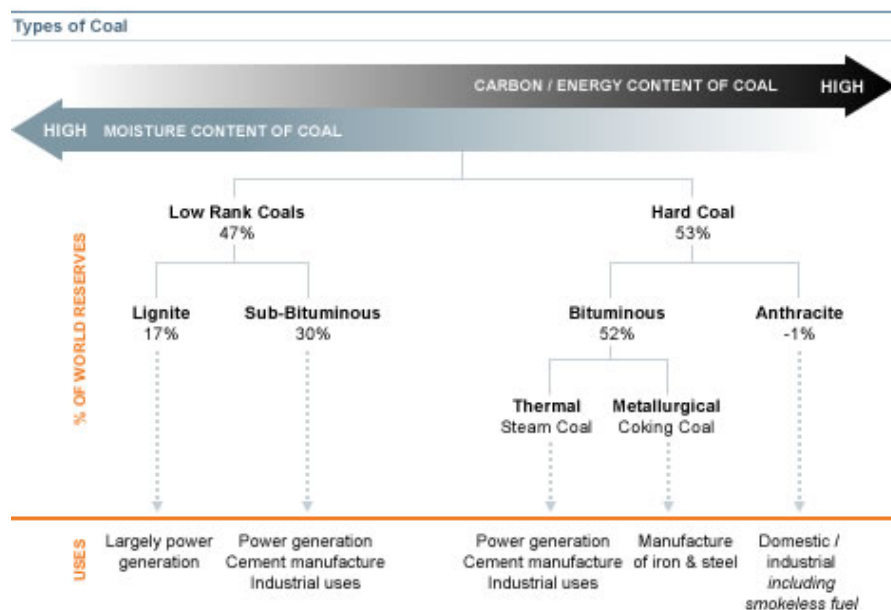
Coal Formation

Coal is a fossil fuel formed from vegetation that has been buried between other rock strata and subjected to high temperatures and pressure, transforming it into coal.

The formation of coal takes place over many millions of years (most mined deposits are over 300m years old) and, as such, different deposits have different 'organic maturities'. As the organic maturity or rank of deposits increases the coal becomes progressively harder and blacker. The final, most mature form of coal is anthracite and is very hard and compact, containing over 92% carbon.

Low rank coal is soft, has high water content and the lowest carbon count (energy content). High rank coal is the opposite, containing more carbon and less moisture.

Figure 115. Types of Coal



Source: World Coal Institute

Coal Processing

Extracted coal contains rock and dirt and is a mixture of size fragments. The treatment method used to convert coal into a consistent quality depends on the type (organic maturity) and intended end-use. Methods can range from simple crushing to more complex treatment:

- Dense medium separation is used to treat larger coal pieces. The raw coal is added to a liquid that is denser than coal but less dense than the impurities; the impurities sink to the bottom whilst the coal floats and is then separated.
- Smaller coal pieces are usually treated in centrifuges where the material is spun very quickly causing the constituent parts to separate. Froth floatation cells can also be used to treat smaller fragments.

Coal Supply and Demand

Production and Reserves

Coal is mined in over 50 countries, however, production and reserves are concentrated around five key nations, together accounting for over 70% of global reserves and production. At current production levels, proven reserves will be exhausted in approximately 123 years, far exceeding the estimated remaining life of oil and gas at 41 and 65 years respectively.

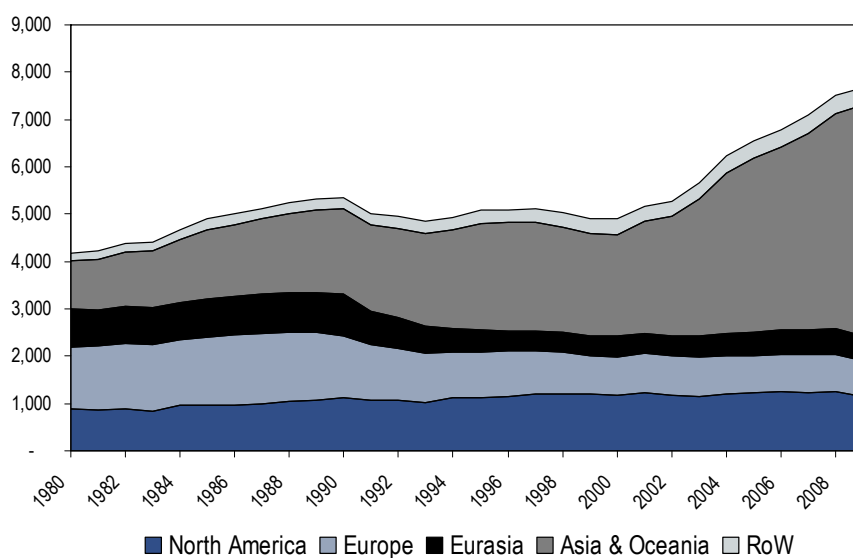
Figure 116. Global Coal Reserves

	2008 Coal Reserves (mt)	2009 Mine Production (mt)	Mine Life (yrs)
United States	260,551	1,073	243
Russia	173,074	327	529
China	126,215	3,362	38
Australia	84,217	440	191
India	66,800	611	109
Germany	44,863	204	220
Ukraine	37,339	61	616
Kazakhstan	37,038	112	331
South Africa	33,241	273	122
Serbia	15,179	42	360
Total	948,000	7,680	123

Source: US EIA

Global production increased at a compound rate of more than 5% between 2000 and 2009; however, there are divergent trends between Europe, where production has declined by 10%, and Asia, where production has more than doubled over the same period.

Figure 117. Global Coal Production by Region – 1980 to 2009 (Mt)



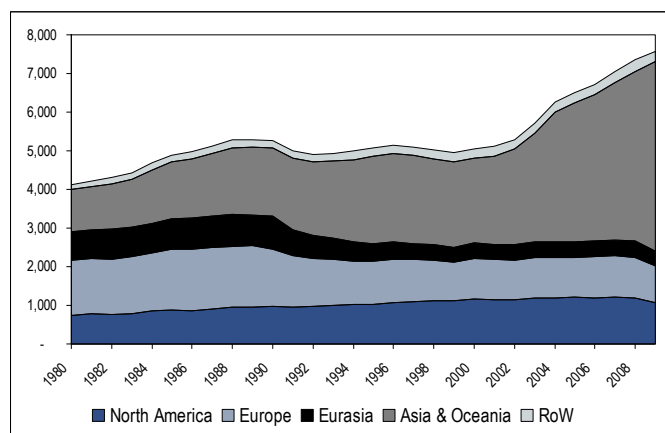
Source: Citi Investment Research and Analysis

Applications and Consumption

The primary uses of coal are in power generation or iron and steel making, but it is also used in cement manufacturing, liquid fuel, paper manufacturing and the chemical and pharmaceutical industries.

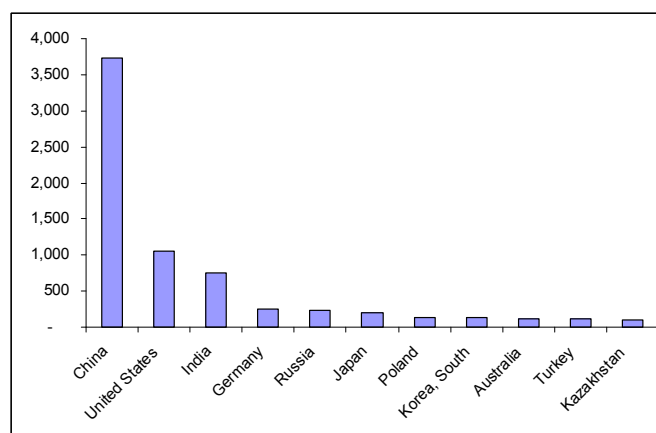
Coal provides 25% of global primary energy needs and fuels almost 40% of electricity globally and it is expected to remain at this percentage for the foreseeable future. It is estimated that the steel industry currently uses c.664 Mt of coal. Since 2000, consumption trends have mirrored production trends with minimal growth in Western regions and compound growth in Asia of close to 10%.

Figure 118. Global Consumption – 1980 to 2009



Source: EIA

Figure 119. Top 10 Consumers 2010 (Mt)



Source: EIA

Coal Imports and Exports

The majority of global coal produced is consumed internally. However, there is an active global market in coal, with overall international trade in coal reaching 775 Mt in 2005. Australia is the world's largest exporter, principally supplying South-east Asia, but also Europe and South America. Australia exported over 75% of its total hard coal production in 2005, accounting for over 50% of global coking coal exports. The US market largely acts in isolation with total imports and exports both representing less than 3% of consumption and production respectively.

Figure 120. Thermal coal exports (mt)

Exports	2009	2010e	2011e	2012e	2013e	2014e	2015e
Australia	139.0	140.9	152.9	177.5	186.6	200.5	215.6
South Africa	59.9	70.0	73.0	78.0	80.0	80.0	82.3
Indonesia	190.2	215.5	242.0	270.0	305.0	320.0	323.6
US	11.6	15.0	18.0	16.0	15.0	14.0	13.0
China	16.9	20.6	17.9	8.9	2.6	2.9	3.0
Columbia	67.9	62.0	64.0	69.1	69.1	69.1	75.2
Canada	6.9	6.9	6.9	6.9	6.9	6.9	6.9
Russia	91.7	80.4	88.1	96.4	102.6	102.6	117.2
Vietnam	17.2	24.0	20.0	25.8	22.0	23.0	20.7
Venezuela	5.0	5.0	5.0	5.0	5.0	5.0	5.0

Source: Brook Hunt, Citi Investment Research and Analysis

Figure 121. Coking coal exports (mt)

Exports	2009	2010e	2011e	2012e	2013e	2014e	2015e
Australia	135.1	158.9	139.7	168.9	180.4	191.9	192.9
US	31.4	47.8	53.0	50.0	50.0	52.5	55.0
Canada	20.6	26.1	26.1	26.1	25.0	25.0	25.0
China	0.6	1.1	0.6	0.6	0.6	0.6	0.6
Russia	10.5	10.0	14.0	12.0	12.0	12.0	13.0
Mozambique	1.0	0.0	4.4	6.0	10.0	12.2	12.2
Other	1.4	3.7	3.0	3.0	3.0	3.0	3.0
Total	200.6	247.7	240.8	266.6	281.0	297.2	301.7

Source: Brook Hunt, Citi Investment Research and Analysis

Gold

The Gold Process

Gold occurs in its natural form as nuggets or grains in rocks, underground veins and alluvial deposits (sediments deposited by water), and is very rare. The earth's crust contains approximately 4 ppbn.

Gold can be commercially extracted from ore grades as low as 0.5 ppm from open pit mines (3 ppm for underground or hard rock operations), although typical ore grades are between 1-5 ppm.

Gold mining techniques have largely moved on from the panning and dredging methods used in various gold rushes throughout history, although both are still used today to identify new deposits.

The majority of gold is produced from hard rock mining. Gold treatment involves a combination of methods depending on the ore type and grade. Low grade oxide deposits are treated directly using cyanide leaching, which we discuss below. However, higher grade ores, or ores resistant to leaching (ores containing sulfides), require pre-treatment (roasting, bio-oxidation, pressure oxidation or grinding) before leaching in order to concentrate and / or oxidise the ore.

Ore Processing:

- The ore is crushed and screened and dumped onto a leach pad. The finely ground gold ore is sprayed or irrigated with a dilute cyanide solution, dissolving the gold. After several weeks, the pregnant solution runs from the heap and is collected in ponds. Heap leaching extracts c.50-65% of the gold from the ore and takes 1-2 months; the remaining ore can either be leached again or re-used in the milling process.
- The solution is passed through a series of tanks containing activated carbon columns that absorb the gold onto the surface of the carbon. Resins can be used in place of carbon and have several advantages such as removing all cyanide complexes and enabling cyanide recycling.
- The carbon columns are then washed with a heated caustic soda (sodium hydroxide) and cyanide solution, dissolving the gold.
- The gold bearing solution is then passed through electrowinning cells where the gold plates onto stainless steel wool cathodes.
- The gold is washed from the cathodes using high pressure sprays, dried and then melted in a furnace at over 2,100°F and cast into dore bars or ingots. These are unrefined gold bars containing between 60-95% gold. The bars are then sent to a refinery for further processing into pure gold (99.9% or 99.99%). Refined 99.9% gold is termed London Good Delivery and can be sold in the bullion market.

Gold Supply and Demand

Production and Reserves

Gold mine production increased significantly over the 1980s and 1990s from less than 1.3 Kt in 1980 to over 2.5 Kt by 1998. The increase has been fuelled by increased production from the majority of gold producing nations, offsetting decreasing production from South Africa. Annual production has now leveled off at c.2.5 Kt per annum. The production of gold has little impact on price since the majority of gold ever mined still exists and dwarfs the amounts produced by mining annually. The total volume of gold ever mined is estimated at over 155 Kt which compares to annual production of c.2.5 Kt.

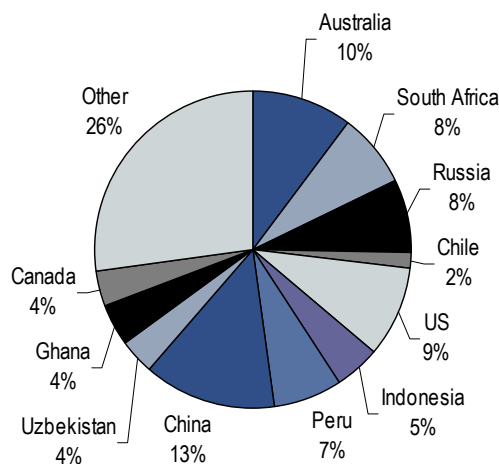
Figure 122. Global Gold Reserves and 2010 Production

	Gold Reserves (t)	Mine Production (t)	Mine Life (yrs)
Australia	7,300	255	29
South Africa	6,000	190	32
Russia	5,000	190	26
Chile	3,400	40	85
US	3,000	230	13
Indonesia	3,000	120	25
Brazil	2,400	65	37
Peru	2,000	170	12
China	1,900	345	6
Uzbekistan	1,700	90	19
Ghana	1,400	100	14
Mexico	1,400	60	23
PNG	1,200	60	20
Canada	990	90	11
Other	10,000	500	20
Total	50,690	2,505	20

Source: USGS

South Africa is the world's largest gold producer, however, volumes have declined in recent years due to the increasing difficulty of extraction (some gold mines in South Africa are now close to 4,000m deep). In the early 1970s, South Africa was producing c.1000 tonnes of gold p.a. (over 70% of global production); however, in 2010 production was down to 190 tonnes or c.8% of global production.

Figure 123. 2010 Global Gold Production



Source: GFMS Ltd

Gold Mines

Figure 124. 10 Largest Gold Mines

Rank	Name	Country	Ownership	Production (t)
1	Yanacocha	Peru	Newmont 51%, Buenaventura 5%, BRGM 44%	83
2	Newmont Nevada	US	Newmont 100%	71
3	PT Freeport Indonesia	Indonesia	Freeport 82.6%, RIO 7.4%	62
4	Goldstrike (inc Meikle)	US	Barrick 100%	59
5	Driefontein Consolidated	South Africa	Goldfields 85%, Mvelaphanda Resources 15%	35
6	Lagunas Norte	Peru	Barrick 100%	31
7	Kloof	South Africa	Goldfields 85%, Mvelaphanda Resources 15%	30
8	Kalgoorlie Superpit	Australia	Barrick 50%, Newmont 50%	26
9	Olympiada	Russian Federation	Norilsk 100%	26
10	Tarkwa	Ghana	Goldfields 71%, Iamgold 19%, Private 10%	23

Source: Brook Hunt, Citi Investment Research and Analysis

Gold Mining Companies

Figure 125. Top 15 Gold Producers 2010

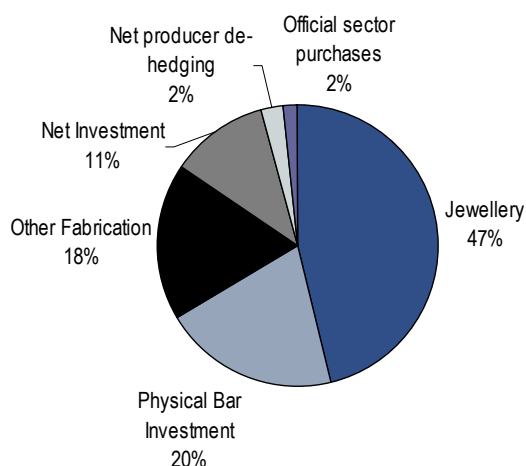
Company	Domicile	2010 Gold Production (t)
Barrick	Canada	242
Newmont	USA	168
Anglogold Ashanti	S. Africa	140
Gold Fields	S. Africa	102
Goldcorp	Canada	78
Newcrest	Australia	73
Kinross	Canada	68
Navoi MMC	Uzbekistan	63
Freeport McMoRan	USA	53
Polyus	Russia	43
Harmony	S. Africa	42
Cia. De M. Buenaventura	Peru	34
Agnico-Eagle	Canada	31
Zijin	China	30
IAMGOLD	Canada	30

Source: Citi Investment Research and Analysis

Gold Demand

Gold, more so than any other commodity, is subject to hoarding and dis-hoarding based on investor sentiment, and this, rather than industrial applications or jewellery demand, is the key determinant of the gold price. The price of gold has surged in the past seven years fuelled by political and economic uncertainty, rising oil prices and the decline in the US\$.

Figure 126. 2010 Global Gold Consumption

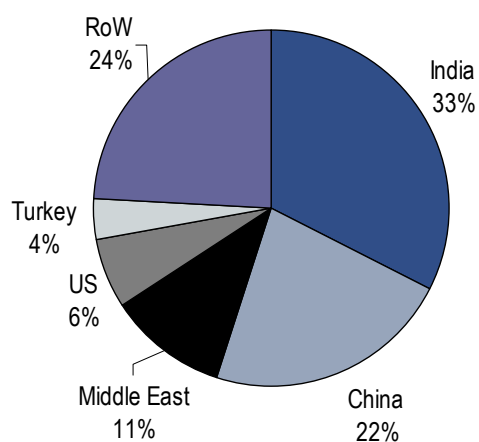


Source: GFMS Ltd

Jewellery

Gold is the most malleable of all the known metals and this property in tandem with its high resistance to corrosion and rarity make it well suited for use in jewellery and coinage. Pure 24 karat gold is too soft for most uses and is hardened by alloying with other metals. To form jewellery, gold can be mixed with silver, copper or PGMs. Over recent years, gold jewellery has faced increased competition for disposable income from other jewellery types, electronics, cosmetic surgery and luxury holidays. Below we provide a regional split of 2010 jewellery demand:

Figure 127. 2010 Gold Jewellery Consumption



Source: Company Reports

PGMs

Overview

There are six platinum group metals, all with strong catalytic properties:

- Platinum and palladium are the two most important PGMs and have the widest use owing to their high resistance to corrosion and high melting point. We focus on these two metals over the following pages.
- Rhodium and iridium are difficult to work with but are valuable alone or as in alloys.
- Osmium and ruthenium are hard and brittle and have little use in their metallic state but are valuable as additions to other metals and as catalysts.

The Mining Process

All of the PGMs occur in close association with one another, together with nickel, gold and copper. They are extremely rare, with platinum occurring as only 1 ppb in the Earth's crust. Platinum can be found uncombined as native platinum, alloyed with other PGMs or as a mineral such as cooperite. The production of a single ounce of platinum requires between 7 and 25 tonnes of ore, depending on ore grade. The extraction process is shown below:

PGMs Supply and Demand

Production and Reserves by Geography

PGMs are extremely scarce, with 75% of global platinum production sourced from South Africa and 44% of Palladium sourced from Russia. Other sources include Zimbabwe, Russia, Canada and the US. South African mining is centered round the Bushveld Complex, with the Merensky reef accounting for approximately half of platinum production in South Africa, whilst Russian operations are based within the Ural Mountains.

Figure 128. Global Reserves and Mine Life

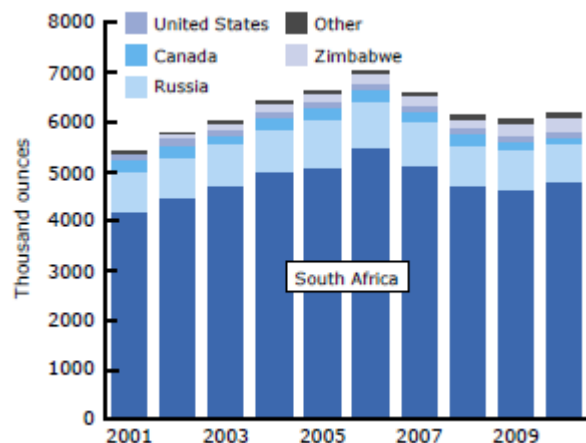
	PGM Reserves (kg)	Platinum Production (kg)	Palladium Production (kg)	Mine Life (yrs)
US	900,000	3,500	11,600	60
Canada	310,000	5,500	9,400	21
Colombia	n/a	1,000	n/a	n/a
Russia	1,100,000	24,000	87,000	10
South Africa	63,000,000	138,000	73,000	299
Zimbabwe	n/a	8,800	6,600	n/a
RoW	800,000	2,400	9,800	66
Total	66,110,000	183,200	197,400	174

Source: US Geological Survey

Platinum and palladium are found in the largest quantities. The other four metals are only produced as bi-products of platinum, palladium and nickel. As such, the supply of these metals is completely inelastic.

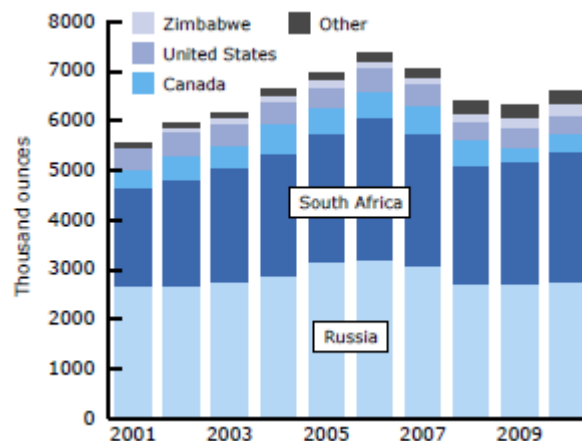
In South Africa, the producers have mined the more profitable and shallower deposits first and future production growth is dependent on the development of deeper mines and the Eastern limb of the Bushveld Complex.

Figure 129. Platinum Mine Production



Source: GFMS

Figure 130. Palladium Mine Production



Source: GFMS

Mining Companies

Figure 131. Top PGM Mining Companies

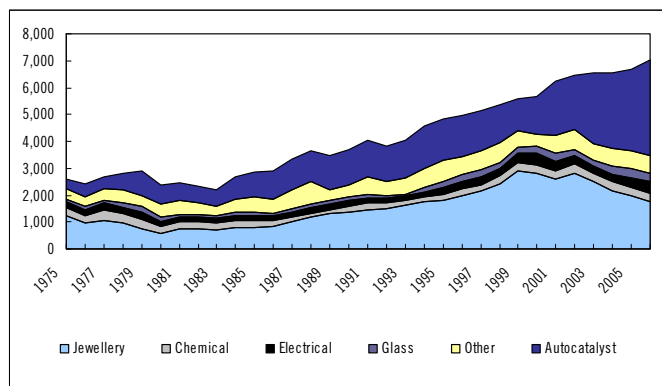
Company	2010 Platinum production (koz)	2010 Palladium production(koz)
Anglo Platinum	1989	1133
Impala Platinum	1265	759
Norilsk Nickel	693	2861
Lonmin	656	305
Aquarius Platinum	266	147
Xstrata	179	220
ARM Platinum	152	131
Northam	150	77
Royal Bafokeng Platinum	126	n/a
Stillwater Mining	111	374

Source: GFMS

Applications and Consumption

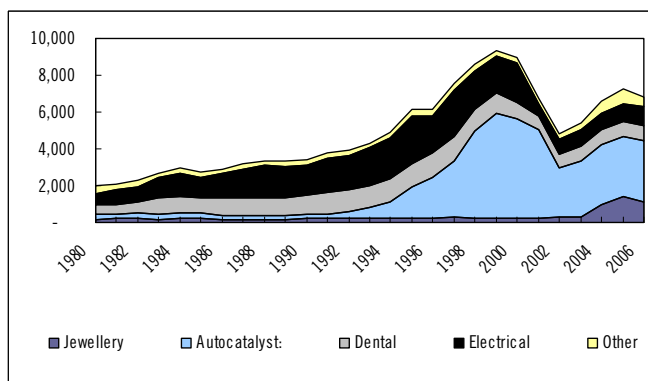
PGMs' high resistance, high melting point and stable electrical properties have been exploited for industrial applications. Platinum, palladium and rhodium are all used in the autocatalyst industry. Demand for platinum from the autocatalysts sector has grown rapidly since 1999 from 1.2m/oz in 1999 to over 3.5m/oz in 2006, driven by stricter emissions legislations favoring diesel engines and supported by the improved performance of modern diesel engines and higher fuel prices.

Figure 132. Platinum consumption by end-user



Source: Johnson Matthey

Figure 133. Palladium consumption by end-user



Source: Johnson Matthey

One in every two cars sold in Europe is a diesel engine and we expect a strong demand structure to remain in place over the coming years supported primarily by the introduction of diesel engine cars in North America and increased demand from Eastern Europe, Russia and Asia.

Other industrial uses for PGMs include the manufacture of nitric acid, hard disks, silicones, dental work and glass. Platinum is also used as an investment metal; however, demand has declined over the last decade to nominal levels.

Jewellery

Platinum's high resistance to tarnish, strength, malleability and rarity make it well suited to the fine jewellery market and more precious than gold (the price of platinum is approximately twice that of gold). In the 1990s jewellery demand was driven by North America and China. However, jewellery demand is elastic and has tailed off since 2000 due to increased prices caused by greater demand from industry. Recent trends show that for every \$100 increase in the platinum price there is a 200 Koz sacrifice in jewellery demand.

Mineral Sands

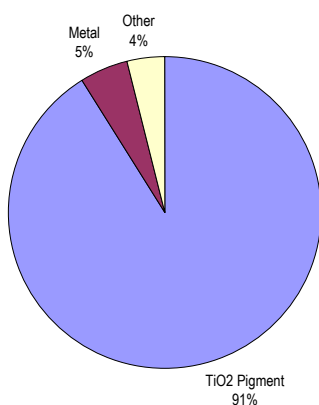
Mineral sands are formed along ancient coastlines where eroded rock is carried downstream before being trapped and concentrated into an economic deposit at a headland by wind and wave action. The mineral sand moniker that covers a range of materials with specific gravities above 2.85. The two major streams are **titanium dioxide (TiO₂)** which is in the form of ilmenite, rutile & leucoxene and **zircon**.

The mineral sands have attributes that make them attractive in a wide range of applications.

Titanium Dioxide

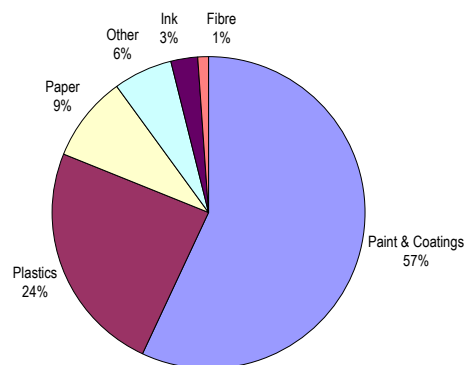
The titanium dioxide stream within the mineral sands complex covers ilmenite, rutile, leucoxene as well as products such as synthetic rutile and titanium slag. The primary use for TiO₂ is in creating white pigment, which accounts for over 90% of all demand. Of the pigment, paint accounts for over 50% of demand with plastics accounting for almost a quarter.

Figure 134. TiO₂ demand by end application



Source: Iluka and TZMI

Figure 135. Pigment demand breakdown



Source: Iluka and TZMI

TiO₂ is suitable for creating pigment as it has a high refractive index which allows it to scatter and bend light strongly. When a threshold of TiO₂ in a product is reached almost all light is refracted which gives the substance the appearance of being opaque and a bright white.

TiO₂ also has the ability to absorb UV light by efficiently turning it into heat. If TiO₂ is added to materials such as paint, plastics and ceramics it prevents degradation from the effects of the UV light making it a great coating for the hot Australian sun.

High grade titanium dioxide is also a feed source for the manufacturing of titanium metal. Titanium is exceptionally strong and light (high strength to weight ratio) which makes it ideal for use in the aerospace industry. Titanium is also corrosive resistant and as such it is also used extensively in chemical applications. Titanium metals accounts for less than 5% of all demand. The other key use for rutile is in the welding rod market.

The total TiO₂ market is ~6mt/year with demand typically growing at ~3%pa. The major pigment markets are Europe and the US, but China will soon become the largest customer given strong demand growth.

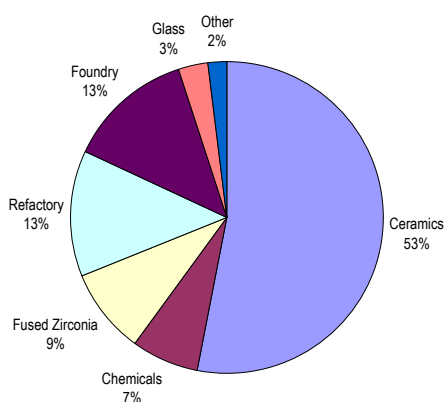
Zircon

The primary use for Zircon is as an agent to make ceramic based products opaque. The ceramic industry is dominated by the tile market with some demand from table ware and sanitary applications (urinals, toilet bowls etc). The high refractive index of zircon makes it perfect for use in making the tiles opaque as well as white. Zircon also has a high hardness making it resistant to scratching and other types wear.

A high heat resistance also makes it suitable for use in foundry and refractory applications, although there has been substitution pressures in these applications over time.

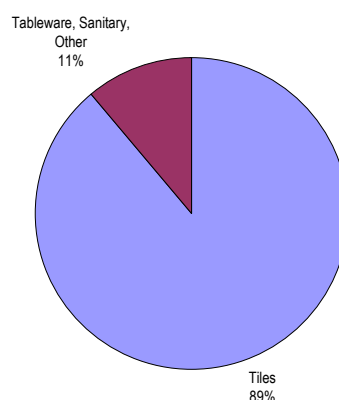
The zircon market is ~1.7mt/year.

Figure 136. Zircon consumption by end use



Source: Iluka and TZMI

Figure 137. Ceramic end use



Source: Iluka and TZMI

Mineral Sands Mining

Unlike other types of mining, the mineral sands process rarely uses blasting. There are effectively 2 types of mines in the industry, although some operations use a combination of both:

- Dry mining – typically used for small, high grade deposits where the amount of material being moved is relatively low.
- Dredge mining – used for large, low grade deposits where a lot of material needs to be moved, and especially for deposits below the water table. The dredge typically sits in a man made pond and contains the “wet” processing plant.

At the “wet” plant, ore from the dry or dredge mining unit is sent through a series of spirals that use gravity to separate the mineral sands from the lighter quartz and clay that they are hosted in. From there the mineral sands are sent to a “dry plant” for processing.

Mineral Sands Processing

The concentrate from the mining stage contains a host of different mineral sands in what is called the assemblage. The heavier mineral sands (ilmenite, rutile and zircon) are separated out from the concentrate using the minerals electrostatic, magnetic and density characteristics. The various components are then further treated and purified. Zircon is put through an acid wash to remove the iron oxide coatings.

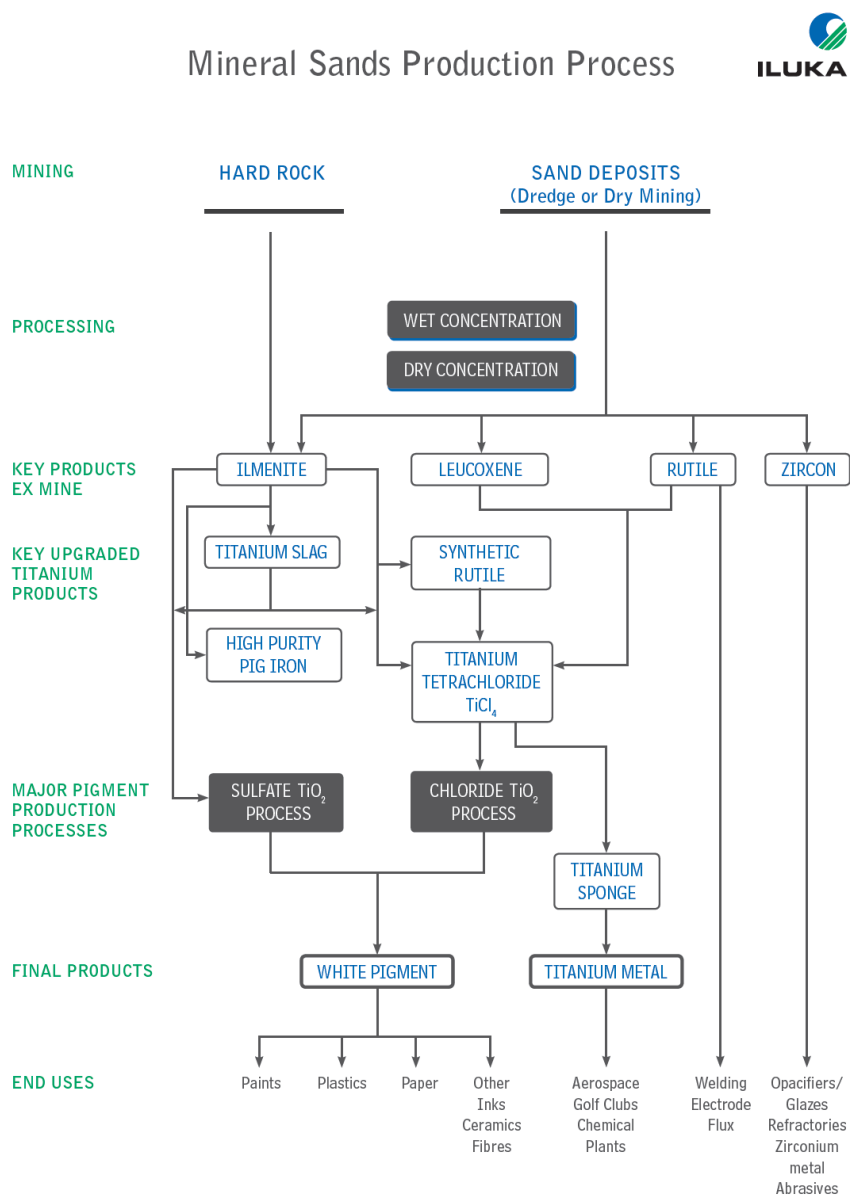
Zircon and leucoxene are generally sold directly to pigment producers, titanium sponge or welding rod manufacturers.

Ilmenite can either be sold directly to a pigment producer, or upgraded into the higher grade TiO_2 products slag or synthetic rutile.

Slag is produced through an electric arc furnace that produces a product with a 80-90% TiO_2 content (rutile ~95% TiO_2 and ilmenite 35-60+% TiO_2). Slag is normally produced where there is availability of cheap power and with low grade ilmenite – iron is the other key component in the ore and the produces a high purity pig iron that is a valuable by-product credit.

Synthetic rutile is produced through a kiln reduction process and typically uses a higher grade ilmenite feed along with coal as the reductant. Iron is a waste product rather than a by-product from this process, but the higher grade end-product (92-94% TiO_2) sells at a premium price to slag.

Figure 138. Mineral Sands Production Process

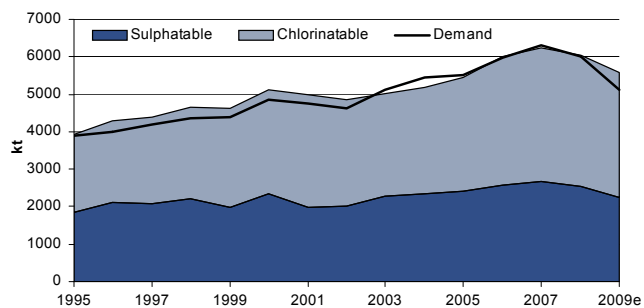


Source: Iluka

The Mineral Sand Market

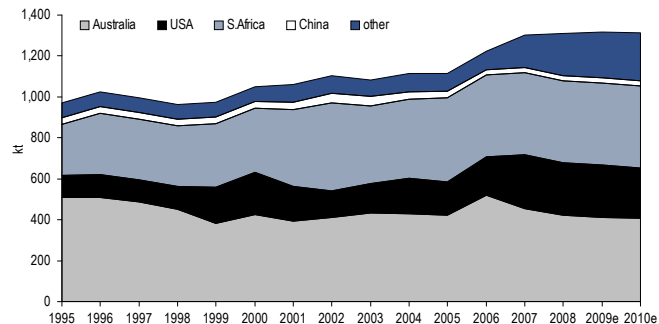
Australia and South Africa are the largest producers of zircon and TiO₂.

Figure 139. Feedstock Production



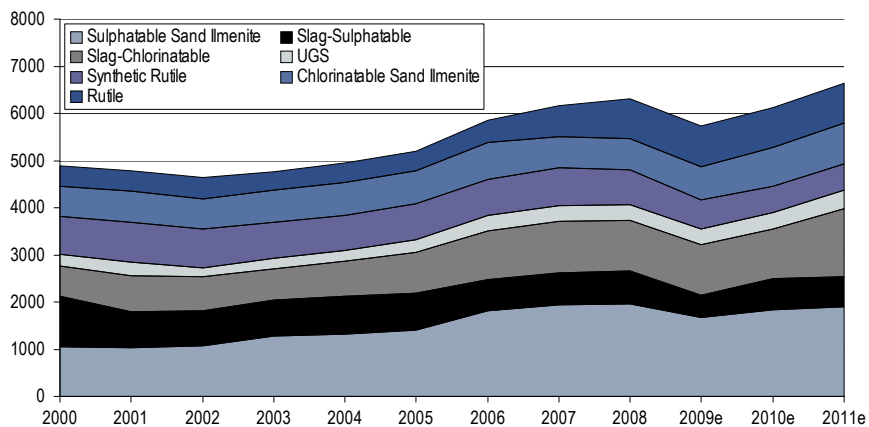
Source: TZMI

Figure 140. Zircon Supply



Source: TZMI

Figure 141. TiO₂ Feedstock Supply



Source: TZMI, Iluka and Citi Investment Research and Analysis

Rare Earths

The REO acronym classifies 15 metallic elements which in turn are further divided into heavy and light rare earths. As the picture below shows, REO's are about as common as cobalt, lead, nickel and uranium. In fact REO's are about 10,000 times more common than platinum. However it is the geographic location of reserves and production (+90% domiciled in China) as well as the acceleration in demand from new technologies that has sent the spotlight focus onto this small sector of the metals market.

REO's are a minor but critical component of many advanced technologies such as hybrid vehicles and mobile electronics as well as being used in the petrochemical and environmental sectors. The distinct properties of the rare earths (namely magnetic, catalytic and electrical) make them a critical component of many technological applications. The evolution of technology and a focus on more environmentally friendly products is leading to a considerable increase in the consumption of rare earths where their properties lead to more advanced, cleaner and efficient products. The rare earth complex produces a range of underlying metals which are used across a broad range of applications. The tables below looks at the major REO metals and their applications.

Figure 142. Applications for REO's

Name	Code	Application use				
		Catalytic	Magnetic	Electrical	Chemical	Optical
Lanthanum	La	√		√	√	√
Cerium	Ce	√		√	√	√
Praseodymium	Pr		√	√	√	√
Neodymium	Nd	√	√	√		√
Samarium	Sm		√			
Europium	Eu					√
Gadolinium	Gd		√			√
Terbium	Tb		√			√
Dysprosium	Dy		√			√
Holmium	Ho					√
Erbium	Er					√
Ytterbium	Yb					√
Yttrium	Y					√

Source: BCC Research and Citi Investment Research and Analysis

Figure 143. Applications for REO's

Periodic code	Name	Usage
La	Lanthanum	High refractive index glass, flint, hydrogen storage, battery-electrode, camera lens
Ce	Cerium	Chemical oxidising agent, polishing powder, yellow colors in glass and ceramics, catalyst for Self-cleaning oven etc.
Pr	Praseodymium	Rare-earth magnets, laser, green colors in glass and ceramics, flint
Nd	Neodymium	Rare-earth magnets, laser, violet colors in glass and ceramics, ceramic capacitor
Pm	Promethium	Nuclear battery
Sm	Samarium	Rare-earth magnets, Laser, neutron capture, maser
Eu	Europium	Red and blue phosphors, laser, mercury-vapor lamp
Gd	Gadolinium	Rare-earth magnets, high refractive index glass or garnets, laser, x-ray tube, computer memory, neutron capture
Tb	Terbium	Green phosphors, laser, fluorescent lamp
Dy	Dysprosium	Rare-earth magnets, laser,
Ho	Holmium	Laser
Er	Erbium	Laser, vanadium steel
Yb	Ytterbium	Infrared Laser, chemical reducing agent, High-temperature superconductors (YBCO)
Lu	Lutetium	.

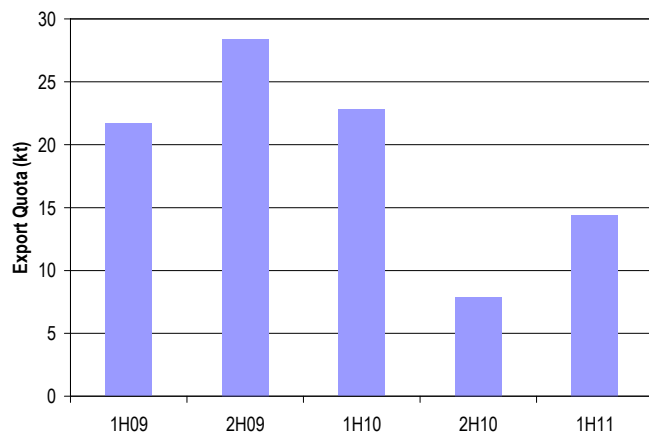
Source: USGS and Citi Investment Research and Analysis

As the charts below indicate that Cerium, Lanthanum and Neodymium account for 90% of the REO split by weight and 86% by price. The pricing of these 3 metals will have the greatest impact on the overall REO basket price.

Chinese supply issues

The squeeze in the REO market has been primarily driven by the emergence of Chinese quotas in restricting exports. Given that the Chinese account for approximately 95% of production, a shift in the export levels significantly alters pricing. The tightening in export levels in 2H10 (chart below left) was the prime driver behind the escalation in the REO price from around US\$15/kg to current prices closer to US\$70/kg.

Figure 144. Chinese REO export quotas



Source: Roskil & Citi Investment Research and Analysis

Figure 145. Chinese REO export quotas (tonnes)

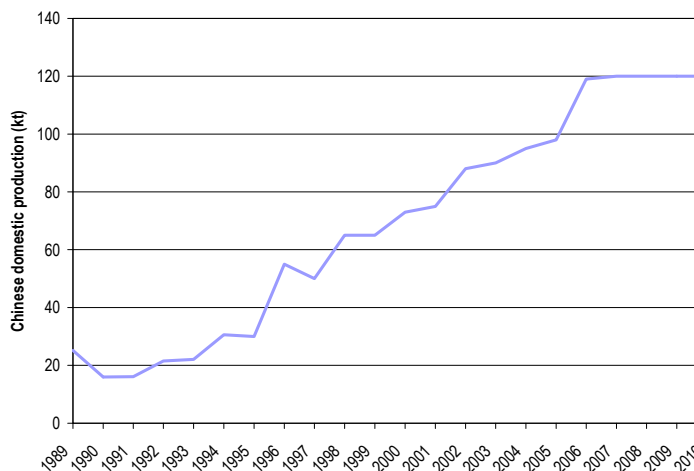
	2004	2005	2006	2007	2008	2009	2010	2011*
Domestic Companies	48040	48040	45752	43754	40987	33300	22513	
Foreign JV's	17569	17569	16069	16069	15950	16845	7746	
Total	65609	65609	61821	59823	56937	50145	30259	28892
YoY change		0%	-6%	-3%	-5%	-12%	-40%	-5%

Source: Roskil & Citi Investment Research and Analysis. * Note: Annualized 1H11 numbers

Chinese production of REO has grown considerably over recent years with production almost doubling from 2000-2010. Current production levels are around the 120ktpa level. The supply comes from 3 main sources in China:

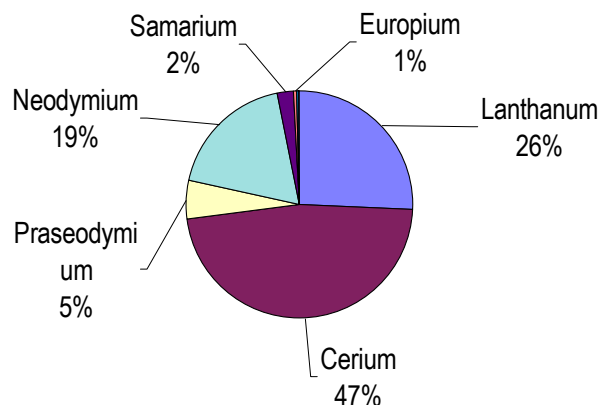
- Baotou (c55kt) – iron ore by product.
- Sichuan (c10kt) – heavy element rare earths.
- Clay (c50kt) – high level of illegal mining and very environmentally unfriendly.

Figure 146. Chinese REO production



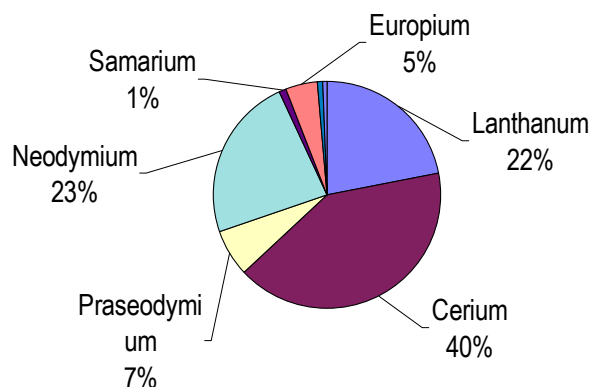
Source: Roskil & Citi Investment Research and Analysis

Figure 147. REO split by \$ weight



Source: Citi Investment Research and Analysis

Figure 148. REO split by \$ value

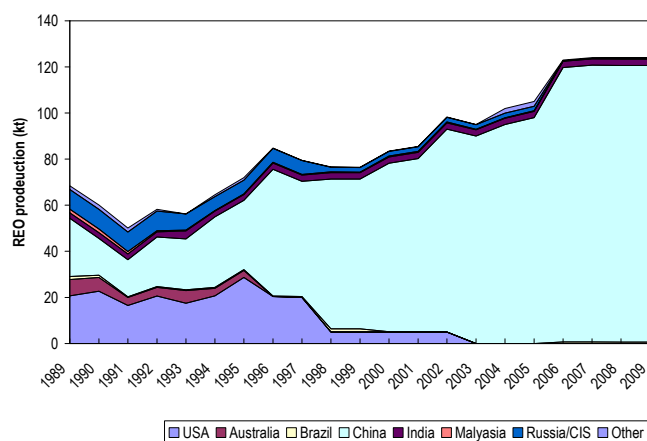


Source: Citi Investment Research and Analysis

The issues with REO's is the location of the reserve base, grade of deposits and the historic low price of the metal not incentivizing new supply. However with high prices and growing demand we believe there will be a supply side response in the REO market which will drive prices lower.

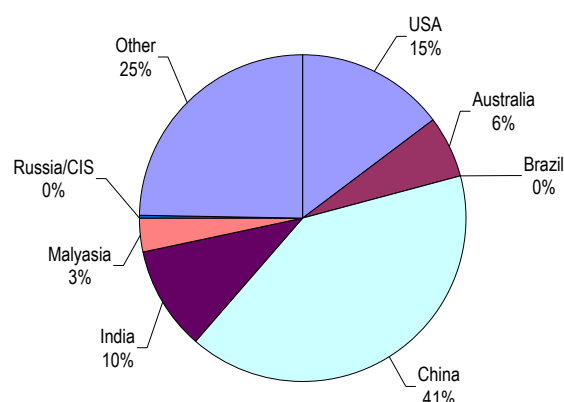
The bulk of the world's reserves and production is based within China which has recently looked to set up export quotas to guarantee enough supply for domestic sources.

Figure 149. REO Supply



Source: USGS and Citi Investment Research and Analysis

Figure 150. REO Reserves



Source: USGS and Citi Investment Research and Analysis

The REO market has seen massive upside in 2010 after years of downward pricing pressure. Demand has been growing for some time as new hi-tech applications have increased their consumption of REO's. However it was the restriction of Chinese exports of REO in the second half of 2010 that really set a spark under the REO price. With China controlling 90% of world REO supply and limited new global supply to come on stream until 2012 many expect the REO price to stay high. Whilst we recognise that the price may indeed be stronger than we have seen previously, we believe that the REO market is potentially a bubble. The major drivers we see that could see a correction in REO prices are:

- Demand destruction (thriftiness, new technology, etc).
- Recycling.
- New supply.

All of the above have the capacity to alter the supply/demand balance and reduce REO prices.

The major positive driver for REO prices is the tightening of Chinese export quotas. Whilst the above could see a significant shift downwards in the REO price, the fact that China controls the bulk of supply can be viewed as a positive driver. The Chinese are currently reducing their export quotas of REO's (1H11 exports down 25% vs 1H10 levels) which has been the main driver for the acceleration in the REO price from the cUS\$15/kg mid year to US\$70/kg today. By limiting exports China is effectively trying to force the manufacturing of goods that use REO into mainland China, further helping to develop Chinese hi-tech manufacturing capabilities. Whilst the restriction of Chinese supply is a massive driver for short term pricing, we believe that the tactic will only fast track demand and supply alternatives and ultimately speed up the decline in REO prices.

Section 7: Modeling Made Easy

Modeling Made Easy

In this section we look at how we construct our models and derive our valuation on a company. Modeling can be as simple or as complicated as the analyst determines. There is no shortage of information that can be put in the model. The key factor is balancing detail versus accuracy versus output. Often a 1% move in a commodity price has a significantly larger impact on earnings and NPV than shifts in cyanide costs, tyre costs and face meters advanced.

More detail does not necessarily equal more accuracy.

For us the core steps in developing a model are determining your key assumptions which in loose order are:

- Commodity and FX assumptions
- Life of asset
- Mill size
- Ore grade through the mill and recovery rates (i.e. production)
- Revenue generated at site
- Cost of production
- Tax rates
- Capital cost for project development and maintenance (stay in business capital)

The above assumptions help generate a profit and cash generation profile for an asset. When the asset profiles have been determined we move on to the corporate assumptions such as:

- Discount rate used for NPV
- SGA, exploration and other head office expenses
- Debt position for the company
- Value of any other exploration assets

The above are the major steps we go through in designing the financial models for the companies we cover. There are different nuances from commodity to commodity and company to company but the generic 'back bone' to model creation remains the same.

To look at how we construct a model we will work through an example using our OZ Minerals (OZL) model to provide a guide to the major stepping stones to model development.

Step 1: Mine life

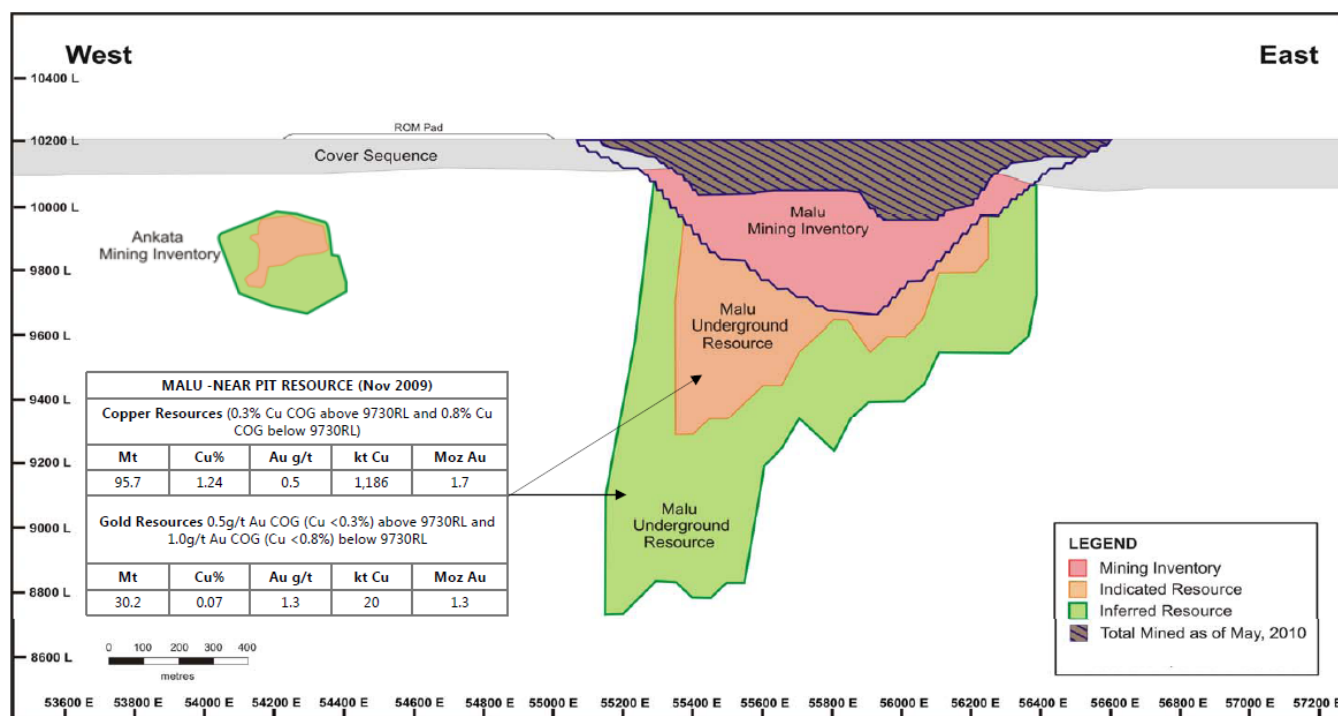
Mine life and production rates go hand in hand as the mine life is a function of the reserve position and the rate of production. Both need to be determined to allow a mine life to be calculated.

As we described in the reserve and resource section, a mine's reserves are the economically mineable part of a measured resource. As such the reserves are effectively the amount of material that the company believes they can mine.

The next step is determining what proportion of resources may potentially be mined in the future. Resources have essentially not been converted into reserves because they have either not had enough drill holes in them to prove up their quality or they fall outside the economic shell of the pit.

For OZL the resources fall outside of the economic pit shell at its Prominent Hill mine site (PH). As such they are unlikely to be mined unless a significantly higher commodity price is assumed. Thus we do not include these resources in our mine life. In the picture below the area in pink, the "mining inventory", is effectively the reserve base.

Figure 151. Prominent Hill pit



Source: OZ Minerals

OZ Minerals has a reserve base of 74.5mt of ore. Its mill has a capacity of 9.6mtpa. Thus its mine life is 7.7 years. As such we model OZ minerals for that explicit period of time.

Figure 152. OZ Minerals simplified reserve and resource statement

RESERVES	TOTAL RESERVES (Attributable)				Metal Contained		
	mt	Cu (%)	Au (g/t)	Ag (g/t)	Cu (kt)	Au (koz)	Ag (koz)
Operation							
Prominent Hill	74.5	1.2%	0.67	3.08	903	1597	7373
					0	0	
TOTAL ATTRIBUTABLE RESERVES	74.5	1.2%	0.67	3.08	903	1597	7373

Source: Company reports and Citi Investment Research and Analysis

Alternatively for an iron ore producer, resources are effectively classified as such because they have not had enough drilling to prove them up, not because they are outside the pit shell. Over time we would expect drilling to occur and resources to be converted into reserves. As such the mine life would not reflect reserve base, but would assume that a proportion of resources are converted into reserves.

An example of this is Fortescue's Cloud Break and Christmas creek operations. Reserves are currently 1547mt of ore. Yet we assume a production over the life of those assets of 2,320mt, due to the fact that we assume conversion of the resources into reserves. Resources currently stand at 2,550mt. so we effectively assume 90% resource to reserve conversion.

Step 2: Milling parameters

The metal production of an operation is dependent on three main parameters:

- Mill throughput
- Grade
- Recovery

At an existing operation, the size of the mill is known. Even at a project stage the company would talk of the conceptual size of the mill. At PH the mill has name plate capacity of 8mtpa. However it has been expanded to over 9mt.

The average grade of a mine should correlate closely to the reserve grade in the reserve and resource statement. The grade profile may change from year to year as the company mines through various parts of the pit, but on a weighted production basis, the grade should at the end of the life of the mine equate to the reserve grade at the start. If the company does not provide any grade profiles, it's often best to just assume the reserve grade for modeling purposes.

Recoveries vary significantly by commodity, by process type and even on a mine by mine basis. Whilst there is no 'industry standard' base metal recoveries tend to be around the 80%-90% range.

The ore through the mill is multiplied by the grade and then multiplied by the recovery rate to deliver the production profile.

Figure 153. Prominent Hill production Profile

Prominent Hill	8yrs	2010	2011e	2012e	2013e	2014e	2015e	2016e	2017e	2018e
Ore Milled	kt	9537	9864	9800	9800	9600	9600	9600	9600	9600
Copper Grade	%	1.33%	1.20%	1.25%	1.25%	1.25%	1.25%	1.25%	1.20%	1.20%
Gold Grade	g/t	0.8	0.7	0.8	0.8	0.6	0.6	0.6	0.6	0.6
Silver Grade	g/t	3.0	2.6	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Copper recovery	%	89%	90%	90%	90%	90%	90%	90%	90%	90%
Gold recovery	%	78%	78%	78%	78%	78%	78%	78%	78%	78%
Silver recovery	%	71%	77%	60%	60%	60%	60%	60%	60%	60%
Copper Con produced	kt	221	207	221	221	216	216	216	207	207
Copper in Con	kt	112	107	110	110	108	108	108	104	104
Gold in Con	koz	196	172	197	197	144	144	144	144	144
Silver in Con	koz	616	631	567	567	556	556	556	556	556

Source: Citi Investment Research and Analysis

PH is a copper mine that produces copper concentrate. This is the intermediary step before the copper is sent to a smelter for further processing and refining into copper metal. Concentrate grades tend to be between 20-40%. As such when production is reported, we state that it is copper in concentrate (or copper in con). At PH the con is higher grade.

Step 3: Revenue

The revenue calculation step is a simple one. Metal volume by price. The only complicating factors are:

- Hedging
- Payable metal
- TCRC's

Hedging: Less common now given the strong balance sheet of many miners. But development projects that have sourced project finance are likely to have hedging in place. The process of calculating revenue doesn't change, it's just that some tonnes will be on spot and some will be at a fixed price.

Payable metal: The smelter pays a percentage of the volume of the metal that is in the concentrate back to the miner. Typically this is c96% for base metals and closer to 98% for precious metals. The payable metal theoretically accounts for losses that the smelter/refiner may occur in the process. Any efficiency gains that can be made allow the refiner to 'keep' or sell back that metal (i.e. 'free metal').

Treatment charge (TC): The price paid for the treatment of the concentrate. It is paid in US\$/t of concentrate processed. **Refining charge (RC):** The price paid for the refining of the metals. It is paid in US\$/lb of copper treated. Whilst TCRC's may seem a cost they are actually a deduction on revenue under accounting standards.

Figure 154. Revenue calculation

		2010	2011e	2012e	2013e	2014e	2015e	2016e	2017e	2018e
Copper in Con	kt	112	107	110	110	108	108	108	104	104
Gold in Con	koz	196	172	197	197	144	144	144	144	144
Silver in Con	koz	616	631	567	567	556	556	556	556	556
Payble - Copper	%	97%	99%	96%	96%	96%	96%	96%	96%	96%
Payble - Gold	%	85%	106%	96%	96%	96%	96%	96%	96%	96%
Payble - Silver	%	92%	90%	96%	96%	96%	96%	96%	96%	96%
Payable Copper in con sold	kt	108	106	106	106	104	104	104	100	100
Payable Gold in con sold	koz	170	182	189	189	139	139	139	139	139
Payable Silver in con sold	koz	559	577	545	545	533	533	533	533	533
Revenue										
LME Copper	US\$/lb	342	413	380	387	365	321	299	306	314
Premium	US\$/lb	-	-	-	-	-	-	-	-	-
Received Copper	US\$/lb	342	413	380	387	365	321	299	306	314
Copper	A\$/m	875	936	861	942	907	824	853	839	860
Gold	A\$/m	229	281	303	295	203	195	207	212	217
Silver	A\$/m	12	19	14	13	11	10	9	9	10
Total	A\$/m	1,116	1,236	1,177	1,249	1,121	1,028	1,069	1,060	1,087
Less Tolling	A\$/m	(21)	(21)	(23)	(25)	(26)	(27)	(50)	(49)	(51)
Total Revenue	A\$/m	1,095	1,144	1,212	1,146	1,152	1,197	1,199	1,154	1,071
Tolling										
TC	A\$/t	51	55	58	63	65	67	128	131	134
TC per lb of copper	A\$/lb	0.11	0.12	0.13	0.14	0.14	0.15	0.28	0.29	0.30
Tolling cost	A\$/m	21	21	23	25	26	27	50	49	51

Source: Citi Investment Research and Analysis

Step 4: Cost of production

Cost modeling tends to be the area where there is the greatest divergence in modeling detail. Some analysts go in to infinite amounts of detail and others prefer a simpler approach and we are in simple is better camp. The complicated modelers would have line items for such costs items as strip ratio's, waste mined, ore mined, ore milled, processing costs, transport costs, SGA costs and we have even seen models where labour levels have been multiplied by average income assumptions to calculate labour costs.

We prefer to model costs on a \$ per tonne of material milled basis. The \$ per tonne of material effectively includes all of the cost parameters. The \$/t of ore milled provides a cost for the milling. It is exclusive of grade. Thus if grade changes but the tonnes milled stays the same, the overall \$ amount of costs stays stable, but the unit cost rises or falls with the amount of end production.

In our experience the more detailed cost modeling does not deliver a greater degree of earnings accuracy. In fact the increased complexity tends to make models unwieldy. Furthermore, many inputs are not disclosed by the company and as such any value is as much guesswork as anything.

To calculate the unit costs we divide the operational costs (pre royalty) by the production volumes. And for a post byproduct credit cash cost, we subtract the revenue of the byproducts (in the case of PH the gold revenue is subtracted) from the total operational cost.

Figure 155. OZ Minerals cost assumptions

Operating Costs		2011e	2012e	2013e	2014e	2015e	2016e	2017e	2018e
Mining Unit costs	US\$/t	41	42	40	35	35	32	33	33
Mining Unit costs	A\$/t								
Mining Costs	A\$m	397	400	409	365	377	381	391	401
Cash Cost - ex by product	US c/lb of copper	182.43	179.26	171.20	151.04	151.04	145.02	154.11	157.97
By Product Credits	US c/lb of copper	121.9	134.0	121.5	82.9	76.4	72.5	77.4	79.3
C1 Production Costs	US c/lb of copper	60.5	45.3	49.7	68.1	74.6	72.5	76.7	78.6
Royalties	US c/lb of copper	7.7	7.5	9.8	10.8	9.6	9.0	9.3	9.5
Other indirect	US c/lb of copper	12.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Cash Cost	US c/lb of copper	80.7	52.7	59.5	78.9	84.2	81.5	86.0	88.2

Source: Citi Investment Research and Analysis

Step 5: Depreciation, royalties and taxes

- Depreciation is the other area where degrees of modeling complexity occur. Depreciation can be modeled on a unit basis where the book value of the asset is depreciated over the units of production in the reserve base. Thus under this method depreciation is a unit item and is a function of production volumes. The simple method is for straight line depreciation over the life of the asset. This is the method we favor.
- Taxes should be simple and be the PTP multiplied by the corporate tax rate of the country. In some instances tax free holidays exist in certain geographies where the governments have curtailed taxes to stimulate investment.
- Royalties are generally set as a percentage of revenue. For PH the revenue is 3% on copper revenues.

Step 6: Capital costs

Capex costs are split into two components.

- Project capex: money spent to build or expand a project
- Maintenance capex: the money that a company needs to spend to keep a mine operating. Also called stay in business (SIB) capex.

Figure 156. OZ Minerals capex profile

		2011e	2012e	2013e	2014e	2015e	2016e	2017e	2018e
Sustaining	A\$m	17	21	21	22	22	23	23	24
Project		100	175	-	-	-	-	-	-
Capex	A\$m	117	196	21	22	22	23	23	24

Source: Citi Investment Research and Analysis

For OZL we expect SIB capex to be c\$20-25mpa. This covers all the maintenance costs associated with the operation. The \$275m in 2011/12 is the capital required for the development of their underground operation.

Step 7: Profit and Loss statement

We model a P/L statement at the mine level. This allows us to generate a stand alone NPV for the asset. Modeling on the asset basis allows us to understand the value of the mine as well as the overall value of the company which is helpful if assets are bought or sold.

Figure 157. PH profit and loss statement

Financials		2010	2011e	2012e	2013e	2014e	2015e	2016e	2017e	2018e
Revenue	A\$m	1,116	1,236	1,177	1,249	1,121	1,028	1,069	1,060	1,087
Operating Costs	A\$m	438	419	423	434	391	404	432	440	451
Royalty/Other Costs	A\$m	17	19	18	25	28	26	27	27	27
Total Costs	A\$m	455	437	441	459	419	430	458	467	479
EBITDA	A\$m	662	799	736	790	703	599	610	593	608
Depreciation	A\$m	140	141	144	148	151	155	159	163	167
EBIT	A\$m	522	658	592	643	551	444	451	431	441
Tax	A\$m	156	197	178	193	165	133	135	129	132
Net Op Profit	A\$m	365	461	415	450	386	310	316	301	309
Sustaining	A\$m	8	17	21	21	22	22	23	23	24
Project		-	-	275	-	-	-	-	-	-
Capex	A\$m	20	16	296	21	22	22	23	23	24
Net Cash Flow	A\$m	485	585	263	576	516	443	452	441	452
NPV	A\$m									
EBITDA Margin		59%	65%	63%	63%	63%	58%	57%	56%	56%

Source: Company reports, Citi Investment Research and Analysis

Step 8: Corporate level modeling

As well as modeling at the mine site level we also model at the corporate level. Here exploration costs, corporate costs and other costs are accounted for.

Figure 158. OZL Corporate P&L statement

Profit and Loss	YE DEC	2010	2011e	2012e	2013e	2014e	2015e	2016e	2017e	2018e
Total Income	A\$m	1,086.5	1,226	1,177	1,249	1,121	1,028	1,069	1,060	1,087
Operating Costs	A\$m	385.2	405	441	459	419	430	458	467	479
SGA	A\$m	54.9	41	21	21	22	22	23	23	24
Exploration	A\$m	50.2	84	10	11	11	-	-	-	-
Other	A\$m	-	25	-	-	-	-	-	-	-
Total Costs	A\$m	490.3	555	472	491	451	452	481	490	502
EBITDA	A\$m	596.2	671	705	759	670	576	588	570	584
Depreciation	A\$m	152.6	154	144	148	151	155	159	163	167
EBIT	A\$m	443.6	517	561	611	519	421	429	407	417
Net Interest Expense	A\$m	(27.6)	(38)	(42)	(42)	(55)	(67)	(79)	(93)	(108)
Operating Profit Before Tax	A\$m	471.2	555	604	653	574	489	508	501	526
Tax	A\$m	65.2	175	181	196	172	147	152	150	158
Profit pre Minorities	A\$m	406.0	380	423	457	402	342	355	351	368
Exceptionals	A\$m	(180.9)	77	-	-	-	-	-	-	-
Reported Profit	A\$m	586.9	303	423	457	402	342	355	351	368
Weighted Issued Shares	m	314	323	323	323	323	323	323	323	323
EPS - reported	\$/share	1.872	0.937	1.307	1.41	1.24	1.06	1.10	1.08	1.14
EPS - Underlying	\$/share	1.252	1.175	1.31	1.41	1.24	1.06	1.10	1.08	1.14

Source: Company reports, Citi Investment Research and Analysis

We also model the balance sheet and cash flow statement at the group level.

Figure 159. OZL cash flow and balance sheet

Cash Flow		2010	2011e	2012e	2013e	2014e	2015e	2016e	2017e	2018e
Net Operating Cash Flow	A\$m	616.1	711	670	638	582	521	540	538	561
Capex - Project	A\$m	-	-	(275)	-	-	-	-	-	-
Capex - Maintenance	A\$m	(54.9)	(52)	(21)	(21)	(22)	(22)	(23)	(23)	(24)
Exploration capitalised	A\$m	(10.4)	(52)	(20)	(21)	(22)	-	-	-	-
Net Investments	A\$m	(131.1)	(247)	-	-	-	-	-	-	-
Free Cash Flow	A\$m	419.7	359	354	595	539	499	517	515	537
Dividends Paid	A\$m	(93.6)	(227)	(96)	(115)	(105)	(90)	(39)	-	-
Equity Raised	A\$m	(0.1)	(389)	-	-	-	-	-	-	-
Change in Borrowings	A\$m	-	-	-	-	-	-	-	-	-
Other Financing	A\$m	-	-	-	-	-	-	-	-	-
Net Cash Flow	A\$m	326.0	(256)	258	480	433	409	478	515	537
Exchange Rate Adjustments	A\$m	(68.0)	(7)	-	-	-	-	-	-	-
Increase in Cash	A\$m	258.0	(263)	258	480	433	409	478	515	537
Cash at Period End	A\$m	1,334.2	1,071	1,100	1,170	1,186	1,329	1,576	1,809	2,037
Balance Sheet		2010	2011e	2012e	2013e	2014e	2015e	2016e	2017e	2018e
Cash/Investments	A\$m	1,334.2	1,071	1,329	1,809	2,243	2,652	3,130	3,645	4,182
Accounts Receivable	A\$m	180.9	155	163	165	146	134	147	146	150
Inventory	A\$m	259.5	135	142	144	127	116	128	127	130
Other Current Assets	A\$m	8.0	14	14	14	14	14	14	14	14
Current Assets	A\$m	1,782.6	1,376	1,649	2,133	2,530	2,917	3,419	3,932	4,477
Net Fixed Assets	A\$m	1,288.1	1,439	1,590	1,464	1,334	1,201	1,065	926	783
Total Financial Assets	A\$m	45.9	31	31	31	31	31	31	31	31
Goodwill	A\$m	-	-	-	-	-	-	-	-	-
Intangible assets	A\$m	-	-	-	-	-	-	-	-	-
Other Non Current Assets	A\$m	270.3	338	338	338	338	338	338	338	338
Total Assets	A\$m	3,386.9	3,183	3,608	3,966	4,233	4,487	4,853	5,227	5,628
Accounts Payable	A\$m	64.6	151	130	137	124	124	134	136	140
Debt - Short Term	A\$m	-	-	-	-	-	-	-	-	-
Other current liabilities	A\$m	3.2	5	5	5	5	5	5	5	5
Current Liabilities	A\$m	67.8	156	134	141	129	129	138	141	144
Debt - Long Term	A\$m	-	-	-	-	-	-	-	-	-
Other non current liabilities	A\$m	14.8	94	94	94	94	94	94	94	94
Provisions	A\$m	13.3	14	14	14	14	14	14	14	14
Total Liabilities	A\$m	95.9	264	242	250	237	237	247	249	252
Minority Interests	A\$m	-	-	-	-	-	-	-	-	-
Preferred Equity	A\$m	-	-	-	-	-	-	-	-	-
Common Equity	A\$m	5,208.8	4,820	4,820	4,820	4,820	4,820	4,820	4,820	4,820
Reserves	A\$m	106.6	110	110	110	110	110	110	110	110
Retained Profit	A\$m	(2,024)	(2,011)	(1,564)	(1,214)	(934)	(680)	(323)	47	445
Share Holders Funds	A\$m	3,291.0	2,919	3,366	3,716	3,996	4,250	4,607	4,978	5,375

Source: Company reports, Citi Investment Research and Analysis

Step 9: Deriving the NPV

Our overall NPV for OZL is a combination of the NPV derived from the operational assets, plus ownership in other assets (such as their stake in Sandfire), plus an exploration upside figure which takes into account OZL exploration package, less net debt, less corporate costs. The combination of the above allows a group NPV to be derived.

Figure 160. OZL NPV analysis

WACC	9.7%	A\$M	Ownership	A\$M	A\$/share
Prominent Hill		2,422	100%	2,422	7.48
Carrapateena		313	100%	313	0.96
Sandfire	7.80	218	100%	218	0.67
Exploration and Investments		100	100%	100	0.31
Corporate Costs		(158)	100%	(158)	(0.49)
Net Cash (Debt)		940	100%	940	2.90
Total		3,835		3,835	11.84

Source: Citi Investment Research and Analysis

For a company with multiple assets, we derive NPV's on each of the individual mines and the sum them up to derive a group NPV. The table below looks at the NPV calculations for Newcrest.

Figure 161. NCM NPV analysis

	A\$M	Ownership	A\$M	A\$/share
Cadia	5,462	100%	5,462	7.14
Telfer	1,744	100%	1,744	2.28
Cracow	163	70%	163	0.21
Gosowong	1,348	82.5%	1,348	1.76
Hidden Valley	680	50%	680	0.89
Wafi/Golpu	3,282	50%	3,282	4.29
Lihir Island	6,527	100%	6,527	8.53
Mount Rawdon	117	100%	117	0.15
Bonikro	319	84%	319	0.42
Exploration	2,000	100%	2,000	2.61
Tax Losses	500	100%	500	0.65
Corporate Costs	(612)	100%	(612)	(0.80)
Net (Debt) / cash	(1,138)	100%	(1,138)	(1.49)
Total	20,393		20,393	26.66

Source: Citi Investment Research and Analysis

Appendix 1: Glossary

Glossary

Agitation: metallurgy, the act or state of being stirred or shaken mechanically, sometimes accomplished by the introduction of compressed air.

Adit: an almost horizontal entrance to a mine.

Alloy: a compound of more than one metal.

Alluvial deposit: sedimentary material deposited by running water.

Anode: rectangular metal plate with a positive charge, cast in a shape suitable for refining by the electrolytic process.

Anomaly: relatively unusual geochemical or geophysical response from a geographical or geological surface – generally positive exploration result.

Ash: the residue remaining after a pulverised sample of the coal is incinerated under standard laboratory conditions.

Assay: a chemical test performed on rock samples to determine the metal content.

Autoclave system: oxidation process in which high temperatures and pressures are applied to convert refractory sulphide mineralisation into amenable oxide ore.

Autogenous grinding: process of grinding ore with the ore itself, in a rotating cylindrical mill. The grinding ore medium is typically large pieces of ore.

Ball mill: steel cylinder containing about 40% volume of steel balls, used to grind ore. The balls in the rotating mill tend to cascade, grinding the ore.

Base metal: non-precious metal, such as copper, lead or zinc. Base metals are commonly used in industry alone rather than in alloys.

Bench: successive steps/horizontal increments mined as an open pit progresses deeper.

Beneficiation: concentrating or enriching of the valuable minerals in ore.

Bio-leaching: metal extraction method that uses bacteria to oxidise refractory sulfide ore, making it amenable to normal oxide ore processing techniques.

Block caving: method of underground mining whereby large blocks of ore are delineated and undercut by drilling, causing the ore to break or cave under its own weight, then extracted through a system of draw points and conduits.

Borehole: a hole drilled through the strata to obtain geological specimens or to provide access for geophysical devices.

Bullion: metal formed into bars or ingots.

By-product: a secondary product recovered in the extraction process (e.g. molybdenum is a common by-product of copper).

Calcine: concentrate type used in smelting; the sulphur has been driven off by oxidation.

Carbon: fine-sized granules of activated carbon, typically derive from ground coconut shell used because of its hardness.

Carbon-in-Column (CIC): method of recovering gold and silver from pregnant solution, using a heap leaching process. The precious metal, suspended by up-flow of solution through a tank, is adsorbed onto fine carbon.

Carbon-in-Leach (CIL): method of recovering gold and silver from fine ground ore by simultaneous dissolution and adsorption of the precious metals onto fine carbon in an agitated tank of ore solids/solution slurry. The carbon flows counter currently to the head of the leaching circuit.

Carbon-in-Pulp (CIP): method of recovering gold and silver from fine ground ore by adsorption of the precious metals onto fine carbon in an agitated tank of ore solids/solution slurry. This recovery step in the process follows the leaching process which is done in similarly agitated tanks, but without carbon.

Cash operating cost: cost per unit that is equivalent to direct operating expense, including mining and processing, waste stripping and mine-site administration, less production royalties, mining taxes and by-product credits for payable metals recovered.

Cathode: rectangular metal plate with a negative charge used in electrolytic refining. The deposited metal recovered onto this plate is called cathode metal and is melted into commercial shapes such as bars or ingots.

Clarifier filter: pressure filter containing cloth leaves which support a porous filter medium and filter out all fine solids from relatively clear feed solution. The sludge collected is periodically flushed from the filter on a regular cycle.

Classifier: mineral processing machine which separates mineral particles according to size and density.

Coking coal: coal which is suitable for making coke, used in steel production.

Complex ore: ore containing a number of minerals of economic value.

Concentrate: fine-textured product separated in the milling process that contains a high percentage of valuable metal.

Concentrator: milling plant that produces a concentrate of the valuable minerals or metals. Further treatment is required to recover the pure metal.

Cone crusher: machine which crushes ore between a gyrating cone or crushing head and an inverted, truncated cone known as a bowl.

Contango: the positive difference between the spot market price and the forward market price.

Core: cylindrical rock sample generally produced by diamond drilling.

Cut-and-fill: method of underground mining whereby the ore is removed in slices, or lifts, and then the excavation is filled with rock or other waste material (backfill), before the subsequent slice is extracted.

Cut-off grade: minimum grade level below which the mineralised geology is not regarded as economic ore. The minimum grade of ore used to establish reserves.

Cyanidation: method of extracting exposed gold or silver grains from crushed or ground ore by dissolving it in a weak cyanide solution, usually in agitation tanks in a mill or on ore heaps.

De-aeration tower: closed tank from which air is evacuated by a vacuum pump on the closed system. As solution flows through a contained media of large surface area, oxygen is removed (boils) from the solution.

Dewatering: process of separating solids from solution by sedimentation in tanks called thickeners, or by filtering the solution through filter cloth in filters.

Diamond drilling: drilling method using a diamond bit to cut a cylindrical hole for taking core samples.

Dilution: mixing of ore grade material with non-ore grade waste material in the mining process.

Disseminated: ore deposit consisting of fine particles of ore mineral dispersed through the enclosing rock.

Doré: unrefined gold or silver bullion bars containing approximately 90% of the precious metal, which will be further refined to near pure gold.

Dragline: large earthmoving machine with a bucket suspended from a crane like boom.

Drift: a horizontal tunnel driven alongside an ore deposit in order to gain access to the deposit.

Electrolysis: process of passing current through a solution containing dissolved metals directing the metals to be deposited onto the negatively charged cathode.

Electrolytic refining: process of purifying metal plates that are suspended as anodes in an electrolytic bath. The targeted metal is progressively plated onto refined starter sheets called cathodes.

Electrowinning: the removal of precious metals from solution by the passage of current through an electrowinning cell. A direct current supply is connected to the anode and cathode. As current passes through the cell, metal is progressively deposited on the cathode. When sufficient metal has been deposited onto the cathode, it is removed from the cell and the sludge rinsed off the plate and dried for further refinement.

Enriched uranium: uranium that has been processed to contain higher-than-natural levels of uranium-235.

Filter press: pressure filter containing cloth leaves which are supported within a closed container to remove solids from a fine-feed slurry. The solid sludge collected is periodically removed from the filter by opening the filter plates, scraping off the solids and replacing the plates with new and clean filter cloth or filter paper for continued filtering.

Flotation: concentrating process in which valuable mineral particles are activated to become attached to bubbles and float above the waste particles, then isolated in a solid-solution pulp.

Flux: mix of chemical substances that react with gangue minerals to form slag, which are liquid at smelting furnace temperature and sufficiently low in density to float on the molten bath of metal.

Fold: any bending or wrinkling of a rock strata.

Free milling: ores of gold or silver from which the precious metals can be recovered by concentrating methods without resorting to pressure leaching or other chemical treatment.

Gangue: collective term for worthless minerals in an ore deposit.

Geophysics: scientific method of prospecting, utilising magnetics, specific gravity, electrical conductivity and radio-activity to detect minerals.

Gold precipitate: fine sludge of gold, silver and zinc. The precious metals are precipitated out of solution by the addition of fine zinc dust.

Grade: ratio measuring the quantity of targeted metal in a given quantity of ore. Head grade is the average grade of ore fed into a mill or heap leach metal recovery operation. Recovered grade is the grade of actual metal content of ore determined after processing.

Grizzly: grating, usually constructed of steel rails, placed over the top of a chute or ore pass for the purpose of stopping large pieces of rock or ore that may hang up in the pass.

Gyratory crusher: machine that crushes ore between an eccentrically mounted crushing cone and a fixed crushing throat. Typically, it has a higher capacity than a jaw crusher.

Heap leach: mineral processing method involving the crushing and stacking of ore on an impermeable liner upon which leach solutions are sprayed to dissolve metals such as gold and copper. The metal in the collected solution flowing from the leach pad is subsequently treated to recover the metals.

High volatile coal: coal with more than 17% volatile matter.

Hydrometallurgical: pertaining to metallurgical processes in which the principal medium is water; can be at temperatures and pressure substantially above ambient conditions when autoclaves are used, generally involves chemical reactions.

Indicated Mineral Resource: that part of a Mineral Resource for which quantity, grade or quality, densities, shape and physical characteristics, can be estimated with a level of confidence sufficient to allow the appropriate application of technical and economic parameters, to support mine planning and evaluation of the economic viability of the deposit. The estimate is based on detailed and reliable exploration and testing information gathered through appropriate techniques from locations such as outcrops, trenches, pits, workings and drill holes that are spaced closely enough for geological and grade continuity to be reasonably assumed.

Inferred Mineral Resource: that part of a Mineral Resource for which quantity and grade or quality can be estimated on the basis of geological evidence and limited sampling and reasonably assumed, but not verified, geological and grade continuity. The estimate is based on limited information and sampling gathered through appropriate techniques from locations such as outcrops, trenches, pits, workings and drill holes.

Ion exchange: the exchange of ions between two electrolytes or between an electrolyte solution and a complex. It is used to denote the processes of purification and separation of aqueous and other ion-containing solutions.

Jaw crusher: a machine in which rock is broken by the action of swinging steel plates.

Jig: milling equipment used to concentrate high density minerals by gravity on a screen submerged in water, either by the reciprocating motion of the screen or by the pulsation of water through it.

Leaching: chemical process used in the extraction of valuable minerals from ore.

Lode: mineral deposit in solid rock usually with well defined boundaries.

Longhole stopping: a method of underground mining involving the drilling of holes up to 30 meters or longer into an ore bearing zone and then blasting a slice of rock which falls into an open space. The broken rock is extracted and the resulting open chamber is not immediately filled with supporting material.

Longwall mining: a high capacity underground mining method which utilises a mechanical shearer to cut the coal.

Magnetic survey: survey of a geological target's magnetic field, either from the ground or from the air.

Measured mineral resource: that part of a Mineral Resource for which quantity, grade or quality, densities, shape, physical characteristics are so well established that they can be estimated with confidence sufficient to allow the appropriate application of technical and economic parameters, to support production planning and evaluation of the economic viability of the deposit. The estimate is based on detailed and reliable exploration, sampling and testing information gathered through appropriate techniques from locations such as outcrops, trenches, pits, workings and drill holes that are spaced closely enough to confirm both geological and grade continuity.

Metallurgical coal: coal used in the production of steel.

Metallurgy: study of extracting metals from their ores.

Mill: processing plant which treats ore for the purpose of upgrading the mineral content into a higher grade product called a concentrate, which is further treated, and disposing of the waste minerals to an impoundment area.

Mineral: naturally occurring substance having distinctive physical properties and chemical composition.

Mineral reserve: the economically mineable part of a Measured or Indicated Mineral Resource demonstrated by at least a preliminary feasibility study. This study must include adequate information on mining, processing, metallurgical, economic and other relevant factors that demonstrate, at the time of reporting, that economic extraction can be justified. A Mineral Reserve includes diluting materials and allowances for losses that may occur when the material is mined.

Mineral resource: a concentration or occurrence of natural, solid, inorganic or fossilized organic material in or on the Earth's crust in such form and quantity and of such a grade or quality that it has reasonable prospects for economic extraction. The location, quantity, grade, geological characteristics and continuity of a Mineral Resource are known, estimated or interpreted from specific geological evidence and knowledge.

Mining claim: that portion of applicable mineral lands that a party has staked or marked out in accordance with applicable mining laws to acquire the right to explore for and exploit the minerals under the surface.

Native metal: metal occurring in nature in pure form, separate from other elements or minerals.

Net profits interest royalty: a royalty based on the profit remaining after recapture of certain operating, capital and other costs.

Net smelter return royalty: a royalty based on a percentage of valuable minerals produced with settlement made either in kind or in currency based on the spot sale proceeds received less all of the offsite smelting, refining and transportation costs associated with the purification of the economic metals.

Non-refractory: ore containing gold or other metal that can be satisfactorily recovered using gravity concentration or cyanidation methods.

Open pit: a mine where the minerals are mined entirely from the surface.

Ore: mixture of valuable minerals and gangue minerals from which at least one of the minerals can be extracted economically. An ore body is a natural concentration of valuable material amenable to economic extraction.

Outcrop: part of rock formation that can be seen naturally at the earth's surface.

Overburden: material which overlies a deposit of useful material.

Oxide ore: mineralised rock in which some of the original minerals have been oxidized.

PCI: pulverised coal injection.

Pig iron: raw iron produced from blast furnaces.

Placer: deposit of sand and gravel containing valuable metals such as gold, tin or diamonds.

Platinum group metals (PGMs): collective term for ruthenium, rhodium, palladium, osmium, iridium and platinum. Platinum metals occur as native metals, invariably associated with each other and gold, silver, copper, nickel and iron.

Precipitation: precipitation in hydrometallurgy involves the chemical precipitation of either metals or their compounds or of the contaminants from aqueous solutions. Precipitation will proceed when, through reagent addition, evaporation, pH change or temperature manipulation, any given species exceeds its limit of solubility.

Probable Mineral Reserve: the economically mineable part of an Indicated and in some circumstances a Measured Mineral Resource demonstrated by at least a preliminary feasibility study. This study must include adequate information on mining, processing, metallurgical, economic, and other relevant factors that demonstrate, at the time of reporting, that economic extraction can be justified.

Prospect: mining property with demonstrable potential for ore discovery.

Proven reserve: the economically mineable part of a measured resource, demonstrated by at least a preliminary feasibility study. This study must include adequate information on mining, processing, metallurgical, economic, and other

relevant factors that demonstrate, at the time of reporting, that economic extraction is justified. A large increase in the price of a particular mineral would, therefore, in all probability, result in an increase in proven reserves.

Pyrometallurgical: pertaining to ore-refining process, such as smelting, dependent on the action of very high levels of heat. Examples include fluid-bed roasting of zinc concentrates, matte smelting of copper concentrates, fire refining of blister copper.

Ramp: an inclined underground tunnel that provides access to and throughout an ore body for exploration, ventilation or exploitation purposes in an underground mine.

Reclamation: the process by which lands disturbed as a result of mining activity are modified to support beneficial land use. Reclamation activity may include the removal of buildings, equipment, machinery and other physical remnants of mining, closure of tailings storage facilities, leach pads and other mine features, and contouring, covering and re-vegetation of waste rock and other disturbed areas.

Recoverable: portion of metal contained in ore that can be extracted by processing.

Recovery rate: a term used in process metallurgy to indicate the proportion of valuable material physically recovered in the processing of ore. It is generally stated as a percentage of the material recovered compared to the total material originally present.

Reef: a South African term for a continuous mineral deposit, especially gold or platinum bearing quartz.

Refining: process of transforming material into its purest state ready for its intended end-use. Precious metal bullion is fire-refined to high purity gold and silver.

Refractory: mineralisation normally requiring more sophisticated processing technology for extraction, such as roasting or autoclaving under pressure.

Reverberatory furnace: long, flat furnace used to slag gangue minerals and produce a matte of sulfide minerals.

Roasting: the treatment of ore by heat and air, or oxygen enriched air, in order to remove sulphur, carbon, antimony or arsenic.

Rod mill: rotating steel cylinder that uses steel rods as a means of grinding ore.

Royalties: government charge for the mining of coal.

Run-of-mine (ROM): mined ore of a size that can be processed without further crushing.

Semi-coking coal: coal which is suitable for making strong coke in its own right but suitable as a component in coke oven blends.

Shaft: a vertical passageway within an underground mine. Shafts can be used for ventilation or transporting workers, equipment ore etc.

Slag: vitreous mass of glass material separated from the fused metals, product of the smelting process.

Smelting furnace: furnace, usually cylindrical, employed to melt concentrated valuable metals and minerals; impurities are isolated and removed in the slag that floats on top of the melt.

Sodium cyanide: chemical used in the milling and heap leaching of gold ores to dissolve gold and silver.

Solutions: includes water solutions in the leaching circuit; fresh make-up water; pregnant solution containing significant amounts of precious metals and leaching reagents (chemicals); barren solution which contains small quantities of valuable metals; leaching reagents.

Solvent extraction-electrowinning (SXEW): metallurgical technique, applied only to copper, in which metal is dissolved from the rock by organic solvents, then recovered from solution by electrolysis.

Stope: an area in an underground mine where ore is mined.

Strata: beds or layers of sedimentary rock.

Strike length: length of a deposit able to be accessed for mining.

Strip ratio: ratio of quantity of overburden waste material that needs to be removed, to tons of ore being accessed, in an open pit mining operation.

Strip vessel: closed container in which loaded carbon is placed in order to remove precious metals from the surface of the activated carbon.

Stripped carbon: activated (coconut shell) carbon, which has had the precious metal values removed by chemical processing, prior to preparation for reactivation and recycling back to the stripping circuit.

Tailings: crushed or finely ground waste rock from which valuable minerals or metals have been extracted.

Tenement: mining lease, exploration permit, or mineral development license.

Thermal coal: coal which is combusted to provide heat for steam generation and subsequent power generation.

Total cash cost: cash operating cost plus production royalties and mining taxes.

Total production cost: total cash cost plus depreciation and amortization.

Troy ounce: measure of weight in which precious metals are sold. One Troy ounce contains 31.1 grams.

Waste: barren rock or mineralised material that is too low in grade to be economically processed.

Zinc dust: finely ground or formed zinc metal used to precipitate gold and silver from solution. The precious metal values precipitate into a fine sludge and the zinc metal replaces these values in solution.

Appendix A-1

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